

A unification of graded and substructural logics

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Abstract

Type systems which account for resource sensitive computations can generally be split into two styles: First, substructural logics such as Linear Logic which seek to restrict weakening and contraction and reintroduce them in a controlled manner; And second, graded systems which allow weakening and contraction by default, but track the use of variables quantitatively in some algebraic structure – usually a semiring. We present GRASS (**G**raded and **S**ubstructural), a type system which incorporates mechanisms from both of these approaches, thus allowing maximally flexible control over variable usage. Furthermore, GRASS allows grades from an arbitrary collection of grade algebras to coexist in the same system, thus allowing different variables to be controlled with respect to different notions of resource within the same program. We develop the categorical semantics of GRASS, and find that, on the level of categorical semantics, it subsumes multiple established systems such as LNL [4], Adjoint Logic [28], and mGL [32].

Keywords: Programming Languages, Substructural Logic, Graded Modal Types, Categorical Semantics

1 Introduction

The treatment of program variables as resources has seen a rise over the last few decades. While classically variables were treated like propositions which could be reused or discarded at will, the resourceful treatment acknowledges that this is not true in many common computing use cases. Treating variables as resources allows programmers to verify interesting correctness properties, like the absence of memory leaks or use-after-free errors.

Two distinct approaches to the resourceful analysis of programs exist in the literature. *Substructural logics* restrict weakening and contraction,³ prohibiting variables from being duplicated or discarded by default, then selectively re-introducing these capabilities where needed. In Linear Logic [15], this is done via the of-course modality. Benton generalized Linear Logic to Linear-non-Linear Logic [4], a system which features two sub-logics, one where assumptions are intuitionistic and one where assumptions can be either intuitionistic or linear. Pruiksma et al. [28] further generalized this system to Adjoint Logic, allowing any number of logics, each with its own set of permitted structural rules, to coexist in the same system. This allows programmers to specify exactly which structural rules are applicable to which variable in a program, independently of the other variables.

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³ Exchange can also be restricted to create non-commutative logics [19], but we do not consider this here.

The second approach is to allow structural rules by default, but to annotate assumptions with so called *grades*, usually drawn from a semiring. In this approach, the control over the usage of assumptions stems from the algebraic structure of the grades with graded contraction being modeled by addition and weakening being modeled by an explicit zero element. Graded systems are commonly parametrized by their grade structure and can therefore model a variety of resource notions. Examples of applications are: Garbage collection [10], liveness analysis [26], security [1,12,21,23,24], dataflow analysis [26,31,30], numerical sensitivity tracking [3,11], and reasoning about probabilistic programs [29].

In this work we present GRASS, a type system which unifies these two approaches. Our system allows grades drawn from any number of grade algebras to be used simultaneously. For example, we can type a function `read_decrypt`: $(k :^{\text{Hi}} \text{Key}) \rightarrow (f :^1 \text{FileHandle}) \rightarrow \text{Text}$, which reads the contents of an encrypted file f using the decryption key k . The grade annotation Hi ensures that the key is used in a secure context, while the grade annotation 1 ensures that the file handle is properly accessed and closed. Our type system further allows control over graded contraction and weakening. For example two file handles used linearly should not be contractible into one which can be used twice.

We develop a categorical semantics for our type system, based on graded linear exponential comonads [17]. Our type system is assembled from a collection of graded logics connected by morphisms, and this design is mirrored in the categorical semantics. We develop a novel notion of morphisms between graded comonads based on morphisms of actegories [sic] [9]. We show that known categorical models for substructural logics give rise to models of GRASS in a straightforward way. Thus, on the level of categorical semantics, GRASS is capable of expressing both graded and substructural logics, as well as combinations thereof.

2 Single-mode system

GRASS is paratemeterized by a collection of so called *modes*, each of which admits different graded structural rules. In this section we explain our novel weakening and contraction rules in the situation when this collection is instantiated to consist of only one mode. In this case, GRASS simplifies to be an ordinary graded type theory, nearly identical to the purely graded fragment of $\mathbf{mGL}$ [32], except with modified weakening and contraction rules. We will focus on the structural rules here; the full set of typing rules is given in §A of the appendix [16]. Before we begin with our treatment of the structural rules, we review weakening and contraction in existing graded systems, then explain how we control them further using modes. For contraction, the idea is to restrict it to a set of mutually contractible grades, subject to some conditions. For weakening the situation is simpler: It can just be toggled on or off. We end this section with a discussion of morphisms of modes, which capture the notion that one mode has more structure than another.

2.1 Weakening and contraction in existing systems

We review the behavior of graded weakening and contraction in existing systems. These structural rules are present in most graded type systems, either as explicit rules [13,25,26,32], as admissible rules [10,12,23], or even a mix of the two with an explicit weakening rule, but admissible graded contraction [24]. In this section we will consider typing judgments of the form $\rho \odot \Gamma \vdash t : T$ where t is a term, T is a type and Γ a typing context. Furthermore, ρ is a list of grades of the same length as Γ . Each entry of ρ dictates the capabilities with which the variable in the corresponding position in Γ may be used by t . The graded contraction and weakening rules are now

$$\begin{array}{c} \text{CONT} \\ \frac{\rho, q_1, q_2 \odot \Gamma, x : A, y : A \vdash e : B}{\rho, q_1 + q_2 \odot \Gamma, z : A \vdash [z/x, z/y]e : B} \end{array} \qquad \begin{array}{c} \text{WEAK} \\ \frac{\rho \odot \Gamma \vdash e : A}{\rho, 0 \odot \Gamma, x : B \vdash e : A} . \end{array}$$

The contraction rule allows sharing a variable across two sub-expressions in a term, but also requires its capabilities to be split between the two expressions. The weakening rule allows the introduction of additional variables into the context, marked by the grade 0. The graded structural rules can be seen as generalizations of the structural rules available using the of-course modality $!A$ in Linear Logic, where they correspond to maps $!A \multimap !A \otimes !A$ and $!A \multimap I$ respectively. The graded structural rules generalize these to $\square_{q_1+q_2} A \multimap \square_{q_1} A \otimes \square_{q_2} A$ and $\square_0 A \multimap I$, where the type $\square_q A$ indicates use at grade q . This allows

fine grained tracking of the usage of each hypothesis. Depending on the grade algebra being used, the graded structural rules may simply correspond to the ones of linear logic, but by choosing different kinds of semirings we can enforce different styles of capability tracking. For example, if we use the natural numbers to grade our types we can track the precise number of times variables are being used rather than simply one or zero.

2.2 Modes

In this section we formalize a novel structure used to control weakening and contraction in the graded setting, which we call *modes*. The main insight is that contraction can be controlled by specifying a set of mutually contractible grades called an *ideal*. The notion of ideals is borrowed from ring theory, where ideals are a well-known concept (see e.g. Lang [20]).

Definition 2.1 A *semiring* is a set R together with binary operations $(+), (\cdot): R \times R \rightarrow R$ and elements $0, 1 \in R$ such that $(R, +, 0)$ is a commutative monoid, $(R, \cdot, 1)$ is a monoid and such that for all $x, y, z \in R$ we have $x \cdot (y + z) = (x \cdot y) + (x \cdot z)$ and $(x + y) \cdot z = (x \cdot z) + (y \cdot z)$ and $0 \cdot x = 0 = x \cdot 0$. A *preordered semiring* is a semiring equipped with a preorder $(\leq)$ such that $(+)$ and $(\cdot)$ are monotone with respect to $(\leq)$, i.e. whenever $x \leq x'$ and $y \leq y'$, then $x + y \leq x' + y'$ and likewise for $(\cdot)$. Preordered semirings are also called *grade algebras* and their elements are called *grades*.

- Example 2.2** (i) The set of natural numbers $\mathbb{N}$ with the usual arithmetic operations forms a semiring. There are several natural choices of preorder on $\mathbb{N}$: The usual ordering $0 \leq 1 \leq \dots$, its opposite $0 \geq 1 \geq \dots$, and the *discrete ordering* $n \leq m \iff n = m$.
- (ii) The set $\{0, 1\}$ can be equipped with two distinct semiring structures, by either defining $1 + 1 = 0$ or $1 + 1 = 1$. Only the latter structure will be of interest to us. There are three interesting preorders, defined by either $0 \leq 1$ or $0 \geq 1$, or the discrete ordering.
- (iii) The *none-one-tons* semiring is the set $\{0, 1, \omega\}$. Addition and multiplication are fixed by the semiring axioms and the equations $1 + 1 = 1 + \omega = \omega + \omega = \omega \cdot \omega = \omega$. Again, there are multiple possible preorders, for example, the preorder generated by $0 \leq \omega$ or the one generated by $0 \leq \omega \leq 1$.
- (iv) There is a unique semiring with $0 = 1$, which we denote $\top$.

Definition 2.3 Let R be a semiring. A (*two sided*) *ideal* in R is a subset $I \subseteq R$ such that $0 \in I$, and whenever $x, y \in I$, then $x + y \in I$, and for any $x \in I$ and $r \in R$, we have $r \cdot x \in I$ and $x \cdot r \in I$.

- Example 2.4** (i) Any semiring R admits the trivial ideals $\{0\} \subseteq R$ and $R \subseteq R$.
- (ii) The sets $\{0, \omega\} \subseteq \{0, 1, \omega\}$ and $\mathbb{N} \setminus \{1\} \subseteq \mathbb{N}$ are ideals.
- (iii) Given a family of ideals I_j , the intersection $\bigcap_j I_j$ is again an ideal. Hence, given any subset $X \subseteq R$ there exists a smallest ideal containing X : There exists an ideal containing X since $R \subseteq R$ is an ideal itself. Now consider the set of all such ideals and take their intersection.

Definition 2.5 A mode m is a triple $(R_m, \text{Cont}(m), \text{Weak}(m))$ where R_m is a grade algebra, $\text{Cont}(m) \subseteq R_m$ is an ideal and $\text{Weak}(m) \in \{\text{true}, \text{false}\}$.

Remark 2.6 Our notion of modes is a generalization of the notion of modes given in Adjoint Logic [28]. There, a mode is equipped with two boolean values, indicating whether weakening and contraction are permitted. In the graded setting, we can control weakening and contraction more granularly by enabling them only for a subset of grades. Contraction is modeled by a set of mutually contractible grades. We will see below how the conditions that this set is an ideal arise naturally in the syntax. While it may appear that weakening is still controlled by a boolean, the situation is a bit more subtle: Most graded systems, including GRASS, have a subsumption rule, which states that grades need only be upper bounds on the actual usage of variables. Thus, when weakening is enabled, unused variables can be introduced at any grade $q \geq 0$, by applying weakening followed by subsumption. We do not require 0 to be the bottom element of a grade algebra, so weakening is controlled by the predicate $\text{Weak}(m)$ and the choice of preorder.

Example 2.7 The following modes will serve as running examples: $U = (\top, \top, \text{true})$, $R = (\top, \top, \text{false})$, $A = (\{0, 1\}, \{0\}, \text{true})$ and $L = (\mathbb{N}, \{0\}, \text{false})$. We will see in Example 2.8 how these modes can recover intuitionistic, relevant, affine and linear logic respectively.

2.3 Restricted structural rules

We can now describe the controlled structural of GRASS. Fix a mode $\mathbf{m} = (R_{\mathbf{m}}, \text{Cont}(\mathbf{m}), \text{Weak}(\mathbf{m}))$. The new rules for contraction and weakening are as follows:

$$\begin{array}{c} \text{S.CONT} \\ \frac{q_1, q_2 \in \text{Cont}(\mathbf{m}) \quad \rho, q_1, q_2 \odot \Gamma, x : A, y : A \vdash e : B}{\rho, q_1 + q_2 \odot \Gamma, z : A \vdash [z/x, z/y]e : B} \end{array} \quad \begin{array}{c} \text{S.WEAK} \\ \frac{\text{Weak}(\mathbf{m}) \quad \rho \odot \Gamma \vdash e : A}{\rho, 0 \odot \Gamma, x : B \vdash e : A} \end{array} .$$

The weakening rule is straightforwardly modified: It can be toggled on or off by the mode. Contraction is only allowed when both grades are elements of $\text{Cont}(\mathbf{m})$. The requirement that $\text{Cont}(\mathbf{m})$ is an ideal arises naturally as follows. First, consider a judgment $\rho, q_1, q_2, q_3 \odot \Gamma, x_1 : A, x_2 : A, x_3 : A \vdash t : B$ where $q_1, q_2, q_3 \in \text{Cont}(\mathbf{m})$. We want $\text{Cont}(\mathbf{m})$ to act as a set of mutually contractible grades, so it should be possible to contract this into a term $\rho, q_1 + q_2 + q_3 \odot \Gamma, z : A \vdash [z/x_1, z/x_2, z/x_3]t : B$. Since the contraction rule only allows contracting two variables at a time, we must pick an order: We either contract x_1 with x_2 and then the result with x_3 , or x_2 and x_3 first and then the result with x_1 . The condition that $\text{Cont}(\mathbf{m})$ is closed under addition means that both orderings are always possible, and in fact the resulting derivations should be considered equal. Similarly, the condition that $0 \in \text{Cont}(\mathbf{m})$ ensures that contraction interacts well with weakening in the sense that for $q \in \text{Cont}(\mathbf{m})$ we have the derivation

$$\frac{\frac{\rho, q \odot \Gamma, x : A \vdash e : B}{\rho, q, 0 \odot \Gamma, x : A, y : A \vdash e : B} \text{S.WEAK}}{\rho, q + 0 \odot \Gamma, z : A \vdash [z/x]e : B} \text{S.CONT}$$

and the resulting term should be considered equal to e . Finally, contraction should also interact well with substitution. In graded type systems, substitution takes the form

$$\frac{\rho, r \odot \Gamma, x : A \vdash e : B \quad \sigma \odot \Delta \vdash t : A}{\rho, r \cdot \sigma \odot \Gamma, \Delta \vdash [t/x]e : B} .$$

Given terms $\rho, r \odot \Gamma, x : A \vdash e : B$ and $\sigma, q_1, q_2 \odot \Delta, z_1 : T, z_2 : T \vdash t : A$ with $q_1, q_2 \in \text{Cont}(\mathbf{m})$ we can apply contraction, obtaining $\sigma, q_1 + q_2 \odot \Delta, z : T \vdash [z/z_1, z/z_2]t : A$ then perform substitution. The condition that $\text{Cont}(\mathbf{m})$ is an ideal ensures that the other ordering is possible too: If we substitute first, obtaining $\rho, r \cdot \sigma, r \cdot q_1, r \cdot q_2 \odot \Gamma, \Delta, z_1 : T, z_2 : T \vdash [t/x]e : B$ then we may still apply contraction. Note that both orders of these operations result in the same term.

Example 2.8 We can recover non-graded logics by using appropriate modes.

- (i) Using the mode $\mathbf{U} = (\top, \top, \text{true})$ we recover intuitionistic logic: Contraction and weakening are permitted.
- (ii) Using mode $\mathbf{R} = (\top, \top, \text{false})$ recovers *relevant logic*: There is only one grade that may be contracted with itself, so variables may be reused arbitrarily. However, weakening is disabled, so unused variables are not allowed.
- (iii) Using the mode $\mathbf{A} = (\{0, 1\}, \{0\}, \text{true})$ with ordering $0 \leq 1$, we recover *affine logic*. Variables graded 1 may be unused by applying a combination of the weakening and subsumption rules. Contraction is only allowed for unused variables graded 0, so reuse is impossible.
- (iv) Using the mode $(\{0, 1\}, \{0\}, \text{false})$ we almost recover linear logic. Variables graded 1 are guaranteed to have linear usage. The system we obtain here is not exactly linear logic (hence “almost”), since variables graded 0 can still be introduced explicitly using the graded modality $\Box_0 A$, or by applying the elimination rule for the unit type. (The unit type has no computational content so its terms may be eliminated at any grade.)
- (v) Similarly, grading by the mode $\mathbf{L} = (\mathbb{N}, \{0\}, \text{false})$ with discrete ordering tracks the number of free occurrences of variables in a term exactly. Since contraction and weakening are disabled, variables

graded by $n \neq 1$ must be introduced explicitly via the graded modality $\Box_n A$, or the unit type's elimination rule. Thus, we recover another variant on linear logic.

Example 2.9 Using the mode $(R = \{0, 1, \omega\}, \{0, \omega\}, \text{true})$, the grades mean unused (0), linear (1), and unrestricted (ω). Furthermore, two variables graded 1 may not be contracted into one variable graded $1 + 1 = \omega$. This is useful, for example, when dealing with file handles: Two file handles used linearly cannot be replaced by one file handle with unrestricted use. Similar considerations apply to the mode $(\mathbb{N}, \mathbb{N} \setminus \{1\}, \text{true})$: Contraction is only allowed for variables which are known to be used non-linearly.

2.4 Comparing modes

Definition 2.10 A morphism of semirings $R \rightarrow S$ is a function $f: R \rightarrow S$ such that $f(0) = 0$, $f(1) = 1$, and for all $x, y \in R$, $f(x + y) = f(x) + f(y)$ and $f(x \cdot y) = f(x) \cdot f(y)$. A morphism of grade algebras $f: R \rightarrow S$ is a morphism of semirings which is also monotone, i.e. $x \leq y$ implies $f(x) \leq f(y)$. A morphism of modes $\mathfrak{m} \rightarrow \mathfrak{n}$ is a morphism $f: R_{\mathfrak{m}} \rightarrow R_{\mathfrak{n}}$ of the underlying grade algebras such that for each $x \in \text{Cont}(\mathfrak{m})$, we have $f(x) \in \text{Cont}(\mathfrak{n})$ and such that the boolean formula $\text{Weak}(\mathfrak{m}) \rightarrow \text{Weak}(\mathfrak{n})$ is true. This definition yields a category of modes, denoted **Mode**.

A morphism of modes $\varphi: \mathfrak{m} \rightarrow \mathfrak{n}$ implies that $\mathfrak{n}$ has “more structure” than $\mathfrak{m}$ in an appropriate way. We give some intuition for this: Given a derivable judgment $\rho \odot \Gamma \vdash t : A$ under mode $\mathfrak{m}$ there exists a corresponding judgment $\varphi(\rho) \odot \Gamma \vdash t : A$ which is derivable under mode $\mathfrak{n}$. Here, $\varphi(\rho)$ is the result of applying φ to each element of ρ . In other words, there is a translation function. This translation is not quite the identity on types and terms, since some types and terms feature grade annotations, which also need to be transported along φ . The proof that this translation is sound is straightforward by induction, since morphisms of modes preserve all structure carried by a mode. In particular, each application of the weakening or contraction rules at mode $\mathfrak{m}$ can be mirrored at $\mathfrak{n}$, which we take to mean that $\mathfrak{n}$ has more structure than $\mathfrak{m}$. This intuition has two interesting special cases: The category **Mode** has a terminal object $\mathbf{U}$ and an initial object $\mathbf{L}$. We already saw in Example 2.8 that these modes recover intuitionistic and linear logic respectively. Therefore it makes sense to think of the terminal and initial modes as having maximal and minimal structure respectively.

3 Syntax of Grass

In this section we explain the syntax and typing rules of GRASS. Our system is parametrized by a family of modes, with each type belonging to one mode. Like with Adjoint Logic [28], the modes are arranged in a preorder, with modes higher in the order admitting more structural rules than those lower in the order, as discussed in §2.4. First, we present several definitions used throughout the paper.

Definition 3.1 GRASS is parametrized by a functor $S \rightarrow \mathbf{Mode}$, where S is some preordered set. We usually elide mention of this functor, instead thinking of its image as a preordered set of modes $\mathfrak{m}, \mathfrak{n}, \mathfrak{l}, \dots$ together with a coherent system of morphisms $\varphi_{\mathfrak{m}, \mathfrak{n}}: \mathfrak{m} \rightarrow \mathfrak{n}$, whenever $\mathfrak{m} \leq \mathfrak{n}$. We fix such data for the remainder of this text.

Definition 3.2 Grades are denoted by the letters q, r, s . They may be drawn from any of the grade algebras $R_{\mathfrak{m}}$. *Grade vectors* are finite lists of grades, denoted by the letters ρ, σ, τ . Grade vectors are *heterogenous*, i.e. entries of one grade vector may be drawn from more than one grade algebra.

Definition 3.3 Let $\mathfrak{m} \leq \mathfrak{n}$ be modes and let $q \in R_{\mathfrak{m}}$ and $r \in R_{\mathfrak{n}}$. We define $qr = \varphi_{\mathfrak{m}, \mathfrak{n}}(q) \cdot r \in R_{\mathfrak{n}}$. We extend this notation to scalar multiplication: If $q \in R_{\mathfrak{m}}$ and $\rho = (r_1, \dots, r_k)$ with $r_i \in R_{\mathfrak{m}_i}$ and $\mathfrak{m}_i \geq \mathfrak{m}$ for $i \in \{1, \dots, k\}$, then we write $q\rho = (qr_1, \dots, qr_k) = (\varphi_{\mathfrak{m}, \mathfrak{m}_1}(q) \cdot r_1, \dots, \varphi_{\mathfrak{m}, \mathfrak{m}_k}(q) \cdot r_k)$.

Definition 3.4 Types are denoted A, B, C, T . Each type in GRASS belongs to a unique mode and we write $A \in \text{Type}(\mathfrak{m})$ to indicate that the type A is well-formed and belongs to mode $\mathfrak{m}$. In this case, variables of type A are graded by grades drawn from $R_{\mathfrak{m}}$, and structural rules may be applied to them as prescribed by $\text{Weak}(\mathfrak{m})$ and $\text{Cont}(\mathfrak{m})$. The rules for well-formed types are given in Figure 1.

$$\begin{array}{c}
 \text{TY.UNIT} \\
 \frac{}{\mathbf{I}_m \in \text{Type}(\mathbf{m})}
 \end{array}
 \quad
 \begin{array}{c}
 \text{TY.PAIR} \\
 \frac{A \in \text{Type}(\mathbf{m}) \quad B \in \text{Type}(\mathbf{m})}{A \otimes B \in \text{Type}(\mathbf{m})}
 \end{array}
 \quad
 \begin{array}{c}
 \text{TY.FUN} \\
 \frac{n \leq m \quad q \in R_m \quad A \in \text{Type}(\mathbf{m}) \quad B \in \text{Type}(\mathbf{n})}{A^{q:m} \multimap B \in \text{Type}(\mathbf{n})}
 \end{array}
 \quad
 \begin{array}{c}
 \text{TY.SUM} \\
 \frac{A \in \text{Type}(\mathbf{m}) \quad B \in \text{Type}(\mathbf{m})}{A \oplus B \in \text{Type}(\mathbf{m})}
 \end{array}$$

$$\begin{array}{c}
 \text{TY.DROP} \\
 \frac{n \leq m \quad q \in R_m \quad A \in \text{Type}(\mathbf{m})}{\downarrow_{n \leq m}^q A \in \text{Type}(\mathbf{n})}
 \end{array}
 \quad
 \begin{array}{c}
 \text{TY.RAISE} \\
 \frac{m \leq n \quad A \in \text{Type}(\mathbf{m})}{\uparrow_{m \leq n} A \in \text{Type}(\mathbf{n})}
 \end{array}$$

Fig. 1. GRASS rules for well-formed types

$$\begin{array}{c}
 \text{VAR} \\
 \frac{A \in \text{Type}(\mathbf{m})}{1 \mid \mathbf{m} \odot x : A \vdash_m x : A}
 \end{array}
 \quad
 \begin{array}{c}
 \text{WEAK} \\
 \frac{\text{Weak}(\mathbf{m}) \quad n \leq m \quad B \in \text{Type}(\mathbf{m}) \quad \rho \mid \mathbf{M} \odot \Gamma \vdash_n e : A}{\rho, 0 \mid \mathbf{M}, \mathbf{m} \odot \Gamma, x : B \vdash_n e : A}
 \end{array}
 \quad
 \begin{array}{c}
 \text{CONT} \\
 \frac{q_1, q_2 \in \text{Cont}(\mathbf{m}) \quad \rho, q_1, q_2 \mid \mathbf{M}, \mathbf{m}, \mathbf{m} \odot \Gamma, x : A, y : A \vdash_n e : B}{\rho, q_1 + q_2 \mid \mathbf{M}, \mathbf{m} \odot \Gamma, z : A \vdash_n [z/x, z/y]e : B}
 \end{array}$$

$$\begin{array}{c}
 \text{SUB} \\
 \frac{\rho \leq \sigma \quad \rho \mid \mathbf{M} \odot \Gamma \vdash_m e : A}{\sigma \mid \mathbf{M} \odot \Gamma \vdash_m e : A}
 \end{array}
 \quad
 \begin{array}{c}
 \text{UNIT.I} \\
 \frac{}{\emptyset \mid \emptyset \odot \emptyset \vdash_m \star_m : \mathbf{I}_m}
 \end{array}
 \quad
 \begin{array}{c}
 \text{UNIT.E} \\
 \frac{\sigma \mid \mathbf{N} \odot \Delta \vdash_m e : \mathbf{I}_m \quad \rho \mid \mathbf{M} \odot \Gamma \vdash_m t : A \quad q \in R_m}{\rho, q\sigma \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_m \text{let}_{@q} \star_m = e \text{ in } t : A}
 \end{array}$$

$$\begin{array}{c}
 \text{FUN.I} \\
 \frac{\rho, q \mid \mathbf{M}, \mathbf{m} \odot \Gamma, x : A \vdash_n t : B}{\rho \mid \mathbf{M} \odot \Gamma \vdash_n \lambda x. t : A^{q:m} \multimap B}
 \end{array}
 \quad
 \begin{array}{c}
 \text{FUN.E} \\
 \frac{\rho \mid \mathbf{M} \odot \Gamma \vdash_n t : A^{q:m} \multimap B \quad \sigma \mid \mathbf{N} \odot \Delta \vdash_m e : A}{\rho, q\sigma \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_n t e : B}
 \end{array}
 \quad
 \begin{array}{c}
 \text{PAIR.I} \\
 \frac{\rho \mid \mathbf{M}_1 \odot \Gamma_1 \vdash_m t_1 : A_1 \quad \sigma \mid \mathbf{M}_2 \odot \Gamma_2 \vdash_m t_2 : A_2}{\rho, \sigma \mid \mathbf{M}_1, \mathbf{M}_2 \odot \Gamma_1, \Gamma_2 \vdash_m (t_1, t_2) : A_1 \otimes A_2}
 \end{array}$$

$$\begin{array}{c}
 \text{PAIR.E} \\
 \frac{\sigma \mid \mathbf{N} \odot \Delta \vdash_m e : A_1 \otimes A_2 \quad \rho, q, q \mid \mathbf{M}, \mathbf{m}, \mathbf{m} \odot \Gamma, x_1 : A_1, x_2 : A_2 \vdash_n t : B}{\rho, q\sigma \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_n \text{let}_{@q} (x_1, x_2) = e \text{ in } t : B}
 \end{array}
 \quad
 \begin{array}{c}
 \text{SUM.II} \\
 \frac{B \in \text{Type}(\mathbf{m}) \quad \rho \mid \mathbf{M} \odot \Gamma \vdash_m e : A}{\rho \mid \mathbf{M} \odot \Gamma \vdash_m \text{inl } e : A \oplus B}
 \end{array}$$

$$\begin{array}{c}
 \text{SUM.IR} \\
 \frac{A \in \text{Type}(\mathbf{m}) \quad \rho \mid \mathbf{M} \odot \Gamma \vdash_m e : B}{\rho \mid \mathbf{M} \odot \Gamma \vdash_m \text{inr } e : A \oplus B}
 \end{array}
 \quad
 \begin{array}{c}
 \text{SUM.E} \\
 \frac{q \geq 1 \quad \sigma \mid \mathbf{N} \odot \Delta \vdash_n e : A_1 \oplus A_2 \quad \rho, q \mid \mathbf{M}, \mathbf{n} \odot \Gamma, x_1 : A_1 \vdash_m t_1 : B \quad \rho, q \mid \mathbf{M}, \mathbf{n} \odot \Gamma, x_2 : A_2 \vdash_m t_2 : B}{\rho, q\sigma \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_m \text{case}_q(e; x_1.t_1; x_2.t_2) : B}
 \end{array}
 \quad
 \begin{array}{c}
 \text{DROP.I} \\
 \frac{n \leq m \quad q \in R_m \quad \rho \mid \mathbf{M} \odot \Gamma \vdash_m e : A}{q\rho \mid \mathbf{M} \odot \Gamma \vdash_n \downarrow_{n \leq m}^q e : \downarrow_{n \leq m}^q A}
 \end{array}$$

$$\begin{array}{c}
 \text{DROP.E} \\
 \frac{l \leq n \leq m \quad \sigma \mid \mathbf{N} \odot \Delta \vdash_m e : \downarrow_{n \leq m}^q A \quad \rho, q \mid \mathbf{M}, \mathbf{m} \odot \Gamma, x : A \vdash_1 t : B}{\rho, \sigma \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_1 \text{let}_{@q} \downarrow_{n \leq m} x = e \text{ in } t : B}
 \end{array}
 \quad
 \begin{array}{c}
 \text{RAISE.I} \\
 \frac{m \leq n \leq M \quad \rho \mid \mathbf{M} \odot \Gamma \vdash_m e : A}{\rho \mid \mathbf{M} \odot \Gamma \vdash_n \uparrow_{m \leq n} e : \uparrow_{m \leq n} A}
 \end{array}
 \quad
 \begin{array}{c}
 \text{RAISE.E} \\
 \frac{\rho \mid \mathbf{M} \odot \Gamma \vdash_n e : \uparrow_{m \leq n} A}{\rho \mid \mathbf{M} \odot \Gamma \vdash_m \uparrow_{m \leq n}^{-1} e : A}
 \end{array}$$

Fig. 2. GRASS typing rules

Definition 3.5 Typing judgments in GRASS have the form $\rho \mid \mathbf{M} \odot \Gamma \vdash_m t : T$. The roles of Γ , t , and T are as before. There are two new components in the judgment: The annotation $\mathbf{m}$ on the turnstile indicates that $T \in \text{Type}(\mathbf{m})$, and we say that this typing judgment is *made at mode* $\mathbf{m}$. The other new component is $\mathbf{M}$, which is a list of modes, also called a *mode vector*. If $\rho = (r_1, \dots, r_k)$ and $\mathbf{M} = (\mathbf{m}_1, \dots, \mathbf{m}_k)$ and $\Gamma = x_1 : A_1, \dots, x_k : A_k$, then the above judgment indicates that $A_i \in \text{Type}(\mathbf{m}_i)$ and that the variable x_i is used with capabilities indicated by r_i .

Definition 3.6 Typing contexts are denoted by Γ, Δ . When Γ and Δ are contexts, their concatenation is written Γ, Δ and is only well-defined when the contexts Γ and Δ have no variables in common. Whenever we use context concatenation, we implicitly assume that this condition is satisfied.

In any derivable typing judgment $\rho \mid \mathbf{M} \odot \Gamma \vdash_m t : T$, each mode $\mathbf{n}$ occurring in $\mathbf{M}$ must satisfy $\mathbf{n} \geq \mathbf{m}$, which we write concisely as $\mathbf{M} \geq \mathbf{m}$. A similar requirement is present in Linear-non-Linear Logic [4], where

$$\begin{array}{c}
 \text{UNIT.}\beta \\
 \frac{\sigma \mid \mathbf{N} \odot \Delta \vdash_m \star_m : \mathbf{I}_m \quad \rho \mid \mathbf{M} \odot \Gamma \vdash_m e : A}{\rho, q\sigma \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_m \text{let}_{@q} \star_m = \star_m \text{ in } e \equiv_\beta e : A}
 \end{array}
 \quad
 \begin{array}{c}
 \text{PAIR.}\beta \\
 \frac{\sigma \mid \mathbf{N} \odot \Delta \vdash_m (t_1, t_2) : A_1 \otimes A_2 \quad \rho, q, q \mid \mathbf{M}, \mathbf{m}, \mathbf{m} \odot \Gamma, x_1 : A_1, x_2 : A_2 \vdash_n e : A}{\rho, q\sigma \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_m \text{let}_{@q}(x_1, x_2) = (t_1, t_2) \text{ in } e \equiv_\beta [t_1/x_1, t_2/x_2]e : A}
 \end{array}$$

$$\begin{array}{c}
 \text{FUN.}\beta \\
 \frac{\rho \mid \mathbf{M} \odot \Gamma \vdash_n \lambda x. e : B^{q:m} \multimap A \quad \sigma \mid \mathbf{N} \odot \Delta \vdash_m t : B}{\rho, q\sigma \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_n (\lambda x. e) t \equiv_\beta [t/x]e : A}
 \end{array}
 \quad
 \begin{array}{c}
 \text{SUM.}\beta.\text{L} \\
 \frac{q \geq 1 \quad \sigma \mid \mathbf{N} \odot \Delta \vdash_n \text{inl } t : A_1 \oplus A_2 \quad \rho, q \mid \mathbf{M}, \mathbf{n} \odot \Gamma, x_1 : A_1 \vdash_m e_1 : A \quad \rho, q \mid \mathbf{M}, \mathbf{n} \odot \Gamma, x_2 : A_2 \vdash_m e_2 : A}{\rho, q\sigma \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_m \text{case}_q(\text{inl } t; x_1. e_1; x_2. e_2) \equiv_\beta [t/x_1]e_1 : A}
 \end{array}$$

$$\begin{array}{c}
 \text{SUM.}\beta.\text{R} \\
 \frac{q \geq 1 \quad \sigma \mid \mathbf{N} \odot \Delta \vdash_n \text{inr } t : A_1 \oplus A_2 \quad \rho, q \mid \mathbf{M}, \mathbf{n} \odot \Gamma, x_1 : A_1 \vdash_m e_1 : A \quad \rho, q \mid \mathbf{M}, \mathbf{n} \odot \Gamma, x_2 : A_2 \vdash_m e_2 : A}{\rho, q\sigma \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_m \text{case}_q(\text{inl } t; x_1. e_1; x_2. e_2) \equiv_\beta [t/x_2]e_2 : A}
 \end{array}
 \quad
 \begin{array}{c}
 \text{RAISE.}\beta \\
 \frac{\mathbf{m} \leq \mathbf{n} \quad \rho \mid \mathbf{M} \odot \Gamma \vdash_n \uparrow_{\mathbf{m} \leq \mathbf{n}} t : \uparrow_{\mathbf{m} \leq \mathbf{n}} A}{\rho \mid \mathbf{M} \odot \Gamma \vdash_m \uparrow_{\mathbf{m} \leq \mathbf{n}}^{-1}(\uparrow_{\mathbf{m} \leq \mathbf{n}} t) \equiv_\beta t : A}
 \end{array}$$

$$\begin{array}{c}
 \text{DROP.}\beta \\
 \frac{\mathbf{l} \leq \mathbf{n} \leq \mathbf{m} \quad \sigma \mid \mathbf{N} \odot \Delta \vdash_n \downarrow_{\mathbf{n} \leq \mathbf{m}}^q t : \downarrow_{\mathbf{n} \leq \mathbf{m}}^q A \quad \rho, q \mid \mathbf{M}, \mathbf{m} \odot \Gamma, x : A \vdash_1 e : B}{\rho, \sigma \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_1 \text{let}_{@q} \downarrow_{\mathbf{n} \leq \mathbf{m}} x = \downarrow_{\mathbf{n} \leq \mathbf{m}}^q t \text{ in } e \equiv_\beta [t/x]e : B}
 \end{array}$$

$$\begin{array}{c}
 \text{UNIT.}\eta \\
 \frac{\rho \mid \mathbf{M} \odot \Gamma \vdash_m t : \mathbf{I}_m}{\rho \mid \mathbf{M} \odot \Gamma \vdash_m t \equiv_\eta \text{let}_{@1} \star_m = t \text{ in } \star_m : \mathbf{I}_m}
 \end{array}
 \quad
 \begin{array}{c}
 \text{PAIR.}\eta \\
 \frac{\rho \mid \mathbf{M} \odot \Gamma \vdash_m t : A_1 \otimes A_2}{\rho \mid \mathbf{M} \odot \Gamma \vdash_m t \equiv_\eta \text{let}_{@1}(x_1, x_2) = t \text{ in } (x_1, x_2) : A_1 \otimes A_2}
 \end{array}$$

$$\begin{array}{c}
 \text{FUN.}\eta \\
 \frac{\rho \mid \mathbf{M} \odot \Gamma \vdash_n t : A^{q:m} \multimap B}{\rho \mid \mathbf{M} \odot \Gamma \vdash_m t \equiv_\eta \lambda x. (t x) : A^{q:m} \multimap B}
 \end{array}
 \quad
 \begin{array}{c}
 \text{SUM.}\eta \\
 \frac{\rho \mid \mathbf{M} \odot \Gamma \vdash_m t : A_1 \oplus A_2}{\rho \mid \mathbf{M} \odot \Gamma \vdash_m t \equiv_\eta \text{case}_1(t; x_1. \text{inl } x_1; x_2. \text{inr } x_2) : A_1 \oplus A_2}
 \end{array}$$

$$\begin{array}{c}
 \text{RAISE.}\eta \\
 \frac{\rho \mid \mathbf{M} \odot \Gamma \vdash_m t : \uparrow_{\mathbf{n} \leq \mathbf{m}} A}{\rho \mid \mathbf{M} \odot \Gamma \vdash_m t \equiv_\eta \uparrow_{\mathbf{n} \leq \mathbf{m}}^{-1}(\uparrow_{\mathbf{n} \leq \mathbf{m}} t) : \uparrow_{\mathbf{n} \leq \mathbf{m}} A}
 \end{array}
 \quad
 \begin{array}{c}
 \text{DROP.}\eta \\
 \frac{\rho \mid \mathbf{M} \odot \Gamma \vdash_n t : \downarrow_{\mathbf{n} \leq \mathbf{m}}^q A}{\rho \mid \mathbf{M} \odot \Gamma \vdash_n t \equiv_\eta \text{let}_{@q} \downarrow_{\mathbf{n} \leq \mathbf{m}} x = t \text{ in } \downarrow_{\mathbf{n} \leq \mathbf{m}}^q x : \downarrow_{\mathbf{n} \leq \mathbf{m}}^q A}
 \end{array}$$

 Fig. 3. GRASS $\beta\eta$ -conversions

linear terms may depend on non-linear variables, but not vice-versa, and was also observed by Pruiksma et al. [28], where it is called *independence*. The typing rules of GRASS are set up to ensure that independence holds in all derivable judgments. In GRASS, independence also ensures that all scalar multiplications occurring in the typing rules are well-defined (cf. Definition 3.3). We discuss the typing rules in detail in the following paragraphs.

We begin with the structural rules. The grade 1 corresponds to linear usage, and is thus used in the variable rule (VAR). The rules for weakening and contraction are as before, only modified to account for the presence of different modes. Contraction (rule CONT) is only allowed between grades of the same mode. In the weakening rule (WEAK) the condition $\mathbf{n} \leq \mathbf{m}$ ensures that independence holds in the conclusion, if it holds in the assumption. We assume an implicit exchange rule, which allows reordering of contexts. When this occurs, the grade and mode vectors must be reordered accordingly.

The type $\downarrow_{\mathbf{n} \leq \mathbf{m}}^q A$ is defined for $A \in \text{Type}(\mathbf{m})$, $q \in R_{\mathbf{m}}$ and $\mathbf{n} \leq \mathbf{m}$ (rule TY.DROP). Terms of this type behave like terms of type A , with the added constraint that they must be used at grade q . Furthermore, the type $\downarrow_{\mathbf{n} \leq \mathbf{m}}^q A$ belongs to mode $\mathbf{n}$ and is therefore subject to the structural rules imposed by $\mathbf{n}$. The introduction rule (DROP.I) promotes a term t of type A to a term $\downarrow_{\mathbf{n} \leq \mathbf{m}}^q t$ of type $\downarrow_{\mathbf{n} \leq \mathbf{m}}^q A$. The resources used to produce t are multiplied by q during this promotion, reflecting the q -fold reuse later. The elimination rule (DROP.E) uses a let binding $\text{let}_{@q} \downarrow_{\mathbf{n} \leq \mathbf{m}} x = e$ in t to extract a term of type A from e , and make it available to the body t , which then must use that term at grade q . This type combines two distinct feature

sets from existing literature: First, some graded type theories (e.g. [1,8,10,13,24]) include a graded modality, commonly written $\Box_q A$. We can recover such a type in GRASS by setting $\Box_q A := \downarrow_{m \leq m}^q A$. Second, $\downarrow_{n \leq m}^q A$ is an analogue of FA from LNL [4], and $\text{Grd}_q A$ from mGL [32], since these constructs move types to a mode with fewer structural rules. Our notation mirrors that of Pruikma et al. [28].

The type $\uparrow_{n \leq m} A$ is defined for $A \in \text{Type}(n)$ and $n \leq m$ (rule TY.RAISE). The introduction rule (RAISE.I) constructs a term $\uparrow_{n \leq m} e : \uparrow_{n \leq m} A$ from a term $e : A$. The produced term and its type belong to mode m and thus have fewer structural restrictions imposed upon them. The elimination rule (RAISE.E) allows us to extract a term of type A from a term of type $\uparrow_{n \leq m} A$ directly. The type $\uparrow_{n \leq m} A$ is the analogue of GA from LNL and Lin from mGL.

GRASS includes a subsumption rule SUB . This rule means that a grade vector ρ in a judgment $\rho \mid M \odot \Gamma \vdash_m t : A$ need only be an upper bound on the actual usage of the variables in Γ within the term e . The relation $\rho \leq \sigma$ is defined inductively by

$$\frac{}{\emptyset \leq \emptyset} \quad \frac{\rho \leq \sigma \quad r \in R_m \quad q \in R_m \quad r \leq q}{(\rho, r) \leq (\sigma, q)}.$$

When the preorder is discrete, this rule is effectively disabled.

For each mode m , we have a unit type $\mathbf{I}_m$ (rule TY.UNIT) which admits a closed term $\star_m$ that can be produced without using any resources (rule UNIT.I). Terms of type $\mathbf{I}_m$ hold no computational information, and may thus be eliminated (rule UNIT.E) at any grade q by the pattern matching construct $\text{let}_{@q} \star_m = e \text{ in } t$.

The tensor product type $A_1 \otimes A_2$ is defined only when the types A_1 and A_2 belong to the same mode. Its introduction rule (PAIR.I) is standard, allowing the construction of a pair (t_1, t_2) from terms t_1 and t_2 . The pair elimination rule (PAIR.E) functions by pattern matching. It asserts that using a pair at grade q is the same thing as using each of its components separately at grade q . Since term reuse is modeled by multiplication, the grades used to produce the pair are multiplied by q as part of the elimination rule.

The function type $A^{q:m} \multimap B$ is defined when $A \in \text{Type}(m)$ and $B \in \text{Type}(n)$ with $m \geq n$. The requirement that $m \geq n$ is an instance of independence. The grade annotation q indicates how functions of this type use their parameters. Functions are introduced (rule FUN.I) using λ -abstraction. The grade annotation q corresponds to the use of the abstracted variable in the function body. Elimination (FUN.E) is done by function application. Since grade multiplication corresponds to reuse, the grades used to produce the parameter e are multiplied by q during application.

Like the product, the sum type $A_1 \oplus A_2$ is defined between types of the same mode. Its introduction forms (SUM.IL , SUM.IR) are standard, injecting a term e of type A_1 (resp. A_2) into the sum. These injections do not affect the grades used to produce e . Elimination occurs by cases: to use a term of type $A_1 \oplus A_2$ at grade q , we must specify two branches, which use a term of A_1 (resp. A_2) at grade q .

Example 3.7 Let m_1, m_2 be two modes, and let L be the initial mode as discussed in §2.4. Instantiate GRASS with modes $m_1 > L < m_2$. Write $\Box_m^q A = \downarrow_{L \leq m}^q (\uparrow_{L \leq m} A)$. Then L behaves like linear logic, with two graded modalities $\Box_{m_1}^q A$ and $\Box_{m_2}^q A$ which can account for two distinct notions of resource usage. This is a generalization of mGL allowing grades from two different grade algebras.

Example 3.8 Using the modes $L < U$ of Example 2.8, we obtain LNL [4]. If we also use the modes A and R , we obtain a combined system in the style of Adjoint Logic [28].

Theorem 3.9 (Substitution) *If $\rho, r \mid M, m \odot \Gamma, x : A \vdash_n e : B$ and $\sigma \mid N \odot \Delta \vdash_m t : A$, then*

$$\rho, r\sigma \mid M, N \odot \Gamma, \Delta \vdash_n [t/x]e : B$$

Proof. See §B.1 in the appendix [16]. □

We include a set of β - and η -conversion rules which are given in Figure 3. The β -rules are mostly standard: Pattern matching eliminators reduce when the scrutinee of the match is a constructor for the respective type. In the pattern matching form for the unit type, the grade annotation q gets ignored since the term $\star_m$ carries no computational information. Function applications reduce by substitution into the function body, when the function in question is a lambda. The type $\uparrow_{m \leq n} A$ is a negative type

with an explicit destructor $\uparrow_{m \leq n}^{-1} e$, rather than a pattern matching one, which acts as a left inverse to the constructor $\uparrow_{m \leq n} e$. The η -rules are similarly standard: eliminating then reconstructing a term recovers that term. These conversions preserve typing:

Theorem 3.10 (Type preservation) *The β - and η -conversions preserve typing and grading, i.e.*

- (i) *Whenever $\rho \mid M \odot \Gamma \vdash_m t_1 \equiv_\beta t_2 : A$, then $\rho \mid M \odot \Gamma \vdash_m t_1 : A$, and $\rho \mid M \odot \Gamma \vdash_m t_2 : A$.*
- (ii) *Whenever $\rho \mid M \odot \Gamma \vdash_m t_1 \equiv_\eta t_2 : A$, then $\rho \mid M \odot \Gamma \vdash_m t_1 : A$, and $\rho \mid M \odot \Gamma \vdash_m t_2 : A$.*

Proof. See §§B.2, B.3 in the appendix [16]. □

4 Categorical semantics

We present our categorical semantics. The structure of the categorical models mirrors that of the syntax: GRASS is composed out of several graded logics, one for each mode, tied together by a system of modalities $\downarrow_{n \leq m}^q A$ and $\uparrow_{m \leq n} A$ mediating between them. Similarly, a model for GRASS consists of several models for ordinary graded logic, tied together by an appropriate system of morphisms, here realized as adjunctions with some additional structure. To appropriately capture the modified weakening and contraction rules, we must make some changes relative to the existing literature. Because of this, we set up all necessary terminology from scratch. The categorical semantics presented here do not include the sum type; Adding the necessary structure to model it is straightforward. The following definitions are adapted from Katsumata [17].

Definition 4.1 Let $m = (R_m, \text{Cont}(m), \text{Weak}(m))$ be a mode. The preorder on R_m equips the sets R_m and $\text{Cont}(m)$ with categorical structures. We write $(R_m, \cdot, 1)$ for the symmetric monoidal structure on R_m whose tensor product is multiplication and tensor unit is $1 \in R_m$. Similarly, we write $(\text{Cont}(m), +, 0)$ for the symmetric monoidal structure on $\text{Cont}(m)$ whose tensor product is addition and tensor unit is $0 \in \text{Cont}(m)$.

Definition 4.2 Let $(\mathcal{C}, \otimes, I)$ be a symmetric monoidal category. Write $\mathbf{SMC}(\mathcal{C}, \mathcal{C})$ for the category of strong symmetric monoidal endofunctors on $\mathcal{C}$ and monoidal natural transformations between them. This category can be equipped with a monoidal structure in two ways:

- (i) Pointwise: For $F, G \in \mathbf{SMC}(\mathcal{C}, \mathcal{C})$, their tensor product is $(F \hat{\otimes} G)(X) = F(X) \otimes G(X)$. The tensor unit is the constant functor sending everything to $I \in \mathcal{C}$. We denote this structure $(\mathbf{SMC}(\mathcal{C}, \mathcal{C}), \hat{\otimes}, \hat{I})$.
- (ii) By composition: The tensor product is composition of functors, and the tensor unit is the identity functor. We denote this structure $(\mathbf{SMC}(\mathcal{C}, \mathcal{C}), \circ, \text{id})$.

Definition 4.3 Let $\mathcal{C}$ be a symmetric monoidal category, and let $m = (R_m, \text{Cont}(m), \text{Weak}(m))$ be a mode. An *exponential action* (of m on $\mathcal{C}$) is a functor $D: R_m^{\text{op}} \rightarrow \mathbf{SMC}(\mathcal{C}, \mathcal{C})$ equipped with the following additional structure.

- (i) The structure of a colax monoidal functor with signature $(R_m^{\text{op}}, \cdot, 1) \rightarrow (\mathbf{SMC}(\mathcal{C}, \mathcal{C}), \circ, \text{id})$ on D , i.e. natural transformations $\delta_{r,q}: D(r \cdot q) \rightarrow D(r) \circ D(q)$ and $\epsilon: D(1) \rightarrow \text{id}$.
- (ii) For $r, q \in \text{Cont}(m)$, a natural transformation $c_{r,q}: D(r + q) \rightarrow D(r) \hat{\otimes} D(q)$. Making the following diagram commute.

$$\begin{array}{ccc}
 D((r_1 + r_2) + r_3) & \xrightarrow{c_{r_1+r_2, r_3}} & D(r_1 + r_2) \hat{\otimes} D(r_3) \xrightarrow{c_{r_1, r_2} \hat{\otimes} \text{id}} (D(r_1) \hat{\otimes} D(r_2)) \hat{\otimes} D(r_3) \\
 \parallel & & \downarrow \sim \\
 D(r_1 + (r_2 + r_3)) & \xrightarrow{c_{r_1, r_2+r_3}} & D(r_1) \hat{\otimes} D(r_2 + r_3) \xrightarrow{\text{id} \hat{\otimes} c_{r_2, r_3}} D(r_1) \hat{\otimes} (D(r_2) \hat{\otimes} D(r_3))
 \end{array}$$

Furthermore, if $\text{Weak}(m)$ is true, a morphism $w: D(0) \rightarrow \hat{I}$. In this case, the transformations $c_{r,q}$ and w are required to assemble into a colax monoidal structure on the functor

$$r \mapsto D(r): (\text{Cont}(m)^{\text{op}}, +, 0) \rightarrow \mathbf{SMC}(\mathcal{C}, \mathcal{C}).$$

$$\begin{array}{ccc}
 (q \cdot r_1 + q \cdot r_2) \odot X & \xrightarrow{c_{qr_1, qr_2, X}} & ((q \cdot r_1) \odot X) \otimes ((q \cdot r_2) \odot X) \\
 \parallel & & \downarrow \delta_{q, r_1, X} \otimes \delta_{q, r_2, X} \\
 (q \cdot (r_1 + r_2)) \odot X & & (q \odot (r_1 \odot X)) \otimes (q \odot (r_2 \odot X)) \\
 \downarrow \delta_{q, r_1 + r_2, X} & & \downarrow \tau_{q, r_1 \odot X, r_2 \odot X} \\
 q \odot ((r_1 + r_2) \odot X) & \xrightarrow{q \odot c_{r_1, r_2, X}} & q \odot ((r_1 \odot X) \otimes (r_2 \odot X))
 \end{array}
 \qquad
 \begin{array}{ccc}
 (r_1 \cdot q + r_2 \cdot q) \odot X & \xlongequal{\quad} & ((r_1 + r_2) \cdot q) \odot X \\
 \downarrow c_{r_1 q, r_2 q, X} & & \downarrow \delta_{(r_1 + r_2), q, X} \\
 ((r_1 \cdot q) \odot X) \otimes ((r_2 \cdot q) \odot X) & & (r_1 + r_2) \odot (q \odot X) \\
 \downarrow \delta_{r_1, q, X} \otimes \delta_{r_2, q, X} & \swarrow c_{r_1, r_2, q \odot X} & \\
 (r_1 \odot (q \odot X)) \otimes (r_2 \odot (q \odot X)) & &
 \end{array}$$

$$\begin{array}{ccc}
 0 \odot X & \xlongequal{\quad} & (0 \cdot r) \odot X \\
 \downarrow w_X & & \downarrow \delta_{0, r, X} \\
 I & \xleftarrow{w_{r \odot X}} & 0 \odot (r \odot X)
 \end{array}
 \qquad
 \begin{array}{ccc}
 0 \odot X & \xlongequal{\quad} & (r \cdot 0) \odot X \xrightarrow{\delta_{r, 0, X}} r \odot (0 \odot X) \\
 \downarrow w_X & & \downarrow r \odot w_X \\
 I & \xrightarrow{\iota_r} & r \odot I
 \end{array}$$

Fig. 4. Graded comonad coherence conditions

Note that the requirement that this functor is colax monoidal already includes the above diagram. We include that diagram explicitly, since we cannot require that this functor be colax monoidal in the absence of the morphism w .

These structures are related by the diagrams of Figure 4, which are required to commute whenever all morphisms in them are defined. For the first two diagrams this means that $r_1, r_2 \in \mathbf{Cont}(\mathbf{m})$ and for the last two diagrams this means that $\mathbf{Weak}(\mathbf{m})$ is true. We write $\tau_{r, X, Y} : D(r)(X) \otimes D(r)(Y) \rightarrow D(r)(X \otimes Y)$ and $\iota_r : I \rightarrow D(r)(I)$ for the strong monoidal structure on $D(r)$. Furthermore, the diagrams in Figure 4 use the infix notation $D(r)(X) =: r \odot X$, which we will use for the remainder of this text.

Example 4.4 We consider exponential actions of the modes given in Example 2.8. Let $\mathcal{C}$ be an SMCC.

- (i) Any SMCC admits an exponential action of the initial mode $\mathbf{L}$ by setting $n \odot X = X^{\otimes n}$. Similarly, any SMCC admits an exponential action of the mode $(\{0, 1\}, \{0\}, \text{false})$, by setting $0 \odot X = I$ and $1 \odot X = X$.
- (ii) An exponential action of the terminal mode $\mathbf{U}$ is equivalent to specifying the structure of a Linear Category [7] on $\mathcal{C}$. In other words, an SMCC with an exponential action of the terminal mode is precisely a model of Intuitionistic Linear Logic, with the exponential action modelling the of-course modality. If we choose $1 \odot (-) = \text{id}$, the action equips $\mathcal{C}$ with “projections and diagonals” and hence $\mathcal{C}$ is a cartesian closed category.⁴
- (iii) An exponential action of the mode $\mathbf{R}$ on $\mathcal{C}$ is a functor D together with diagonal maps $DX \rightarrow DX \otimes DX$. If we fix $D = \text{id}$, then $\mathcal{C}$ becomes equipped with the structure of a *relevant category* [27]. Conversely, any relevant category admits an exponential action of $\mathbf{R}$ by setting $1 \odot (-) = \text{id}$.
- (iv) If the tensor unit I is the terminal object of $\mathcal{C}$, then $\mathcal{C}$ admits an exponential action of the mode $\mathbf{A}$ by setting $1 \odot (-) = \text{id}$ and $0 \odot (-) = \hat{I}$. Conversely, an action of $\mathbf{A}$ on $\mathcal{C}$ with $1 \odot (-) = \text{id}$ and $0 \odot (-) = \hat{I}$ exhibits the tensor unit as the terminal object of $\mathcal{C}$.

All in all, we see that known models of intuitionistic, linear, relevant and affine logic admit canonical exponential actions of the modes $\mathbf{U}, \mathbf{L}, \mathbf{R}$ and $\mathbf{A}$ respectively.

Definition 4.5 By a *symmetric lax monoidal adjunction* we mean a symmetric lax monoidal functor $F : \mathcal{C} \rightarrow \mathcal{D}$ between SMCs together with a lax monoidal right adjoint G , such that the unit and counit of the adjunction are monoidal natural transformations. We concisely denote this by $F \dashv G : \mathcal{C} \rightarrow \mathcal{D}$. Recall that F is strong monoidal in this case [18].

Definition 4.6 Let $\varphi : \mathbf{m} \rightarrow \mathbf{n}$ be a morphism of modes, and let $\odot_{\mathbf{m}}$ and $\odot_{\mathbf{n}}$ be exponential actions of $\mathbf{m}, \mathbf{n}$ on the SMCCs $\mathcal{C}_{\mathbf{m}}$ and $\mathcal{C}_{\mathbf{n}}$ respectively. A morphism of actions (over φ) consists of a symmetric lax

⁴ This result is folklore; A proof is given in Mellies [22].

$$\begin{array}{ccc}
 \varphi(1) \odot G(A) & \xrightarrow{\ell} & G(1 \odot A) \\
 \parallel & & \downarrow G(\epsilon) \\
 1 \odot G(A) & \xrightarrow{\epsilon} & G(A)
 \end{array}
 \qquad
 \begin{array}{ccc}
 0 \odot G(A) & \xrightarrow{\ell} & G(0 \odot A) \\
 w_{G(A)} \downarrow & & \downarrow G(w_A) \\
 I & \longrightarrow & G(I)
 \end{array}$$

$$\begin{array}{ccc}
 (\varphi(q) \cdot \varphi(r)) \odot G(A) & \xlongequal{\quad} & \varphi(qr) \odot G(A) \\
 \delta \downarrow & & \downarrow \ell \\
 \varphi(q) \odot \varphi(r) \odot G(A) & & G(qr \odot A) \\
 \varphi(q) \odot \ell \downarrow & & \downarrow G(\delta) \\
 \varphi(q) \odot G(r \odot A) & \xrightarrow{\ell} & G(q \odot r \odot A)
 \end{array}
 \qquad
 \begin{array}{ccc}
 (\varphi(q) + \varphi(r)) \odot G(A) & \xlongequal{\quad} & \varphi(q+r) \odot G(A) \\
 c \downarrow & & \downarrow \ell \\
 \varphi(q) \odot G(A) \otimes \varphi(r) \odot G(A) & & G((q+r) \odot A) \\
 \ell \otimes \ell \downarrow & & \downarrow G(c) \\
 G(q \odot A) \otimes G(r \odot A) & \longrightarrow & G(q \odot A \otimes r \odot A)
 \end{array}$$

Fig. 5. Coherence laws on morphisms of exponential actions. Unlabelled arrows are the lax monoidal structure on G .

monoidal adjunction $F \dashv G: \mathcal{C}_n \rightarrow \mathcal{C}_m$ together with a monoidal natural transformation

$$\ell_{r,A}: \varphi(r) \odot_n G(A) \rightarrow G(r \odot_m A)$$

satisfying the coherence conditions given in Figure 5. We define another natural transformation $\mu_{r,A}: F(\varphi(r) \odot A) \rightarrow r \odot F(A)$ as the transpose of the composite

$$\varphi(r) \odot A \xrightarrow{\varphi(r) \odot \eta} \varphi(r) \odot GFA \xrightarrow{\ell} G(r \odot FA).$$

Morphisms between exponential actions compose in a straightforward way, giving rise to a category **ExpAct** of exponential actions, whose objects are triples $(m, \mathcal{C}, \odot)$ consisting of a mode m , an SMCC $\mathcal{C}$ and an exponential action of m on $\mathcal{C}$, and whose morphisms are quadruples (φ, F, G, ℓ) , where φ is a morphism of modes and (F, G, ℓ) is a morphism of actions over φ . We endow **ExpAct** with the structure of a 2-category: A 2-cell between (φ, F, G, ℓ) and (φ, F', G', ℓ') (note that the underlying morphisms of modes are required to be identical) is a natural isomorphism $\alpha: G \rightarrow G'$ which respects the lineators in the obvious way:

$$\begin{array}{ccc}
 \varphi(r) \odot GA & \xrightarrow{\ell} & G(r \odot A) \\
 \varphi(r) \odot \alpha \downarrow & & \downarrow \alpha \\
 \varphi(r) \odot G'A & \xrightarrow{\ell'} & G'(r \odot A)
 \end{array}$$

Remark 4.7 The terminology *lineator* is borrowed from the theory of actegories [sic]. In fact, our notion of morphisms between actions is adapted from the notion of morphisms between actegories. See e.g. Cappucci and Gavranović [9].

Remark 4.8 Let $(\varphi, F, G, \ell): (m, \mathcal{C}_m, \odot_m) \rightarrow (n, \mathcal{C}_n, \odot_n)$ be a morphism in **ExpAct**. Transposing ℓ along the adjunction $F \dashv G$ yields a natural transformation $F(\varphi(-) \odot_n G(=)) \rightarrow (-) \odot_m (=)$. Katsumata proved that the domain of this transformation satisfies the coherence conditions for exponential actions [17]. In Katsumata's setting, this functor is in fact a graded linear exponential comonad. Here, it may fail to be an exponential action since G is not required to be strong monoidal. Thus, ℓ is equivalent to a comparison morphism of the two actions on $\mathcal{C}_m$ that naturally arise from the adjunction $F \dashv G$.

Lemma 4.9 *The coherence conditions on ℓ given in Figure 5 are equivalent to the analogous coherence conditions on μ given in Figure 6.*

Proof. Straightforward manipulation of diagrams, using the fact that ℓ can be written as the composite $\varphi(r) \odot G(A) \xrightarrow{\eta} GF(\varphi(r) \odot G(A)) \xrightarrow{G(\mu)} G(r \odot FG(A)) \xrightarrow{G(r \odot \epsilon)} G(r \odot A)$. Here ϵ is the counit of the adjunction $F \dashv G$. $\square$

$$\begin{array}{ccc}
 F(\varphi(1) \odot A) & \xrightarrow{\mu} & 1 \odot FA \\
 \parallel & & \downarrow \epsilon \\
 F(1 \odot A) & \xrightarrow{F\epsilon} & F(A)
 \end{array}
 \qquad
 \begin{array}{ccc}
 F(\varphi(0) \odot A) & \xrightarrow{\mu} & 0 \odot FA \\
 Fw \downarrow & & \downarrow w \\
 F(I) & \xrightarrow{u^{-1}} & I
 \end{array}$$

$$\begin{array}{ccc}
 F(\varphi(q)\varphi(r) \odot A) & \equiv & F(\varphi(qr) \odot A) \\
 F(\delta) \downarrow & & \downarrow \\
 F(\varphi(q) \odot \varphi(r) \odot A) & & qr \odot F(A) \\
 \mu \downarrow & & \downarrow \delta \\
 q \odot F(\varphi(r) \odot A) & \xrightarrow{q \odot \mu} & q \odot r \odot FA
 \end{array}
 \qquad
 \begin{array}{ccc}
 F((\varphi(q) + \varphi(r)) \odot A) & \equiv & F(\varphi(q+r) \odot A) \\
 F(c) \downarrow & & \downarrow \mu \\
 F(\varphi(q) \odot A \otimes \varphi(r) \odot A) & & (q+r) \odot FA \\
 m^{-1} \downarrow & & \downarrow c \\
 F(\varphi(q) \odot A) \otimes F(\varphi(r) \odot A) & \xrightarrow{\mu \otimes \mu} & q \odot FA \otimes r \odot FA
 \end{array}$$

 Fig. 6. Coherence laws on μ .

Lemma 4.10 *Let (G, F, ℓ) and (G', F', ℓ') be morphisms of actions over φ , and $\alpha: G \rightarrow G'$ a 2-cell in **ExpAct**. Then α induces a natural isomorphism $\beta: F' \rightarrow F$ which respects μ in the obvious way:*

$$\begin{array}{ccc}
 F'(\varphi(q) \odot A) & \xrightarrow{\mu} & q \odot F'(A) \\
 \beta \downarrow & & \downarrow q \odot \beta \\
 F(\varphi(q) \odot A) & \xrightarrow{\mu'} & q \odot F(A)
 \end{array}$$

Definition 4.11 Recall that GRASS is parametrized by a functor $d: S \rightarrow \mathbf{Mode}$ where S is some preordered set. We have a projection functor $p: \mathbf{ExpAct} \rightarrow \mathbf{Mode}$ given by $(\mathbf{m}, \mathcal{C}, \odot) \mapsto \mathbf{m}$. A categorical model for GRASS consists of a pseudofunctor $D: S \rightarrow \mathbf{ExpAct}$ such that $p \circ D = d$. Spelling this out, this amounts to specifying the following data: For each mode $\mathbf{m}$, an SMCC $\mathcal{C}_{\mathbf{m}}$ together with an exponential action $\odot_{\mathbf{m}}$ of $\mathbf{m}$ on $\mathcal{C}_{\mathbf{m}}$. And whenever $\mathbf{m} \leq \mathbf{n}$, a morphism of exponential actions $(\varphi_{\mathbf{m}, \mathbf{n}}, F_{\mathbf{m}}^{\mathbf{n}}, G_{\mathbf{m}}^{\mathbf{n}}, \ell_{\mathbf{m}}^{\mathbf{n}}): (\mathbf{m}, \mathcal{C}_{\mathbf{m}}, \odot_{\mathbf{m}}) \rightarrow (\mathbf{n}, \mathcal{C}_{\mathbf{n}}, \odot_{\mathbf{n}})$ where $\varphi_{\mathbf{m}, \mathbf{n}}$ is the morphism of grade algebras specified in Definition 3.1 along with a coherent system of natural isomorphisms $G_{\mathbf{m}_2}^{\mathbf{m}_3} \circ G_{\mathbf{m}_1}^{\mathbf{m}_2} \cong G_{\mathbf{m}_1}^{\mathbf{m}_3}$ whenever $\mathbf{m}_1 \leq \mathbf{m}_2 \leq \mathbf{m}_3$ which are compatible with the lineators.

We write $u: I \rightarrow F_{\mathbf{n}}^{\mathbf{m}}(I)$ and $m: F_{\mathbf{n}}^{\mathbf{m}}(A) \otimes F_{\mathbf{n}}^{\mathbf{m}}(B) \rightarrow F_{\mathbf{n}}^{\mathbf{m}}(A \otimes B)$ for the strong monoidal structure on the functors $F_{\mathbf{n}}^{\mathbf{m}}$. We denote internal hom-objects by $X \multimap Y$. We write $ev: (X \multimap Y) \otimes X \rightarrow Y$ for the counit of the adjunction $(-) \otimes X \dashv X \multimap (-)$. Given $e: X \otimes Y \rightarrow Z$ we write $\lambda e: X \rightarrow (Y \multimap Z)$ for the transpose of e along this adjunction.

Example 4.12 Let $F \dashv G: \mathcal{C} \rightarrow \mathcal{L}$ be a symmetric lax monoidal adjunction. Suppose that $\mathbf{L}$ acts on $\mathcal{L}$ as described in Example 4.4, and that $\mathcal{C}$ admits an exponential action by a mode $\mathbf{m} = (R, R, \text{true})$ for some grade algebra R . We promote this adjunction to a morphism of exponential actions with lineator

$$\varphi(n) \odot G(A) = (\varphi(1) + \dots + \varphi(1)) \odot G(A) \xrightarrow{l} G(A)^{\otimes n} \xrightarrow{z} G(A^{\otimes n})$$

where the first arrow, l , is c (resp. w if $n = 0$) and the second arrow, z , is the monoidal structure on G . Therefore, any model of **mGL** [32] is a model of GRASS instantiated with the modes $\mathbf{L} < \mathbf{m}$. When $\mathbf{m} = \mathbf{U}$, $\mathcal{C}$ is cartesian closed, and $\mathbf{U}$ acts on $\mathcal{C}$ by $1 \odot (-) = \text{id}$, as described in Example 4.4, we recover the categorical situation of LNL [4].

Definition 4.13 Recall the rules for well-formed types from Figure 1. We assign each type $A \in \text{Type}(\mathbf{m})$

Name	Type	Isomorphism?
δ	$(rq) \odot X \rightarrow r \odot (q \odot X)$	no
ϵ	$1 \odot X \rightarrow X$	no
τ	$(r \odot X) \otimes (r \odot Y) \rightarrow r \odot (X \otimes Y)$	yes
ι	$I \rightarrow r \odot I$	yes
c	$(r + q) \odot X \rightarrow (r \odot X) \otimes (q \odot X)$	no
w	$0 \odot X \rightarrow I$	no
m	$F(X) \otimes F(Y) \rightarrow F(X \otimes Y)$	yes
u	$I \rightarrow F(I)$	yes
μ	$F(\varphi_{n,m}(q) \odot X) \rightarrow q \odot F_n^m(X)$	no

Fig. 7. Summary of named natural transformations in the categorical model for GRASS

an object of $\llbracket A \rrbracket \in \mathcal{C}_m$ by induction on the proof of $A \in \text{Type}(m)$.

$$\begin{array}{c}
\text{UNIT} \\
\hline
\llbracket \mathbf{I}_m \rrbracket = I \in \mathcal{C}_m
\end{array}
\quad
\begin{array}{c}
\text{PAIR} \\
\hline
\frac{A \in \text{Type}(m) \quad B \in \text{Type}(m)}{\llbracket A \otimes B \rrbracket = \llbracket A \rrbracket \otimes \llbracket B \rrbracket \in \mathcal{C}_m}
\end{array}
\quad
\begin{array}{c}
\text{FUN} \\
\hline
\frac{n \leq m \quad q \in R_m \quad A \in \text{Type}(m) \quad B \in \text{Type}(n)}{\llbracket A^{q:m} \multimap B \rrbracket = F_n^m(q \odot \llbracket A \rrbracket) \multimap \llbracket B \rrbracket \in \mathcal{C}_n}
\end{array}$$

$$\begin{array}{c}
\text{DROP} \\
\hline
\frac{n \leq m \quad q \in R_m \quad A \in \text{Type}(m)}{\llbracket \downarrow_{n \leq m}^q A \rrbracket = F_n^m(q \odot \llbracket A \rrbracket) \in \mathcal{C}_n}
\end{array}
\quad
\begin{array}{c}
\text{DROP} \\
\hline
\frac{m \leq n \quad A \in \text{Type}(m)}{\llbracket \uparrow_{m \leq n} A \rrbracket = G_m^n \llbracket A \rrbracket \in \mathcal{C}_n}
\end{array}$$

We will often omit the brackets $\llbracket - \rrbracket$ writing A instead of $\llbracket A \rrbracket$.

Definition 4.14 Let $\rho = r_1, \dots, r_k$ and $M = m_1, \dots, m_k$ and $\Gamma = x_1 : A_1, \dots, x_k : A_k$, and $M \geq m$. Define

$$\llbracket \rho \mid M \odot \Gamma \rrbracket_m := \bigotimes_{i=1}^k \llbracket \downarrow_{m \leq m_i}^{r_i} A_i \rrbracket = \bigotimes_{i=1}^k F_m^{m_i}(r_i \odot A_i)$$

Note that there is a canonical isomorphism $F_n^m(\llbracket \rho \mid M \odot \Gamma \rrbracket_m) \cong \llbracket \rho \mid M \odot \Gamma \rrbracket_n$ whenever $n \leq m$.

Definition 4.15 In the situation of Definition 4.14, let $q \in R_m$ and let $t : \llbracket \rho \mid M \odot \Gamma \rrbracket_m \rightarrow A$ be a morphism. Define $q \cdot t : \llbracket q\rho \mid M \odot \Gamma \rrbracket_m \rightarrow q \odot A$ as

$$\bigotimes_{i=1}^k F_m^{m_i}(q \cdot r_i) \odot A_i \xrightarrow{\tau \odot \bigotimes_i (\mu \odot F_m^{m_i} \delta)} q \odot \bigotimes_{i=1}^k F_m^{m_i}(r_i \odot A_i) \xrightarrow{q \odot t} q \odot A$$

for $k > 0$. When $k = 0$, define $q \cdot t = (q \odot t) \circ \iota : I_m \rightarrow q \odot I_m \rightarrow q \odot A$.

Definition 4.16 We assign each judgment $\rho \mid M \odot \Gamma \vdash_m t : A$ a morphism $\llbracket t \rrbracket : \llbracket \rho \mid M \odot \Gamma \rrbracket \rightarrow \llbracket A \rrbracket$ in $\mathcal{C}_m$. This assignment is specified in Figure 8 by induction on the derivation of t .

Theorem 4.17 (Soundness) β - and η -conversions are sound with respect to the categorical semantics:

- (i) If $\rho \mid M \odot \Gamma \vdash_m t_1 \equiv_\beta t_2 : A$ then $\llbracket t_1 \rrbracket = \llbracket t_2 \rrbracket$ in $\mathcal{C}_m$.
- (ii) If $\rho \mid M \odot \Gamma \vdash_m t_1 \equiv_\eta t_2 : A$ then $\llbracket t_1 \rrbracket = \llbracket t_2 \rrbracket$ in $\mathcal{C}_m$.

Proof. Given in §§C.2, C.3 of the appendix [16]. □

5 Conclusion, related and future work

We defined a graded logic which allows grades from multiple modes to exist in a single system. Furthermore, we have generalized the graded weakening and contraction rules to obtain more fine-grained control in the graded setting. This allowed us to recover non-graded substructural logics in an entirely graded setting. We

$$\begin{array}{c}
 \text{VAR} \\
 \hline
 F_m^m(1 \odot A) \cong 1 \odot A \xrightarrow{e} A \\
 \\
 \text{WEAK} \\
 \text{Weak(I)} \quad m \leq l \quad B \in \text{Type}(l) \quad \llbracket \rho \mid M \odot \Gamma \rrbracket_m \xrightarrow{t} A \\
 \hline
 \llbracket \rho \mid M \odot \Gamma \rrbracket_m \otimes F_m^l(0 \odot B) \xrightarrow{t \otimes (u^{-1} \circ F(w))} A \otimes I \cong A \\
 \\
 \text{SUB} \\
 \hline
 (r_i \leq q_i)_i \quad \bigotimes_i F_m^{m_i}(r_i \odot X_i) \xrightarrow{t} A \\
 \hline
 \bigotimes_i F_m^{m_i}(q_i \odot X_i) \xrightarrow{F_m^{m_i}((r_i \leq q_i) \odot X_i)} \bigotimes_i F_m^{m_i}(r_i \odot X_i) \xrightarrow{t} A \\
 \\
 \text{CONT} \\
 \hline
 r_1, r_2 \in \text{Cont}(m) \quad \llbracket \rho \mid M \odot \Gamma \rrbracket_n \otimes F_n^m(r_1 \odot A) \otimes F_n^m(r_2 \odot A) \xrightarrow{t} B \\
 \hline
 \llbracket \rho \mid M \odot \Gamma \rrbracket_n \otimes F_n^m((r_1 + r_2) \odot A) \xrightarrow{\text{id} \otimes (m^{-1} \circ F_n^m(c))} \llbracket \rho \mid M \odot \Gamma \rrbracket_n \otimes F_n^m(r_1 \odot A) \otimes F_n^m(r_2 \odot A) \xrightarrow{t} B \\
 \\
 \text{UNIT.E} \\
 \hline
 \llbracket \sigma \mid N \odot \Delta \rrbracket_m \xrightarrow{e} I_m \quad \llbracket \rho \mid M \odot \Gamma \rrbracket_m \xrightarrow{t} A \\
 \hline
 \llbracket \rho \mid M \odot \Gamma \rrbracket_m \otimes \llbracket q\sigma \mid N \odot \Delta \rrbracket_m \xrightarrow{t \otimes (\iota^{-1} \circ (q \cdot e))} A \otimes I_m \cong A \\
 \\
 \text{FUN.I} \\
 \hline
 \llbracket \rho \mid M \odot \Gamma \rrbracket_n \otimes F_n^m(r \odot A) \xrightarrow{e} B \\
 \hline
 \llbracket \rho \mid M \odot \Gamma \rrbracket_m \xrightarrow{\lambda_e} F_n^m(r \odot A) \multimap B \\
 \\
 \text{FUN.E} \\
 \hline
 \llbracket \rho \mid M \odot \Gamma \rrbracket_n \xrightarrow{t} F_n^m(q \odot A) \multimap B \quad \llbracket \sigma \mid N \odot \Delta \rrbracket_m \xrightarrow{e} A \\
 \hline
 \llbracket \rho \mid M \odot \Gamma \rrbracket_n \otimes F_n^m \llbracket \sigma \mid N \odot \Delta \rrbracket_m \xrightarrow{t \otimes F_n^m(q \cdot e)} (F_n^m(q \odot A) \multimap B) \otimes F_n^m(q \odot A) \xrightarrow{ev} B \\
 \\
 \text{PAIR.I} \\
 \hline
 \llbracket \rho_1 \mid M_1 \odot \Gamma_1 \rrbracket_m \xrightarrow{a_1} A_1 \quad \llbracket \rho_2 \mid M_2 \odot \Gamma_2 \rrbracket_m \xrightarrow{a_2} A_2 \\
 \hline
 \llbracket \rho_1 \mid M_1 \odot \Gamma_1 \rrbracket_m \otimes \llbracket \rho_2 \mid M_2 \odot \Gamma_2 \rrbracket_m \xrightarrow{a_1 \otimes a_2} A_1 \otimes A_2 \\
 \\
 \text{PAIR.E} \\
 \hline
 \llbracket \sigma \mid N \odot \Delta \rrbracket_m \xrightarrow{e} A_1 \otimes A_2 \quad \llbracket \rho \mid M \odot \Gamma \rrbracket_n \otimes F_n^m(q \odot A_1) \otimes F_n^m(q \odot A_2) \xrightarrow{t} B \\
 \hline
 \llbracket \rho \mid M \odot \Gamma \rrbracket_n \otimes F_n^m \llbracket q\sigma \mid N \odot \Delta \rrbracket_m \xrightarrow{\text{id} \otimes (m^{-1} \circ F_n^m(\tau^{-1} \circ (q \cdot e)))} \llbracket \rho \mid M \odot \Gamma \rrbracket_n \otimes F_n^m(q \odot A_1) \otimes F_n^m(q \odot A_2) \xrightarrow{t} B \\
 \\
 \text{RAISE.I} \\
 \hline
 m \leq n \leq M \quad \llbracket \rho \mid M \odot \Gamma \rrbracket_m \xrightarrow{e} A \\
 \hline
 \llbracket \rho \mid M \odot \Gamma \rrbracket_n \xrightarrow{\text{unit}} G_m^n F_m^n \llbracket \rho \mid M \odot \Gamma \rrbracket_n \cong G_m^n \llbracket \rho \mid M \odot \Gamma \rrbracket_m \xrightarrow{G_m^n e} G_m^n A \\
 \\
 \text{RAISE.E} \\
 \hline
 m \leq n \quad \llbracket \rho \mid M \odot \Gamma \rrbracket_n \xrightarrow{e} G_m^n A \\
 \hline
 F_n^m \llbracket \rho \mid M \odot \Gamma \rrbracket_n \xrightarrow{F_n^m e} F_n^m G_m^n A \xrightarrow{\text{counit}} A \\
 \\
 \text{DROP.I} \\
 \hline
 n \leq m \quad q \in R_m \quad \llbracket \rho \mid M \odot \Gamma \rrbracket_m \xrightarrow{e} A \\
 \hline
 F_n^m \llbracket q\rho \mid M \odot \Gamma \rrbracket_m \xrightarrow{F_n^m(q \cdot e)} F_n^m(q \odot A) \\
 \\
 \text{DROP.E} \\
 \hline
 l \leq n \leq m \quad \llbracket \sigma \mid N \odot \Delta \rrbracket_n \xrightarrow{e} F_n^m(q \odot A) \quad \llbracket \rho \mid M \odot \Gamma \rrbracket_l \otimes F_l^m(q \odot A) \xrightarrow{t} B \\
 \hline
 \llbracket \rho \mid M \odot \Gamma \rrbracket_l \otimes F_l^m \llbracket \sigma \mid N \odot \Delta \rrbracket_n \xrightarrow{\text{id} \otimes F_l^m e} \llbracket \rho \mid M \odot \Gamma \rrbracket_l \otimes F_l^m F_n^m(q \odot A) \cong \llbracket \rho \mid M \odot \Gamma \rrbracket_l \otimes F_l^m(q \odot A) \xrightarrow{t} B
 \end{array}$$

Fig. 8. GRASS interpretation into a categorical model

gave an account of the categorical semantics of our logic based on graded comonads. Our logic consists of multiple graded systems connected by morphisms of modes, and this design is mirrored in the categorical semantics. We introduced a novel notion of morphisms of graded comonads, and showed how it connects to the existing literature. We saw that categorical models of substructural logics provide categorical models for the modes which recover such logics at the syntactic level.

Combined systems

Systems which combine logics permitting different sets of structural rules originate in Benton's Linear-non-Linear Logic (LNL) [4], a system which combines linear and intuitionistic logics. A similar result was

achieved for graded and linear logics in **mGL** of Vollmer et al. [32]. In the non-graded world, LNL was further generalized by Pruiksmas et al. into Adjoint Logic [28], which allows an arbitrary collection of logics with different structural rules (relevant, affine, linear, intuitionistic) to exist in one system. The present work initially started out as a “graded Adjoint Logic,” and thus bears many similarities to Adjoint Logic. For example, our mode system and modal operators $\uparrow_{n \leq m} A$ and $\downarrow_{n \leq m}^q A$ are direct adaptations of the mode system and modal operators presented in Adjoint Logic. All of the systems mentioned here have one observation in common: Terms from logics with fewer structural rules may depend on variables permitting more structural rules, but not vice-versa. This invariant is also present in our work, and we leverage it to ensure that reuse of more structured variables is well-behaved in less structured modes.

Heterogenously graded systems

Systems where grades can be drawn from more than one grade algebra have been studied for subsets of Java by Bianchini et al. [5,6] as well as Giannini and Duso [14]. We refer to these works collectively as *Graded Java*. Modes in GRASS correspond to *kinds* of coefficients in Graded Java. Both modes and kinds are arranged in a preorder, and both systems use morphisms of grade algebras to combine (i.e. add or multiply) grades from different algebras.

We highlight some differences: Graded Java reduces the problem of tracking multiple kinds of grades in one program to the case of a single grade algebra by equipping the disjoint union $\bigsqcup_k R_k$ with an appropriate grade algebra structure, where k ranges over the set of kinds of coefficients. The set of kinds is required to have joins, and if $r_1 \in R_{k_1}$ and $r_2 \in R_{k_2}$, then $r_1 + r_2 \in R_{k_1 \vee k_2}$ and similarly for multiplication. This can lead to a loss of information, e.g. when $k_1 \vee k_2$ is the terminal kind $k_\top$ with $R_{k_\top} = \top$ the terminal semiring. This means that a variable can be marked unrestricted in a term, even though it should have non-trivial usage restrictions in each of its subterms. In contrast to this, GRASS’s typing rules ensure that additions only happen between grades from the same mode, and multiplications only happen when the modes are comparable in the preorder, thus avoiding the need for joins. Furthermore each variable in GRASS belongs to a unique mode, and its grades are always guaranteed to be drawn from that mode. Hence there is no risk of the loss of information mentioned above. We leave it to future work to investigate if/how our approach can be implemented in a system like Graded Java.

Categorical semantics of graded modal types

Many existing graded type systems have been given categorical semantics [8,12,13,25,26]. These semantics were all somewhat similar, employing a notion of *graded linear exponential comonads* (though terminology varied). Later, Katsumata [17] gave a comprehensive account of graded linear exponential comonads. Katsumata’s work also serves as the basis for the exponential actions defined in the present text. Beyond the changes that were necessary to account for our modified weakening and contraction rules, there is one other difference between our presentation and that of Katsumata: We require that $r \odot (-)$ be a strong monoidal functor, while the categorical semantics in the work cited here all require this functor only to be lax monoidal. We attribute this difference to the fact that strong monoidality of $r \odot (-)$ is required for the interpretation of the tensor product type, which was not included in the above works. Categorical semantics were given to **mGL** by Vollmer et al. [32]. This type system does have a tensor product, and they require the functors $r \odot (-)$ to be strict monoidal, i.e. the morphism $(r \odot A) \otimes (r \odot B) \rightarrow r \odot (A \otimes B)$ must be an identity. We have shown here that the strictness is not necessary for the interpretation of the graded tensor product. Finally, Atkey [2] has given categorical semantics to a dependently typed graded system using *Quantitative Categories with Families*, an approach which is not based on graded comonads.

Future work

One future direction is to use the lessons learned here about the categorical semantics of graded types towards a categorical semantics of graded *dependent* types which is based on graded comonads. One problem in this area is that uses of terms inside of types are considered computationally irrelevant, and therefore exempt from grading. In other words, types are treated as intuitionistic, but terms must still be graded. The categorical semantics presented here are able to capture graded and intuitionistic variables existing in the same system, by instantiating GRASS with any one mode for the graded system, together

with the terminal mode to recover intuitionistic logic. We hope to use this as a starting point for the categorical semantics of graded dependent types.

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A Specification of the single mode system

This section contains the specification for the single mode specialization of GRASS. The syntax for types and terms, and the corresponding typing rules are

$$\begin{aligned}
 (\text{Types}) \quad A, B, C, T &::= \mathbf{I} \mid A \otimes B \mid A \oplus B \mid A^q \multimap B \mid \Box_q A \\
 (\text{Terms}) \quad e, f, t &::= x \mid \star \mid \text{let}_{@q} \star = t \text{ in } e \mid (e_1, e_2) \mid \text{let}_{@q}(x, y) = t \text{ in } e \mid \text{inl } t \mid \text{inr } t \\
 &\mid \text{case}_q(t; x_1.e_1; x_2.e_2) \mid \lambda x.t \mid f t \mid \Box_q t \mid \text{let}_{@q} \Box x = t \text{ in } e
 \end{aligned}$$

$$\begin{array}{c}
 \text{S.VAR} \\
 \frac{}{1 \odot x : A \vdash x : A} \\
 \\
 \text{S.WEAK} \\
 \text{Weak(m)} \quad \frac{\rho \odot \Gamma \vdash e : A}{\rho, 0 \odot \Gamma, x : B \vdash e : A} \\
 \\
 \text{S.CONT} \\
 \frac{q_1, q_2 \in \text{Cont(m)} \quad \rho, q_1, q_2 \odot \Gamma, x : A, y : A \vdash e : B}{\rho, q_1 + q_2 \odot \Gamma, z : A \vdash [z/x, z/y]e : B} \\
 \\
 \text{S.SUB} \quad \frac{\rho \leq \sigma \quad \rho \odot \Gamma \vdash e : A}{\sigma \odot \Gamma \vdash e : A} \quad \text{S.UNIT.I} \quad \frac{}{\emptyset \odot \emptyset \vdash \star : \mathbf{I}} \quad \text{S.UNIT.E} \quad \frac{\sigma \odot \Delta \vdash t : \mathbf{I} \quad \rho \odot \Gamma \vdash e : A}{\rho, q \cdot \sigma \odot \Gamma, \Delta \vdash \text{let}_{@q} \star = t \text{ in } e : A} \\
 \\
 \text{S.PAIR.I} \quad \frac{\rho \odot \Gamma_1 \vdash t_1 : A_1 \quad \sigma \odot \Gamma_2 \vdash t_2 : A_2}{\rho, \sigma \odot \Gamma_1, \Gamma_2 \vdash (t_1, t_2) : A_1 \otimes A_2} \quad \text{S.PAIR.E} \quad \frac{\sigma \odot \Delta \vdash t : A_1 \otimes A_2 \quad \rho, q, q \odot \Gamma, x_1 : A_1, x_2 : A_2 \vdash e : B}{\rho, q \cdot \sigma \odot \Gamma, \Delta \vdash \text{let}_{@q}(x_1, x_2) = t \text{ in } e : B} \quad \text{S.SUM.IL} \quad \frac{\rho \odot \Gamma \vdash e : A}{\rho \odot \Gamma \vdash \text{inl } e : A \oplus B} \\
 \\
 \text{S.SUM.IR} \quad \frac{\rho \odot \Gamma \vdash e : B}{\rho \odot \Gamma \vdash \text{inr } e : A \oplus B} \quad \text{S.SUM.E} \quad \frac{\sigma \odot \Delta \vdash t : A_1 \oplus A_2 \quad \rho, q \odot \Gamma, x_1 : A_1 \vdash e_1 : B \quad \rho, q \odot \Gamma, x_2 : A_2 \vdash e_2 : B}{\rho, q \cdot \sigma \odot \Gamma, \Delta \vdash \text{case}_q(t; x_1.e_1; x_2.e_2) : B} \quad \text{S.FUN.I} \quad \frac{\rho, q \odot \Gamma, x : A \vdash t : B}{\rho \odot \Gamma \vdash \lambda x.t : A^q \multimap B} \\
 \\
 \text{S.FUN.E} \quad \frac{\rho \odot \Gamma \vdash t : A^q \multimap B \quad \sigma \odot \Delta \vdash e : A}{\rho, q \cdot \sigma \odot \Gamma, \Delta \vdash t e : B} \quad \text{S.BOX.I} \quad \frac{\rho \odot \Gamma \vdash t : A}{q \cdot \rho \odot \Gamma \vdash \Box_q t : \Box_q A} \quad \text{S.BOX.E} \quad \frac{\rho, q \odot \Gamma, x : A \vdash t : B \quad \sigma \odot \Delta \vdash e : \Box_q A}{\rho, \sigma \odot \Gamma, \Delta \vdash \text{let}_{@q} \Box x = e \text{ in } t : B}
 \end{array}$$

B Syntactic properties of Grass

B.1 Substitution theorem

Theorem B.1 *If $\rho, r \mid \mathbf{M}, \mathbf{m} \odot \Gamma, x : A \vdash t : B$ and $\sigma \mid \mathbf{N} \odot \Delta \vdash_{\mathbf{m}} e : A$, then $\rho, r\sigma \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash [e/x]t : B$.*

Due to the presence of an explicit contraction rule, we cannot prove this theorem directly. Instead we must strengthen the inductive hypothesis to make the structural induction work out.

Theorem B.2 (Simultaneous Substitution) *Suppose that $\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t : T$ with $\rho = r_1, \dots, r_k$, $\mathbf{M} = \mathbf{m}_1, \dots, \mathbf{m}_k$ and $\Gamma = x_1 : A_1, \dots, x_k : A_k$. Suppose that for each $i \in \{1, \dots, k\}$ we have a term $\sigma_i \mid \mathbf{N}_i \odot \Delta_i \vdash_{\mathbf{m}_i} e_i : A_i$. Then*

$$r_1\sigma_1, \dots, r_k\sigma_k \mid \mathbf{N}_1, \dots, \mathbf{N}_k \odot \Delta_1, \dots, \Delta_k \vdash_{\mathbf{m}} [e_1/x_1, \dots, e_k/x_k]t : T$$

We may abbreviate this term as $[e_i/x_i]t$.

Proof. By induction on the derivation of $\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t : T$.

Case 1 (VAR) In this case $k = 1$ and $T = A_1$ and the derivation is

$$\frac{}{1 \mid \mathbf{m}_1 \odot x_1 : A_1 \vdash_{\mathbf{m}} x_1 : A_1}.$$

We must show that $\sigma_1 \mid \mathbf{N}_1 \odot \Delta_1 \vdash_{\mathbf{m}_1} e_1 : A_1$ which is an assumption.

Case 2 (WEAK) In this case the derivation is

$$\frac{\text{Weak}(\mathbf{m}_{k+1}) \quad \mathbf{m} \leq \mathbf{m}_{k+1} \quad A_{k+1} \in \text{Type}(\mathbf{m}_{k+1}) \quad \rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t : T}{\rho, 0 \mid \mathbf{M}, \mathbf{m}_{k+1} \odot \Gamma, x_{k+1} : A_{k+1} \vdash_{\mathbf{m}} t : T}$$

The inductive hypothesis is

$$r_1 \cdot \sigma_1, \dots, r_k \cdot \sigma_k \mid \mathbf{N}_1, \dots, \mathbf{N}_k \odot \Delta_1, \dots, \Delta_k \vdash [e_1/x_1, \dots, e_k/x_k]t : T$$

and we need to show that

$$r_1 \cdot \sigma_1, \dots, r_k \cdot \sigma_k, 0 \cdot \sigma_{k+1} \mid \mathbf{N}_1, \dots, \mathbf{N}_k, \mathbf{N}_{k+1} \odot \Delta_1, \dots, \Delta_k, \Delta_{k+1} \vdash [e_1/x_1, \dots, e_k/x_k, e_{k+1}/x_{k+1}]t : T.$$

From $\rho_{k+1} \mid \mathbf{N}_{k+1} \odot \Delta_{k+1} \vdash_{\mathbf{m}_{k+1}} e_{k+1} : A_{k+1}$ it follows that $\mathbf{N} \geq \mathbf{m}_{k+1}$. Together with **Weak** $\mathbf{m}_{k+1}$, this implies that **Weak**($\mathbf{n}$) holds for each $\mathbf{n}$ in $\mathbf{N}_{k+1}$. Hence we may apply a sequence of weakenings to to weaken by each type occurring in Δ_{k+1} to obtain

$$r_1 \cdot \sigma_1, \dots, r_k \cdot \sigma_k, 0 \cdot \sigma_{k+1} \mid \mathbf{N}_1, \dots, \mathbf{N}_k, \mathbf{N}_{k+1} \odot \Delta_1, \dots, \Delta_k, \Delta_{k+1} \vdash [e_1/x_1, \dots, e_k/x_k]t : T.$$

which is the desired judgment since x_{k+1} does not occur freely in t .

Case 3 (SUB) In this case we have $\rho' = r'_1, \dots, r'_k$ and $r'_i \leq r_i$ for each i and the derivation is

$$\frac{\rho' \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t : T}{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t : T}$$

By monotonicity of multiplication, we have $r'_i\sigma_i \leq r_i\sigma_i$ for each i and hence $r'_1\sigma_1, \dots, r'_k\sigma_k \leq r_1\sigma_1, \dots, r_k\sigma_k$. The following application of the SUB rule

$$\frac{r'_1\sigma_1, \dots, r'_k\sigma_k \mid \mathbf{N}_1, \dots, \mathbf{N}_k \odot \Delta_1, \dots, \Delta_k \vdash_{\mathbf{m}} [e_1/x_1, \dots, e_k/x_k]t : T \quad r'_1\sigma_1, \dots, r'_k\sigma_k \leq r_1\sigma_1, \dots, r_k\sigma_k}{r_1\sigma_1, \dots, r_k\sigma_k \mid \mathbf{N}_1, \dots, \mathbf{N}_k \odot \Delta_1, \dots, \Delta_k \vdash_{\mathbf{m}} [e_1/x_1, \dots, e_k/x_k]t : T}$$

yields the desired result.

Case 4 (CONT) The derivation has the shape

$$\frac{\rho, r_k, r_{k+1} \mid \mathbf{M}, \mathbf{m}_k, \mathbf{m}_k \odot \Gamma, x_k : A_k, x_{k+1} : A_k \vdash_{\mathbf{m}} t : T \quad r_k, r_{k+1} \in \text{Cont } \mathbf{m}}{\rho, r_k + r_{k+1} \mid \mathbf{M}, \mathbf{m}_k \odot \Gamma, z : A_k \vdash_{\mathbf{m}} [z/x_k, z/x_{k+1}]t : T}$$

We apply the inductive hypothesis with $e_{k+1} = e_k$ to obtain

$$r_1 \cdot \sigma_1, \dots, r_k \cdot \sigma_k, r_{k+1} \cdot \sigma_k \mid \mathbf{N}_1, \dots, \mathbf{N}_k, \mathbf{N}_k \odot \Delta_1, \dots, \Delta_k, \Delta_k \vdash_{\mathbf{m}} [e_1/x_1, \dots, e_k/x_k, e_k/x_{k+1}]t : T$$

Since $\text{Cont } \mathbf{m}$ is an ideal, each entry in $r_k \cdot \sigma_k$ and $r_{k+1} \cdot \sigma_k$ is in $\text{Cont } \mathbf{m}$. A sequence of exchanges followed by a sequence of contractions yields

$$r_1 \cdot \sigma_1, \dots, (r_k + r_{k+1}) \cdot \sigma_k \mid \mathbf{N}_1, \dots, \mathbf{N}_k \odot \Delta_1, \dots, \Delta_k \vdash_{\mathbf{m}} [e_1/x_1, \dots, e_k/x_k, e_k/x_{k+1}]t : T.$$

The term in the desired result $[e_1/x_1, \dots, e_{k-1}/x_{k-1}, e_k/z][z/x_k, z/x_{k+1}]t$ which is equal to the one above.

Case 5 (UNIT.I) There is nothing to prove in this case.

Case 6 (UNIT.E) In this case the derivation is

$$\frac{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t : T \quad \rho' \mid \mathbf{M}' \odot \Gamma' \vdash_{\mathbf{m}} t' : \mathbf{I}_{\mathbf{m}}}{\rho, q\rho' \mid \mathbf{M}, \mathbf{M}' \odot \Gamma, \Gamma' \vdash_{\mathbf{m}} \text{let}_{@q} \star_{\mathbf{m}} = t' \text{ in } t : T}$$

Write $\rho = r_1, \dots, r_k$ and $\rho' = r'_1, \dots, r'_\ell$; Write $\Gamma = x_1 : A_1, \dots, x_k : A_k$ and $\Gamma' = x'_1 : A'_1, \dots, x'_\ell : A'_\ell$; Write $\mathbf{M} = \mathbf{m}_1, \dots, \mathbf{m}_k$ and $\mathbf{M}' = \mathbf{m}'_1, \dots, \mathbf{m}'_\ell$. Given terms $\sigma_i \mid \mathbf{N}_i \odot \Delta_i \vdash_{\mathbf{m}_i} e_i : A_i$ ($i \in \{1, \dots, k\}$) and $\sigma'_i \mid \mathbf{N}_i \odot \Delta'_i \vdash_{\mathbf{m}'_i} e'_i : A'_i$ ($i \in \{1, \dots, \ell\}$), we must show that

$$\begin{aligned} r_1 \sigma_1, \dots, r_k \sigma_k, q r'_1 \cdot \sigma'_1, \dots, q r'_\ell \cdot \sigma'_\ell \mid \mathbf{N}_1, \dots, \mathbf{N}_k, \mathbf{N}'_1, \dots, \mathbf{N}'_\ell \odot \Delta_1, \dots, \Delta_k, \Delta'_1, \dots, \Delta'_\ell \\ \vdash_{\mathbf{m}} \text{let}_{@q} \star_{\mathbf{m}} = [e'_i/x'_i]t' \text{ in } [e_i/x_i]t : T \end{aligned}$$

By induction we have

$$\begin{aligned} r_1 \sigma_1, \dots, r_k \sigma_k \mid \mathbf{N}_1, \dots, \mathbf{N}_k \odot \Delta_1, \dots, \Delta_k \vdash_{\mathbf{m}} [e_i/x_i]t : T \\ r'_1 \sigma'_1, \dots, r'_\ell \sigma'_\ell \mid \mathbf{N}'_1, \dots, \mathbf{N}'_\ell \odot \Delta'_1, \dots, \Delta'_\ell \vdash_{\mathbf{m}} [e'_i/x'_i]t' : \mathbf{I}_{\mathbf{m}} \end{aligned}$$

applying the rule UNIT.E to these judgments, we obtain the desired result, since $q(r'_1 \sigma'_1, \dots, r'_\ell \sigma'_\ell) = q r'_1 \sigma'_1, \dots, q r'_\ell \sigma'_\ell$.

Case 7 (FUN.I) The derivation is

$$\frac{\rho, q \mid \mathbf{M}, \mathbf{n} \odot \Gamma, y : B \vdash_{\mathbf{m}} t : T}{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} \lambda y. t : B^{q:\mathbf{n}} \multimap T}$$

With $\rho = r_1, \dots, r_k$ $\mathbf{M} = \mathbf{m}_1, \dots, \mathbf{m}_k$ and $\Gamma = x_1 : A_1, \dots, x_k : A_k$. $\sigma_i \mid \mathbf{N}_i \odot \Delta_i \vdash_{\mathbf{m}_i} e_i : A_i$ ($i \in \{1, \dots, k\}$) and we must prove

$$r_1 \sigma_1, \dots, r_k \sigma_k \mid \mathbf{N}_1, \dots, \mathbf{N}_k \odot \Delta_1, \dots, \Delta_k \vdash_{\mathbf{m}} \lambda y. [e_i/x_i]t : B^{q:\mathbf{n}} \multimap T$$

We apply the inductive hypothesis with $A_{k+1} = B$ and $e_{k+1} = (1 \mid \mathbf{n} \odot y : B \vdash_{\mathbf{n}} y : B)$ to obtain

$$r_1 \sigma_1, \dots, r_k \sigma_k, q \mid \mathbf{N}_1, \dots, \mathbf{N}_k, \mathbf{n} \odot \Delta_1, \dots, \Delta_k, y : B \vdash_{\mathbf{m}} [e_i/x_i]t : T.$$

Applying rule FUN.I yields the desired result.

Case 8 (FUN.E) In this case the derivation has the shape

$$\frac{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t : B^{q:\mathbf{m}'} \multimap T \quad \rho' \mid \mathbf{M}' \odot \Gamma' \vdash_{\mathbf{m}'} t' : B}{\rho, q\rho' \mid \mathbf{M}, \mathbf{M}' \odot \Gamma, \Gamma' \vdash_{\mathbf{m}} t t' : T}$$

Write $\rho = r_1, \dots, r_k$ and $\rho' = r'_1, \dots, r'_\ell$; Write $\Gamma = x_1 : A_1, \dots, x_k : A_k$ and $\Gamma' = x'_1 : A'_1, \dots, x'_\ell : A'_\ell$; Write $\mathbf{M} = \mathbf{m}_1, \dots, \mathbf{m}_k$ and $\mathbf{M}' = \mathbf{m}'_1, \dots, \mathbf{m}'_\ell$. Given terms $\sigma_i \mid \mathbf{N}_i \odot \Delta_i \vdash_{\mathbf{m}_i} e_i : A_i$ ($i \in \{1, \dots, k\}$) and $\sigma'_i \mid \mathbf{N}_i \odot \Delta'_i \vdash_{\mathbf{m}'_i} e'_i : A'_i$ ($i \in \{1, \dots, \ell\}$), we must show that

$$r_1\sigma_1, \dots, r_k\sigma_k, qr'_1 \cdot \sigma'_1, \dots, qr_k \cdot \sigma'_\ell \mid \mathbf{N}_1, \dots, \mathbf{N}_k, \mathbf{N}'_1, \dots, \mathbf{N}'_\ell \odot \Delta_1, \dots, \Delta_k, \Delta'_1, \dots, \Delta'_\ell \vdash_{\mathbf{m}} [e_i/x_i]t [e'_i/x'_i]t' : T$$

The inductive hypotheses are

$$r_1\sigma_1, \dots, r_k\sigma_k, \mid \mathbf{N}_1, \dots, \mathbf{N}_k, \odot \Delta_1, \dots, \Delta_k, \vdash_{\mathbf{m}} [e_i/x_i]t : B^{q:\mathbf{m}'} \multimap T$$

$$r'_1 \cdot \sigma'_1, \dots, r_k \cdot \sigma'_\ell \mid \mathbf{N}'_1, \dots, \mathbf{N}'_\ell \odot \Delta'_1, \dots, \Delta'_\ell \vdash_{\mathbf{m}} [e'_i/x'_i]t' : B$$

Applying the rule FUN.E yields the desired result.

Case 9 (PAIR.I) In this case the derivation has the shape

$$\frac{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t : T \quad \rho' \mid \mathbf{M}' \odot \Gamma' \vdash_{\mathbf{m}} t' : T'}{\rho, \rho' \mid \mathbf{M}, \mathbf{M}' \odot \Gamma, \Gamma' \vdash_{\mathbf{m}} (t, t') : T \otimes T'}$$

Write $\rho = r_1, \dots, r_k$ and $\rho' = r'_1, \dots, r'_\ell$; Write $\Gamma = x_1 : A_1, \dots, x_k : A_k$ and $\Gamma' = x'_1 : A'_1, \dots, x'_\ell : A'_\ell$; Write $\mathbf{M} = \mathbf{m}_1, \dots, \mathbf{m}_k$ and $\mathbf{M}' = \mathbf{m}'_1, \dots, \mathbf{m}'_\ell$. Given terms $\sigma_i \mid \mathbf{N}_i \odot \Delta_i \vdash_{\mathbf{m}_i} e_i : A_i$ ($i \in \{1, \dots, k\}$) and $\sigma'_i \mid \mathbf{N}_i \odot \Delta'_i \vdash_{\mathbf{m}'_i} e'_i : A'_i$ ($i \in \{1, \dots, \ell\}$), we must show that

$$r_1\sigma_1, \dots, r_k\sigma_k, r'_1 \cdot \sigma'_1, \dots, r'_\ell \cdot \sigma'_\ell \mid \mathbf{N}_1, \dots, \mathbf{N}_k, \mathbf{N}'_1, \dots, \mathbf{N}'_\ell \odot \Delta_1, \dots, \Delta_k, \Delta'_1, \dots, \Delta'_\ell \vdash_{\mathbf{m}} ([e_i/x_i]t, [e'_i/x'_i]t') : T \otimes T'$$

By induction we have

$$r_1\sigma_1, \dots, r_k\sigma_k, \mid \mathbf{N}_1, \dots, \mathbf{N}_k, \odot \Delta_1, \dots, \Delta_k, \vdash_{\mathbf{m}} [e_i/x_i]t : T$$

$$r'_1 \cdot \sigma'_1, \dots, r'_\ell \cdot \sigma'_\ell \mid \mathbf{N}'_1, \dots, \mathbf{N}'_\ell \odot \Delta'_1, \dots, \Delta'_\ell \vdash_{\mathbf{m}} [e'_i/x'_i]t' : T'$$

Applying the rule PAIR.I yields the desired result.

Case 10 (PAIR.E) In this case the derivation is

$$\frac{\rho, q, q \mid \mathbf{M}, \mathbf{n}, \mathbf{n} \odot \Gamma, y_1 : T_1, y_2 : T_2 \vdash_{\mathbf{m}} t : T \quad \rho' \mid \mathbf{M}' \odot \Gamma' \vdash_{\mathbf{n}} t' : T_1 \otimes T_2}{\rho, q\rho' \mid \mathbf{M}, \mathbf{M}' \odot \Gamma, \Gamma' \vdash_{\mathbf{m}} \text{let}_{@q}(y_1, y_2) = t' \text{ in } t : T}$$

Write $\rho = r_1, \dots, r_k$ and $\rho' = r'_1, \dots, r'_\ell$; Write $\Gamma = x_1 : A_1, \dots, x_k : A_k$ and $\Gamma' = x'_1 : A'_1, \dots, x'_\ell : A'_\ell$; Write $\mathbf{M} = \mathbf{m}_1, \dots, \mathbf{m}_k$ and $\mathbf{M}' = \mathbf{m}'_1, \dots, \mathbf{m}'_\ell$. Given terms $\sigma_i \mid \mathbf{N}_i \odot \Delta_i \vdash_{\mathbf{m}_i} e_i : A_i$ ($i \in \{1, \dots, k\}$) and $\sigma'_i \mid \mathbf{N}_i \odot \Delta'_i \vdash_{\mathbf{m}'_i} e'_i : A'_i$ ($i \in \{1, \dots, \ell\}$), we must show that

$$\begin{aligned} r_1\sigma_1, \dots, r_k\sigma_k, qr'_1 \cdot \sigma'_1, \dots, qr_k \cdot \sigma'_\ell \mid \mathbf{N}_1, \dots, \mathbf{N}_k, \mathbf{N}'_1, \dots, \mathbf{N}'_\ell \odot \Delta_1, \dots, \Delta_k, \Delta'_1, \dots, \Delta'_\ell \\ \vdash_{\mathbf{m}} \text{let}_{@q}(y_1, y_2) = [e'_i/x'_i]t' \text{ in } [e_i/x_i]t : T \end{aligned}$$

Applying the inductive hypothesis to $\rho' \mid \mathbf{M}' \odot \Gamma' \vdash_{\mathbf{n}} t' : T_1 \otimes T_2$, we obtain

$$r'_1 \cdot \sigma'_1, \dots, r_k \cdot \sigma'_\ell \mid \mathbf{N}'_1, \dots, \mathbf{N}'_\ell \odot \Delta'_1, \dots, \Delta'_\ell \vdash_{\mathbf{m}} [e'_i/x'_i]t' : T_1 \otimes T_2$$

Furthermore, applying the inductive hypothesis to the judgment $\rho, q, q \mid \mathbf{M}, \mathbf{n}, \mathbf{n} \odot \Gamma, y_1 : T_1, y_2 : T_2 \vdash_{\mathbf{m}} t : T$ with the terms $e_1, \dots, e_k$ and $1 \mid \mathbf{n} \odot y_i : T_i \vdash_{\mathbf{n}} y_i : T_i$ ($i \in \{1, 2\}$) yields the judgment

$$r_1\sigma_1, \dots, r_k\sigma_k, q, q \mid \mathbf{N}_1, \dots, \mathbf{N}_k, \mathbf{n}, \mathbf{n} \odot \Delta_1, \dots, \Delta_k, y_1 : T_1, y_2 : T_2 \vdash_{\mathbf{m}} [e_i/x_i]t : T$$

Now, applying the rule PAIR.E to the two inductive hypotheses yields the desired result.

Case 11 (SUM.IL) The derivation has the form

$$\frac{T_2 \in \text{Type}(\mathbf{m}) \quad \rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t : T_1}{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} \text{inl } t : T_1 \oplus T_2}$$

The inductive hypothesis is

$$r_1\sigma_1, \dots, r_k\sigma_k \mid \mathbf{N}_1, \dots, \mathbf{N}_k \odot \Delta_1, \dots, \Delta_k \vdash_{\mathbf{m}} [e_i/x_i]t : T_1$$

applying the rule SUM.IL yields the desired result.

Case 12 (SUM.IR) Analogous to SUM.IL.

Case 13 (SUM.E) The derivation has the following shape

$$\frac{q \geq 1 \quad \rho, q \mid \mathbf{M}, \mathbf{n} \odot \Gamma, y_1 : T_1 \vdash_{\mathbf{m}} t_1 : T \quad \rho, q \mid \mathbf{M}, \mathbf{n} \odot \Gamma, y_2 : T_2 \vdash_{\mathbf{m}} t_2 : T \quad \rho' \mid \mathbf{M}' \odot \Gamma' \vdash_{\mathbf{n}} t : T_1 \oplus T_2}{\rho, q\sigma \mid \mathbf{M}, \mathbf{M}' \odot \Gamma, \Gamma' \vdash_{\mathbf{m}} \text{case}_q(t; y_1.t_1; y_2.t_2) : T}$$

Write $\rho = r_1, \dots, r_k$ and $\rho' = r'_1, \dots, r'_\ell$; Write $\Gamma = x_1 : A_1, \dots, x_k : A_k$ and $\Gamma' = x'_1 : A'_1, \dots, x'_\ell : A'_\ell$; Write $\mathbf{M} = \mathbf{m}_1, \dots, \mathbf{m}_k$ and $\mathbf{M}' = \mathbf{m}'_1, \dots, \mathbf{m}'_\ell$. Given terms $\sigma_i \mid \mathbf{N}_i \odot \Delta_i \vdash_{\mathbf{m}_i} e_i : A_i$ ($i \in \{1, \dots, k\}$) and $\sigma'_i \mid \mathbf{N}_i \odot \Delta'_i \vdash_{\mathbf{m}'_i} e'_i : A'_i$ ($i \in \{1, \dots, \ell\}$), we must show that

$$r_1\sigma_1, \dots, r_k\sigma_k, qr'_1 \cdot \sigma'_1, \dots, qr'_\ell \cdot \sigma'_\ell \mid \mathbf{N}_1, \dots, \mathbf{N}_k, \mathbf{N}'_1, \dots, \mathbf{N}'_\ell \odot \Delta_1, \dots, \Delta_k, \Delta'_1, \dots, \Delta'_\ell \vdash_{\mathbf{m}} \text{case}_q([e'_i/x'_i]t; y_1.[e_i/x_i]t_1; y_2.[e_i/x_i]t_2) : T$$

Applying the inductive hypothesis to t' yields the judgment

$$r'_1 \cdot \sigma'_1, \dots, r_k \cdot \sigma'_k \mid \mathbf{N}'_1, \dots, \mathbf{N}'_\ell \odot \Delta'_1, \dots, \Delta'_\ell \vdash_{\mathbf{m}} [e'_i/x'_i]t : T_1 \oplus T_2$$

Applying the inductive hypothesis to t_i ($i \in \{1, 2\}$) with $e_1, \dots, e_k$ and $e_{k+1} = 1 \mid \mathbf{n} \odot y_i : T_i \vdash_{\mathbf{n}} y_i : T_i$ yields the judgments

$$r_1\sigma_1, \dots, r_k\sigma_k, q \mid \mathbf{N}_1, \dots, \mathbf{N}_k, \mathbf{n} \odot \Delta_1, \dots, \Delta_k, y_1 : T_1 \vdash_{\mathbf{n}} [e_i/x_i]t_1 : T$$

$$r_1\sigma_1, \dots, r_k\sigma_k, q \mid \mathbf{N}_1, \dots, \mathbf{N}_k, \mathbf{n} \odot \Delta_1, \dots, \Delta_k, y_2 : T_2 \vdash_{\mathbf{n}} [e_i/x_i]t_2 : T.$$

Now applying the rule SUM.E yields the desired result.

Case 14 (RAISE.I) The derivation has the shape

$$\frac{\mathbf{m} \leq \mathbf{n} \quad \mathbf{n} \leq \mathbf{M} \quad \rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t : T}{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{n}} \uparrow_{\mathbf{m} \leq \mathbf{n}} t : \uparrow_{\mathbf{m} \leq \mathbf{n}} T}$$

The inductive hypothesis is

$$r_1\sigma_1, \dots, r_k\sigma_k \mid \mathbf{N}_1, \dots, \mathbf{N}_k \odot \Delta_1, \dots, \Delta_k \vdash_{\mathbf{m}} [e_i/x_i]t : T.$$

If we can apply the rule RAISE.I, we will obtain the desired result. But first we must verify that $\mathbf{n} \leq \mathbf{N}_1, \dots, \mathbf{N}_k$. To do so, it is sufficient to verify $\mathbf{n} \leq \mathbf{N}_i$ for each i . But this follows from transitivity of $\leq$ and the fact that $\mathbf{n} < \mathbf{m}_i < \mathbf{N}_i$. Where the second inequality is a consequence of the assumption $\sigma_i \mid \mathbf{N}_i \odot \Delta_i \vdash_{\mathbf{m}_i} e_i : A_i$.

Case 15 (RAISE.E) The derivation has the form

$$\frac{\rho \mid \mathbf{M} \odot \Gamma \vdash_n t : \uparrow_{\mathbf{m} \leq n} T}{\rho \mid \mathbf{M} \odot \Gamma \vdash_m \uparrow_{\mathbf{m} \leq n}^{-1} t : T}$$

The inductive hypothesis is

$$r_1 \sigma_1, \dots, r_k \sigma_k \mid \mathbf{N}_1, \dots, \mathbf{N}_k \odot \Delta_1, \dots, \Delta_k \vdash_n [e_i/x_i]t : \uparrow_{\mathbf{m} \leq n} T$$

and applying the rule RAISE.E yields the desired result.

Case 16 (DROP.I) The derivation has the shape

$$\frac{n \leq m \quad q \in R_m \quad \rho \mid \mathbf{M} \odot \Gamma \vdash_m t : T}{q\rho \mid \mathbf{M} \odot \Gamma \vdash_n \downarrow_{n \leq m}^q t : \downarrow_{n \leq m}^q T}$$

The inductive hypothesis is

$$r_1 \sigma_1, \dots, r_k \sigma_k \mid \mathbf{N}_1, \dots, \mathbf{N}_k \odot \Delta_1, \dots, \Delta_k \vdash_m [e_i/x_i]t : T$$

and applying the rule DROP.I yields the desired result.

Case 17 (DROP.E) The derivation has the shape

$$\frac{l \leq n \quad n \leq m \quad \rho, q \mid \mathbf{M}, \mathbf{m} \odot \Gamma, y : T' \vdash_l t : T \quad \rho' \mid \mathbf{M}' \odot \Gamma' \vdash_n t' : \downarrow_{n \leq m}^q T'}{\rho, \rho' \mid \mathbf{M}, \mathbf{M}' \odot \Gamma, \Gamma' \vdash_l \text{let}_{@q} \downarrow_{n \leq m} y = t' \text{ in } t : T}$$

Write $\rho = r_1, \dots, r_k$ and $\rho' = r'_1, \dots, r'_\ell$; Write $\Gamma = x_1 : A_1, \dots, x_k : A_k$ and $\Gamma' = x'_1 : A'_1, \dots, x'_\ell : A'_\ell$; Write $\mathbf{M} = \mathbf{m}_1, \dots, \mathbf{m}_k$ and $\mathbf{M}' = \mathbf{m}'_1, \dots, \mathbf{m}'_\ell$. Given terms $\sigma_i \mid \mathbf{N}_i \odot \Delta_i \vdash_{\mathbf{m}_i} e_i : A_i$ ($i \in \{1, \dots, k\}$) and $\sigma'_i \mid \mathbf{N}_i \odot \Delta'_i \vdash_{\mathbf{m}'_i} e'_i : A'_i$ ($i \in \{1, \dots, \ell\}$), we must show that

$$r_1 \sigma_1, \dots, r_k \sigma_k, r'_1 \sigma'_1, \dots, r'_\ell \sigma'_\ell \mid \mathbf{N}_1, \dots, \mathbf{N}_k, \mathbf{N}'_1, \dots, \mathbf{N}'_\ell \odot \Delta_1, \dots, \Delta_k, \Delta'_1, \dots, \Delta'_\ell \vdash_m \text{let}_{@q} \downarrow_{n \leq m} y = [e'_i/x'_i]t' \text{ in } [e_i/x_i]t : T$$

Applying the inductive hypothesis to t' , we obtain

$$r'_1 \sigma'_1, \dots, r'_\ell \sigma'_\ell \mid \mathbf{N}'_1, \dots, \mathbf{N}'_\ell \odot \Delta'_1, \dots, \Delta'_\ell \vdash_m [e'_i/x'_i]t' : T'.$$

Furthermore, applying the inductive hypothesis to t with $e_1, \dots, e_k$ and $e_{k+1} = 1 \mid \mathbf{m} \odot y : T' \vdash_m y : T'$ yields

$$r_1 \sigma_1, \dots, r_k \sigma_k, q \mid \mathbf{N}_1, \dots, \mathbf{N}_k, \mathbf{m} \odot \Delta_1, \dots, \Delta_k, y : T' \vdash_m [e_i/x_i]t : T.$$

Applying the rule DROP.E to the two inductive hypotheses yields the desired result. $\square$

B.2 Syntactic soundness of beta-conversions

Definition B.3 β -conversions are defined by

$$\begin{array}{c}
 \text{UNIT.}\beta \\
 \frac{\sigma \mid \mathbf{N} \odot \Delta \vdash_m \star_m : \mathbf{I}_m \quad \rho \mid \mathbf{M} \odot \Gamma \vdash_m e : A}{\rho, q\sigma \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_m \text{let}_{@q} \star_m = \star_m \text{ in } e \equiv_\beta e : A} \\
 \\
 \text{PAIR.}\beta \\
 \frac{\sigma \mid \mathbf{N} \odot \Delta \vdash_m (t_1, t_2) : A_1 \otimes A_2 \quad \rho, q, q \mid \mathbf{M}, \mathbf{m}, \mathbf{m} \odot \Gamma, x_1 : A_1, x_2 : A_2 \vdash_n e : A}{\rho, q\sigma \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_m \text{let}_{@q}(x_1, x_2) = (t_1, t_2) \text{ in } e \equiv_\beta [t_1/x_1, t_2/x_2]e : A} \\
 \\
 \text{FUN.}\beta \\
 \frac{\rho \mid \mathbf{M} \odot \Gamma \vdash_n \lambda x. e : B^{q:m} \multimap A \quad \sigma \mid \mathbf{N} \odot \Delta \vdash_m t : B}{\rho, q\sigma \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_n (\lambda x. e) t \equiv_\beta [t/x]e : A} \\
 \\
 \text{SUM.}\beta.\text{L} \\
 \frac{q \geq 1 \quad \sigma \mid \mathbf{N} \odot \Delta \vdash_n \text{inl } t : A_1 \oplus A_2 \quad \rho, q \mid \mathbf{M}, \mathbf{n} \odot \Gamma, x_1 : A_1 \vdash_m e_1 : A \quad \rho, q \mid \mathbf{M}, \mathbf{n} \odot \Gamma, x_2 : A_2 \vdash_m e_2 : A}{\rho, q\sigma \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_m \text{case}_q(\text{inl } t; x_1.e_1; x_2.e_2) \equiv_\beta [t/x_1]e_1 : A} \\
 \\
 \text{SUM.}\beta.\text{R} \\
 \frac{q \geq 1 \quad \sigma \mid \mathbf{N} \odot \Delta \vdash_n \text{inr } t : A_1 \oplus A_2 \quad \rho, q \mid \mathbf{M}, \mathbf{n} \odot \Gamma, x_1 : A_1 \vdash_m e_1 : A \quad \rho, q \mid \mathbf{M}, \mathbf{n} \odot \Gamma, x_2 : A_2 \vdash_m e_2 : A}{\rho, q\sigma \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_m \text{case}_q(\text{inl } t; x_1.e_1; x_2.e_2) \equiv_\beta [t/x_2]e_2 : A} \\
 \\
 \text{RAISE.}\beta \\
 \frac{\mathbf{m} \leq \mathbf{n} \quad \rho \mid \mathbf{M} \odot \Gamma \vdash_n \uparrow_{\mathbf{m} \leq \mathbf{n}} t : \uparrow_{\mathbf{m} \leq \mathbf{n}} A}{\rho \mid \mathbf{M} \odot \Gamma \vdash_m \uparrow_{\mathbf{m} \leq \mathbf{n}}^{-1}(\uparrow_{\mathbf{m} \leq \mathbf{n}} t) \equiv_\beta t : A} \\
 \\
 \text{DROP.}\beta \\
 \frac{\mathbf{l} \leq \mathbf{n} \leq \mathbf{m} \quad \sigma \mid \mathbf{N} \odot \Delta \vdash_n \downarrow_{\mathbf{n} \leq \mathbf{m}}^q t : \downarrow_{\mathbf{n} \leq \mathbf{m}}^q A \quad \rho, q \mid \mathbf{M}, \mathbf{m} \odot \Gamma, x : A \vdash_1 e : B}{\rho, \sigma \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_1 \text{let}_{@q} \downarrow_{\mathbf{n} \leq \mathbf{m}} x = \downarrow_{\mathbf{n} \leq \mathbf{m}}^q t \text{ in } e \equiv_\beta [t/x]e : B}
 \end{array}$$

Theorem B.4 β -conversion preserves types and grades, i.e. whenever $\rho \mid \mathbf{M} \odot \Gamma \vdash_m t_1 \equiv_\beta t_2 : A$ then $\rho \mid \mathbf{M} \odot \Gamma \vdash_m t_1 : A$ and $\rho \mid \mathbf{M} \odot \Gamma \vdash_m t_2 : A$

Proof. In each rule, t_1 is obtained from the assumptions of that rule by applying the corresponding elimination rule. It remains to show that $\rho \mid \mathbf{M} \odot \Gamma \vdash_m t_2 : A$. We proceed by case analysis.

Case 1 (UNIT. β) We proceed by induction on the derivation of $\sigma \mid \mathbf{N} \odot \Delta \vdash_m \star_m : \mathbf{I}_m$. We must consider the cases UNIT.I and WEAK, SUB, CONT. All other subcases do not apply, since their derivations do not produce a term of the form $\star_m$.

SUBCASE (UNIT.I). In this case the assumptions are $\emptyset \mid \emptyset \odot \emptyset \vdash_m \star_m : \mathbf{I}_m$ and $\rho \mid \mathbf{M} \odot \Gamma \vdash_m e : A$ and we must prove $\rho \mid \mathbf{M} \odot \Gamma \vdash_m e : A$ which is an assumption.

SUBCASE (WEAK). The derivation has the form

$$\frac{\text{Weak}(\mathbf{n}) \quad \mathbf{n} \geq \mathbf{m} \quad B \in \text{Type}(\mathbf{n}) \quad \sigma \mid \mathbf{N} \odot \Delta \vdash_m \star_m : \mathbf{I}_m}{\sigma, 0 \mid \mathbf{N}, \mathbf{n} \odot \Delta, x : B \vdash_m \star_m : \mathbf{I}_m}.$$

The inductive hypothesis is

$$\rho, q\sigma \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_m e : A$$

The assumptions $\text{Weak}(\mathbf{n})$, $\mathbf{n} \geq \mathbf{m}$ and $B \in \text{Type}(\mathbf{n})$ guarantee that we can apply weakening to the IH to obtain the desired result, noticing that $q(\sigma, 0) = (q\sigma), 0$.

SUBCASE (SUB). The derivation is

$$\frac{\sigma \leq \sigma' \quad \sigma \mid \mathbf{N} \odot \Delta \vdash_m \star_m : \mathbf{I}_m}{\sigma' \mid \mathbf{N} \odot \Delta \vdash_m \star_m : \mathbf{I}_m}$$

with IH $\rho, q\sigma \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_m \star_m : \mathbf{I}_m$. From $\sigma \leq \sigma'$ we obtain $\rho, q\sigma \leq \rho, q\sigma'$, so applying the rule SUB to the IH yields the desired result.

SUBCASE (CONT). In this case the derivation has the form

$$\frac{r_1, r_2 \in \text{Cont}(\mathbf{n}) \quad \sigma, r_1, r_2 \mid \mathbf{N}, \mathbf{n}, \mathbf{n} \odot \Delta, y_1 : B, y_2 : B \vdash_{\mathbf{m}} \star_{\mathbf{m}} : \mathbf{I}_{\mathbf{m}}}{\sigma, r_1 + r_2 \mid \mathbf{N}, \mathbf{n} \odot \Delta, z : B \vdash_{\mathbf{m}} \star_{\mathbf{m}} : \mathbf{I}_{\mathbf{m}}}$$

The IH is

$$\rho, q\sigma, qr_1, qr_2 \mid \mathbf{M}, \mathbf{N}, \mathbf{n}, \mathbf{n} \odot \Gamma, \Delta, y_1 : B, y_2 : B \vdash_{\mathbf{m}} e : A$$

The variables y_1, y_2 do not occur freely in e , because we have the derivation $\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} e : A$. Therefore, it suffices to show that

$$\rho, q\sigma, q(r_1 + r_2) \mid \mathbf{M}, \mathbf{N}, \mathbf{n} \odot \Gamma, \Delta, z : B \vdash_{\mathbf{m}} e : A$$

Since $\text{Cont}(\mathbf{n})$ is an ideal, we have $qr_1, qr_2 \in \text{Cont}(\mathbf{n})$, and hence may apply the rule CONT to obtain the desired result.

Case 2 (PAIR. β) Given

$$\rho, q, q \mid \mathbf{M}, \mathbf{m}, \mathbf{m} \odot \Gamma, x_1 : A_1, x_2 : A_2 \vdash_{\mathbf{n}} e : A \quad \sigma \mid \mathbf{N} \odot \Delta \vdash_{\mathbf{m}} (t_1, t_2) : A_1 \otimes A_2,$$

we must show that

$$\rho, q\sigma \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_{\mathbf{n}} [t_1/x_1, t_2/x_2]e : A.$$

Proceed by induction on the derivation of $\sigma \mid \mathbf{N} \odot \Delta \vdash_{\mathbf{m}} (t_1, t_2) : A_1 \otimes A_2$. We must consider the cases PAIR.I and WEAK, SUB, CONT. All other subcases do not apply since they don't produce a term of the form (t_1, t_2) .

SUBCASE (PAIR.I). In this case the derivation is

$$\frac{\sigma_1 \mid \mathbf{N}_1 \odot \Delta_1 \vdash_{\mathbf{m}} t_1 : A_1 \quad \sigma_2 \mid \mathbf{N}_2 \odot \Delta_2 \vdash_{\mathbf{m}} t_2 : A_2}{\sigma_1, \sigma_2 \mid \mathbf{N}_1, \mathbf{N}_2 \odot \Delta_1, \Delta_2 \vdash_{\mathbf{m}} (t_1, t_2) : A_1 \otimes A_2}$$

Applying the substitution theorem with t_1 and t_2 , we obtain

$$\rho, q\sigma_1, q\sigma_2 \mid \mathbf{M}, \mathbf{N}_1, \mathbf{N}_2 \odot \Gamma, \Delta_1, \Delta_2 \vdash_{\mathbf{n}} [t_1/x_1, t_2/x_2]e : A$$

as desired.

SUBCASE (WEAK). In this case the derivation is

$$\frac{\mathbf{n}' \geq \mathbf{m} \quad \text{Weak}(\mathbf{n}') \quad B \in \text{Type}(\mathbf{n}') \quad \sigma \mid \mathbf{N} \odot \Delta \vdash_{\mathbf{m}} (t_1, t_2) : A_1 \otimes A_2}{\sigma, 0 \mid \mathbf{N}, \mathbf{n}' \odot \Delta, z : B \vdash_{\mathbf{m}} (t_1, t_2) : A_1 \otimes A_2}$$

By induction we have

$$\rho, q\sigma \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_{\mathbf{n}} [t_1/x_1, t_2/x_2]e : A.$$

We have $\mathbf{n} \leq \mathbf{m} \leq \mathbf{n}'$ and hence may apply the rule WEAK to the IH to get

$$\rho, q\sigma, 0 \mid \mathbf{M}, \mathbf{N}, \mathbf{n}' \odot \Gamma, \Delta, z : B \vdash_{\mathbf{n}} [t_1/x_1, t_2/x_2]e : A$$

as desired.

SUBCASE (SUB). The derivation is

$$\frac{\sigma' \leq \sigma \quad \sigma' \mid \mathbf{N} \odot \Delta \vdash_{\mathbf{m}} (t_1, t_2) : A_1 \otimes A_2}{\sigma \mid \mathbf{N} \odot \Delta \vdash_{\mathbf{m}} (t_1, t_2) : A_1 \otimes A_2}$$

and the inductive hypothesis is

$$\rho, q\sigma' \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_{\mathbf{n}} [t_1/x_1, t_2/x_2]e : A.$$

The desired result follows by applying the rule SUB using the fact that multiplication is monotone. SUBCASE (CONT). The derivation is

$$\frac{r_1, r_2 \in \text{Cont}(\mathbf{n}') \quad \sigma, r_1, r_2 \mid \mathbf{N}, \mathbf{n}', \mathbf{n}' \odot \Delta, y_1 : B, y_2 : B \vdash_{\mathbf{m}} (t_1, t_2) : A_1 \otimes A_2}{\sigma, r_1 + r_2 \mid \mathbf{N}, \mathbf{n}', \mathbf{n}' \odot \Delta, z : B \vdash_{\mathbf{m}} ([z/y_1, z/y_2]t_1, [z/y_1, z/y_2]t_2) : A_1 \otimes A_2}$$

The IH is

$$\rho, q\sigma, qr_1, qr_2 \mid \mathbf{M}, \mathbf{N}, \mathbf{n}', \mathbf{n}' \odot \Gamma, \Delta, y_1 : B, y_2 : B \vdash_{\mathbf{n}} [t_1/x_1, t_2/x_2]e : A$$

Since $\text{Cont}(\mathbf{n}')$ is an ideal, we have $qr_1, qr_2 \in \text{Cont}(\mathbf{n}')$. We thus apply the rule CONT to obtain

$$\rho, q\sigma, qr_1 + qr_2 \mid \mathbf{M}, \mathbf{N}, \mathbf{n}' \odot \Gamma, \Delta, z : B \vdash_{\mathbf{n}} [z/y_1, z/y_2][t_1/x_1, t_2/x_2]e : A$$

Since the variables y_1, y_2 only occur freely in t_1 and t_2 , the resulting term is equal to

$$[[z/y_1, z/y_2]t_1/x_1, [z/y_1, z/y_2]t_2/x_2]e$$

as desired.

Case 3 (FUN. β) Given

$$\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{n}} \lambda x.e : B^{q:\mathbf{m}} \multimap A \qquad \sigma \mid \mathbf{N} \odot \Delta \vdash_{\mathbf{m}} t : B$$

we must prove that

$$\rho, q\sigma \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_{\mathbf{n}} [t/x]e : A$$

We proceed by induction on the derivation of

$$\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{n}} \lambda x.e : B^{q:\mathbf{m}} \multimap A.$$

We must analyse the cases FUN.I and WEAK, SUB, CONT. The other cases do not apply, since they do not produce a term of the form $\lambda x.e$.

SUBCASE (FUN.I). In this case we have

$$\frac{\rho, q \mid \mathbf{M}, \mathbf{n} \odot \Gamma, x : B \vdash_{\mathbf{n}} e : A}{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{n}} \lambda x.e : B^{q:\mathbf{m}} \multimap A}$$

The claim follows by applying substitution to the assumption $\rho, q \mid \mathbf{M}, \mathbf{n} \odot \Gamma, x : B \vdash_{\mathbf{n}} e : A$

SUBCASE (WEAK). In this case the derivation is

$$\frac{\text{Weak}(\mathbf{m}') \quad \mathbf{m}' \geq \mathbf{n} \quad T \in \text{Type}(\mathbf{m}') \quad \rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{n}} \lambda x.e : B^{q:\mathbf{m}} \multimap A}{\rho, 0 \mid \mathbf{M}, \mathbf{m}' \odot \Gamma, y : T \vdash_{\mathbf{n}} \lambda x.e : B^{q:\mathbf{m}} \multimap A}$$

By induction we have

$$\rho, q\sigma \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_{\mathbf{n}} [t/x]e : A$$

and we may apply the rule WEAK to obtain

$$\rho, q\sigma, 0 \mid \mathbf{M}, \mathbf{N}, \mathbf{m}' \odot \Gamma, \Delta, y : T \vdash_{\mathbf{n}} [t/x]e : A$$

Since we treat contexts as unordered, this is equivalent to the desired judgment

$$\rho, 0, q\sigma \mid \mathbf{M}, \mathbf{m}', \mathbf{N} \odot \Gamma, y : T, \Delta \vdash_{\mathbf{n}} [t/x]e : A$$

SUBCASE (CONT). In this case the derivation is

$$\frac{r_1, r_2 \in \text{Cont}(\mathbf{m}') \quad \rho, r_1, r_2 \mid \mathbf{M}, \mathbf{m}', \mathbf{m}' \odot \Gamma, y_1 : T, y_2 : T \vdash_n \lambda x. e : B^{q:\mathbf{m}} \multimap A}{\rho, r_1 + r_2 \mid \mathbf{M}, \mathbf{m}', \mathbf{m}' \odot \Gamma, z : T \vdash_n (\lambda x. [z/y_1, z/y_2]e) : B^{q:\mathbf{m}} \multimap A}$$

By induction we have

$$\rho, r_1, r_2, q\sigma \mid \mathbf{M}, \mathbf{m}', \mathbf{m}', \mathbf{N} \odot \Gamma, y_1 : T, y_2 : T, \Delta \vdash_n [t/x]e : A$$

Since contexts are unordered, we may apply the rule CONT to obtain

$$\rho, r_1 + r_2, q\sigma \mid \mathbf{M}, \mathbf{m}', \mathbf{N} \odot \Gamma, z : T, \Delta \vdash_n [z/y_1, z/y_2][t/x]e : A$$

Since the variables y_1, y_2 are not free in t this term is equal to $[t/x][z/y_1, z/y_2]e$ which is the desired term.

SUBCASE (SUB). In this case we have the derivation

$$\frac{\rho' \leq \rho \quad \rho' \mid \mathbf{M} \odot \Gamma \vdash_n \lambda x. e : B^{q:\mathbf{m}} \multimap A}{\rho \mid \mathbf{M} \odot \Gamma \vdash_n \lambda x. e : B^{q:\mathbf{m}} \multimap A}$$

The inductive hypothesis is

$$\rho', q\sigma \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_n [t/x]e : A$$

and since $\rho', q\sigma \leq \rho, q\sigma$ we may apply the rule SUB to obtain the desired result.

Case 4 (RAISE. β) For modes $\mathbf{m} \leq \mathbf{n}$, given $\rho \mid \mathbf{M} \odot \Gamma \vdash_n \uparrow_{\mathbf{m} \leq \mathbf{n}} t : \uparrow_{\mathbf{m} \leq \mathbf{n}} A$, we must prove that $\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t : A$. We proceed by induction on the derivation of $\rho \mid \mathbf{M} \odot \Gamma \vdash_n \uparrow_{\mathbf{m} \leq \mathbf{n}} t : \uparrow_{\mathbf{m} \leq \mathbf{n}} A$. We must analyse the cases RAISE.I and WEAK, SUB, CONT.

SUBCASE (RAISE.I). In this case the derivation is

$$\frac{\mathbf{m} \leq \mathbf{n} \quad \rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t : A}{\rho \mid \mathbf{M} \odot \Gamma \vdash_n \uparrow_{\mathbf{m} \leq \mathbf{n}} t : \uparrow_{\mathbf{m} \leq \mathbf{n}} A}.$$

We are done, since our goal is an assumption.

SUBCASE (WEAK). In this case the derivation is

$$\frac{\text{Weak}(\mathbf{m}') \quad \mathbf{n} \leq \mathbf{m}' \quad B \in \text{Type}(\mathbf{m}') \quad \rho \mid \mathbf{M} \odot \Gamma \vdash_n \uparrow_{\mathbf{m} \leq \mathbf{n}} t : \uparrow_{\mathbf{m} \leq \mathbf{n}} A}{\rho, 0 \mid \mathbf{M}, \mathbf{m}' \odot \Gamma, x : B \vdash_n \uparrow_{\mathbf{m} \leq \mathbf{n}} t : \uparrow_{\mathbf{m} \leq \mathbf{n}} A}$$

The IH is $\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t : A$. Because of $\mathbf{m} \leq \mathbf{n} \leq \mathbf{m}'$ we may apply WEAK to the IH to obtain $\rho, 0 \mid \mathbf{M}, \mathbf{m}' \odot \Gamma, x : B \vdash_{\mathbf{m}} t : A$ as desired.

SUBCASE (CONT). In this case the derivation is

$$\frac{r_1, r_2 \in \text{Cont}(\mathbf{m}') \quad \rho, r_1, r_2 \mid \mathbf{M}, \mathbf{m}', \mathbf{m}' \odot \Gamma, x_1 : B, x_2 : B \vdash_n \uparrow_{\mathbf{m} \leq \mathbf{n}} t : \uparrow_{\mathbf{m} \leq \mathbf{n}} A}{\rho, r_1 + r_2 \mid \mathbf{M}, \mathbf{m}' \odot \Gamma, z : B \vdash_n \uparrow_{\mathbf{m} \leq \mathbf{n}} [z/x_1, z/x_2]t : \uparrow_{\mathbf{m} \leq \mathbf{n}} A}.$$

The IH is $\rho, r_1, r_2 \mid \mathbf{M}, \mathbf{m}', \mathbf{m}' \odot \Gamma, x_1 : B, x_2 : B \vdash_{\mathbf{m}} t : A$ and we still have $r_1, r_2 \in \text{Cont}(\mathbf{m}')$ and thus may apply the rule CONT to obtain $\rho, r_1 + r_2 \mid \mathbf{M}, \mathbf{m}' \odot \Gamma, z : B \vdash_n [z/x_1, z/x_2]t : A$ which is the desired term.

SUBCASE (SUB). The derivation has the form

$$\frac{\rho' \leq \rho \quad \rho' \mid \mathbf{M} \odot \Gamma \vdash_n \uparrow_{\mathbf{m} \leq \mathbf{n}} t : \uparrow_{\mathbf{m} \leq \mathbf{n}} A}{\rho \mid \mathbf{M} \odot \Gamma \vdash_n \uparrow_{\mathbf{m} \leq \mathbf{n}} t : \uparrow_{\mathbf{m} \leq \mathbf{n}} A}$$

The IH is $\rho' \mid \mathbf{M} \odot \Gamma \vdash_n t : A$ and applying rule SUB with $\rho' \leq \rho$ produces the desired term.

Case 5 (DROP. β) Given $l \leq n \leq m$ and

$$\sigma \mid \mathbf{N} \odot \Delta \vdash_n \downarrow_{n \leq m}^q t : \downarrow_{n \leq m}^q A \qquad \rho, q \mid \mathbf{M}, m \odot \Gamma, x : A \vdash_l e : B,$$

we must show that

$$\rho, \sigma \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_l [t/x]e : B.$$

We proceed by induction on the derivation of $\sigma \mid \mathbf{N} \odot \Delta \vdash_n \downarrow_{n \leq m}^q t : \downarrow_{n \leq m}^q A$. We must analyse the cases DROP.I and WEAK, SUB, CONT.

SUBCASE (DROP.I). In this case there exists a grade vector σ' with $q\sigma' = \sigma$ such that the derivation is

$$\frac{\sigma' \mid \mathbf{N} \odot \Delta \vdash_m t : A}{q\sigma' \mid \mathbf{N} \odot \Delta \vdash_n \downarrow_{n \leq m}^q t : \downarrow_{n \leq m}^q A}$$

Apply substitution with the judgment $\sigma' \mid \mathbf{N} \odot \Delta \vdash_m t : A$ to obtain the desired term.

SUBCASE (WEAK). In this case the derivation is

$$\frac{\text{Weak}(n') \quad n' \geq n \quad T \in \text{Type}(n') \quad \sigma \mid \mathbf{N} \odot \Delta \vdash_n \downarrow_{n \leq m}^q t : \downarrow_{n \leq m}^q A}{\sigma, 0 \mid \mathbf{N}, n' \odot \Delta, y : T \vdash_n \downarrow_{n \leq m}^q t : \downarrow_{n \leq m}^q A}$$

The inductive hypothesis is

$$\rho, \sigma \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_l [t/x]e : B.$$

and since we have $n' \geq n \geq l$ we may apply weakening to obtain

$$\rho, \sigma, 0 \mid \mathbf{M}, \mathbf{N}, n' \odot \Gamma, \Delta, y : T \vdash_l [t/x]e : B.$$

SUBCASE (SUB). In this case the derivation is

$$\frac{\sigma' \leq \sigma \quad \sigma' \mid \mathbf{N} \odot \Delta \vdash_n \downarrow_{n \leq m}^q t : \downarrow_{n \leq m}^q A}{\sigma \mid \mathbf{N} \odot \Delta \vdash_n \downarrow_{n \leq m}^q t : \downarrow_{n \leq m}^q A}$$

From $\sigma' \leq \sigma$ we obtain $\rho, \sigma' \leq \rho, \sigma$ and hence we may apply the rule SUB to the inductive hypothesis

$$\rho, \sigma' \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_l [t/x]e : B.$$

to obtain the desired term.

SUBCASE (CONT). In this case the derivation is

$$\frac{r_1, r_2 \in \text{Cont}(n') \quad \sigma, r_1, r_2 \mid \mathbf{N}, n', n' \odot \Delta, y_1 : T, y_2 : T \vdash_n \downarrow_{n \leq m}^q t : \downarrow_{n \leq m}^q A}{\sigma, r_1 + r_2 \mid \mathbf{N}, n' \odot \Delta, z : T \vdash_n \downarrow_{n \leq m}^q [z/y_1, z/y_2]t : \downarrow_{n \leq m}^q A}$$

The inductive hypothesis is

$$\rho, \sigma, r_1, r_2 \mid \mathbf{M}, \mathbf{N}, n', n' \odot \Gamma, \Delta, y_1 : T, y_2 : T \vdash_l [t/x]e : B.$$

Applying CONT we obtain

$$\rho, \sigma, r_1 + r_2 \mid \mathbf{M}, \mathbf{N}, n' \odot \Gamma, \Delta, z : T \vdash_l [z/y_1, z/y_2][t/x]e : B.$$

Since the variables y_1, y_2 are not free in e , this is equal to the term $[[z/y_1, z/y_2]t/x]e$, which is the desired term.

Case 6 (SUM.β.L) Given

$$q \geq 1 \quad \rho, q \mid \mathbf{M}, \mathbf{n} \odot \Gamma, x_1 : A_1 \vdash_{\mathbf{m}} e_1 : A \quad \sigma \mid \mathbf{N} \odot \Delta \vdash_{\mathbf{n}} \text{inl } t : A_1 \oplus A_2$$

we must show that

$$\rho, q\sigma \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_{\mathbf{m}} [t/x_1]e_1 : A$$

We proceed by induction on the derivation of $\sigma \mid \mathbf{N} \odot \Delta \vdash_{\mathbf{n}} \text{inl } t : A_1 \oplus A_2$. We must analyse the cases SUM.IL and WEAK, SUB, CONT.

SUBCASE (SUM.IL). In this case the derivation is

$$\frac{\sigma \mid \mathbf{N} \odot \Delta \vdash_{\mathbf{n}} t : A_1}{\sigma \mid \mathbf{N} \odot \Delta \vdash_{\mathbf{n}} \text{inl } t : A_1 \oplus A_2}$$

Apply substitution to obtain the desired result.

SUBCASE (WEAK). In this case the derivation is

$$\frac{\text{Weak}(\mathbf{n}') \quad \mathbf{n}' \geq \mathbf{n} \quad T \in \text{Type}(\mathbf{n}') \quad \sigma \mid \mathbf{N} \odot \Delta \vdash_{\mathbf{n}} \text{inl } t : A_1 \oplus A_2}{\sigma, 0 \mid \mathbf{N}, \mathbf{n}' \odot \Delta, y : T \vdash_{\mathbf{n}} \text{inl } t : A_1 \oplus A_2}$$

We have $\mathbf{n}' \geq \mathbf{n} \geq \mathbf{m}$ and hence may apply the rule WEAK to the inductive hypothesis, obtaining

$$\frac{\rho, q\sigma \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_{\mathbf{m}} [t/x_1]e_1 : A}{\rho, q\sigma, 0 \mid \mathbf{M}, \mathbf{N}, \mathbf{n}' \odot \Gamma, \Delta, y : T \vdash_{\mathbf{m}} [t/x_1]e_1 : A}$$

as desired.

SUBCASE (SUB). In this case the derivation is

$$\frac{\sigma' \leq \sigma \quad \sigma' \mid \mathbf{N} \odot \Delta \vdash_{\mathbf{n}} \text{inl } t : A_1 \oplus A_2}{\sigma \mid \mathbf{N} \odot \Delta \vdash_{\mathbf{n}} \text{inl } t : A_1 \oplus A_2}$$

By monotonicity of multiplication we have $\rho, q\sigma' \leq \rho, q\sigma$ and hence we may apply the rule SUB to the inductive hypothesis

$$\rho, q\sigma' \mid \mathbf{M}, \mathbf{N} \odot \Gamma, \Delta \vdash_{\mathbf{m}} [t/x_1]e_1 : A$$

obtaining the desired result.

SUBCASE (CONT). In this case the derivation is

$$\frac{r_1, r_2 \in \text{Cont}(\mathbf{n}') \quad \sigma, r_1, r_2 \mid \mathbf{N}, \mathbf{n}', \mathbf{n}' \odot \Delta, y_1 : T, y_2 : T \vdash_{\mathbf{n}} \text{inl } t : A_1 \oplus A_2}{\sigma, r_1 + r_2 \mid \mathbf{N}, \mathbf{n}' \odot \Delta, z : T \vdash_{\mathbf{n}} \text{inl } [z/y_1, z/y_2]t : A_1 \oplus A_2}$$

The inductive hypothesis is

$$\rho, q(\sigma, r_1, r_2) \mid \mathbf{M}, \mathbf{N}, \mathbf{n}', \mathbf{n}' \odot \Gamma, \Delta, y_1 : T, y_2 : T \vdash_{\mathbf{m}} [t/x_1]e_1 : A.$$

Since $\text{Cont}(\mathbf{n}')$ is an ideal, we have $qr_1, qr_2 \in \text{Cont}(\mathbf{n}')$ and thus may apply the rule CONT to obtain

$$\rho, q(\sigma, r_1 + r_2) \mid \mathbf{M}, \mathbf{N}, \mathbf{n}' \odot \Gamma, \Delta, z : T \vdash_{\mathbf{m}} [z/y_1, z/y_2][t/x_1]e_1 : A.$$

Since the variables y_1, y_2 are not free in e_1 , this term is equal to $[[z/y_1, z/y_2]t/x_1]e_1$, concluding the proof.

Case 7 (SUM.β.R) Analogous to the case SUM.β.L.

□

B.3 Syntactic soundness eta-conversions

We define η -conversion as a set of typed equalities as follows:

$$\begin{array}{c}
\text{UNIT.}\eta \quad \frac{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t : \mathbf{I}_{\mathbf{m}}}{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t \equiv_{\eta} \text{let}_{@1} \star_{\mathbf{m}} = t \text{ in } \star_{\mathbf{m}} : \mathbf{I}_{\mathbf{m}}} \\
\text{FUN.}\eta \quad \frac{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{n}} t : A^{q:\mathbf{m}} \multimap B}{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t \equiv_{\eta} \lambda x.(tx) : A^{q:\mathbf{m}} \multimap B} \\
\text{RAISE.}\eta \quad \frac{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t : \uparrow_{n \leq \mathbf{m}} A}{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t \equiv_{\eta} \uparrow_{n \leq \mathbf{m}}(\uparrow_{n \leq \mathbf{m}}^{-1} t) : \uparrow_{n \leq \mathbf{m}} A} \\
\text{PAIR.}\eta \quad \frac{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t : A_1 \otimes A_2}{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t \equiv_{\eta} \text{let}_{@1}(x_1, x_2) = t \text{ in } (x_1, x_2) : A_1 \otimes A_2} \\
\text{SUM.}\eta \quad \frac{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t : A_1 \oplus A_2}{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t \equiv_{\eta} \text{case}_1(t; x_1.\text{inl } x_1; x_2.\text{inr } x_2) : A_1 \oplus A_2} \\
\text{DROP.}\eta \quad \frac{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{n}} t : \downarrow_{n \leq \mathbf{m}}^q A}{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{n}} t \equiv_{\eta} \text{let}_{@q} \downarrow_{n \leq \mathbf{m}} x = t \text{ in } \downarrow_{n \leq \mathbf{m}}^q x : \downarrow_{n \leq \mathbf{m}}^q A}
\end{array}$$

Theorem B.5 *η -conversion preserves types and grades: Whenever $\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t_1 \equiv_{\eta} t_2 : A$ then $\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t_1 : A$ and $\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t_2 : A$.*

Proof. The assertion that $\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t_1 : A$ is an assumption in each rule for η -expansion.

For the assertion that $\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t_2 : A$ we have the following derivations:

$$\begin{array}{c}
\text{UNIT.I} \quad \frac{}{\emptyset \mid \emptyset \odot \emptyset \vdash_{\mathbf{m}} \star_{\mathbf{m}} : \mathbf{I}_{\mathbf{m}}} \quad \text{UNIT.E} \quad \frac{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t : \mathbf{I}_{\mathbf{m}}}{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} \text{let}_{@1} \star_{\mathbf{m}} = t \text{ in } \star_{\mathbf{m}} : \mathbf{I}_{\mathbf{m}}} \\
\text{PAIR.I} \quad \frac{\text{VAR} \quad \frac{}{1 \mid \mathbf{m} \odot x_1 : A_1 \vdash_{\mathbf{m}} x_1 : A_1} \quad \text{VAR} \quad \frac{}{1 \mid \mathbf{m} \odot x_2 : A_2 \vdash_{\mathbf{m}} x_2 : A_2}}{1, 1 \mid \mathbf{m}, \mathbf{m} \odot x_1 : A_1, x_2 : A_2 \vdash_{\mathbf{m}} (x_1, x_2) : A_1 \otimes A_2} \quad \text{PAIR.E} \quad \frac{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t : A_1 \otimes A_2}{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} \text{let}_{@1}(x_1, x_2) = t \text{ in } (x_1, x_2) : A_1 \otimes A_2} \\
\text{FUN.E} \quad \frac{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{n}} t : A^{q:\mathbf{m}} \multimap B \quad \text{VAR} \quad \frac{}{1 \mid \mathbf{m} \odot x : A \vdash_{\mathbf{m}} x : A}}{\rho, q \mid \mathbf{M}, \mathbf{m} \odot \Gamma, x : A \vdash_{\mathbf{n}} tx : B} \quad \text{FUN.I} \quad \frac{}{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{n}} \lambda x.(tx) : A^{q:\mathbf{m}} \multimap B} \\
\text{RAISE.E} \quad \frac{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t : \uparrow_{n \leq \mathbf{m}} A}{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{n}} \uparrow_{n \leq \mathbf{m}}^{-1} t : A} \quad \text{RAISE.I} \quad \frac{}{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} \uparrow_{n \leq \mathbf{m}}(\uparrow_{n \leq \mathbf{m}}^{-1} t) : \uparrow_{n \leq \mathbf{m}} A} \\
\text{DROP.I} \quad \frac{\text{VAR} \quad \frac{}{1 \mid \mathbf{m} \odot z : A \vdash_{\mathbf{m}} z : A}}{q \mid \mathbf{m} \odot z : A \vdash_{\mathbf{n}} \downarrow_{n \leq \mathbf{m}}^q z : \downarrow_{n \leq \mathbf{m}}^q A} \quad \text{DROP.E} \quad \frac{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{n}} t : \downarrow_{n \leq \mathbf{m}}^q A}{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{n}} \text{let}_{@q} \downarrow_{n \leq \mathbf{m}} z = t \text{ in } \downarrow_{n \leq \mathbf{m}}^q z : \downarrow_{n \leq \mathbf{m}}^q A} \\
\text{SUM.II} \quad \frac{\text{VAR} \quad \frac{}{1 \mid \mathbf{m} \odot x_1 : A_1 \vdash_{\mathbf{m}} x_1 : A_1}}{1 \mid \mathbf{m} \odot x_1 : A_1 \vdash_{\mathbf{m}} \text{inl } x_1 : A_1 \oplus A_2} \quad \text{SUM.IR} \quad \frac{\text{VAR} \quad \frac{}{1 \mid \mathbf{m} \odot x_2 : A_2 \vdash_{\mathbf{m}} x_2 : A_2}}{1 \mid \mathbf{m} \odot x_2 : A_2 \vdash_{\mathbf{m}} \text{inr } x_1 : A_1 \oplus A_2} \quad \text{SUM.E} \quad \frac{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t : A_1 \oplus A_2}{\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t \equiv_{\eta} \text{case}_1(t; x_1.\text{inl } x_1; x_2.\text{inr } x_2) : A_1 \oplus A_2}
\end{array}$$

□

C Semantic soundness of the equational theory

C.1 Substitution is composition

Lemma C.1 *In the situation of Theorem B.2, suppose the terms t and $e_1, \dots, e_k$ have categorical semantics*

$$\llbracket \rho \mid M \odot \Gamma \rrbracket_m = \bigotimes_i F_m^{m_i}(r_i \odot A_i) \xrightarrow{t} B \quad \llbracket \sigma_i \mid N_i \odot \Delta_i \rrbracket_{m_i} = \bigotimes_j F_{m_i}^{n_{ij}}(q_{ij} \odot A_{ij}) \xrightarrow{e_i} A_i \quad (i \in \{1, \dots, k\})$$

The interpretation of the substitution $r_1\sigma_1, \dots, r_k\sigma_k \mid N_1, \dots, N_k \odot \Delta_1, \dots, \Delta_k \vdash [e_i/x_i]t : B$ is given by the morphism

$$\bigotimes_i \llbracket r_i\sigma_i \mid N_i \odot \Delta_i \rrbracket_m \cong \bigotimes_i F_m^{m_i} \llbracket r_i\sigma_i \mid N_i \odot \Delta_i \rrbracket_{m_i} \xrightarrow{\bigotimes_i F_m^{m_i}(r_i \cdot e_i)} \llbracket \rho \mid M \odot \Gamma \rrbracket_m \xrightarrow{t} B$$

or, equivalently:

$$\bigotimes_i \bigotimes_j F_m^{n_{ij}}(r_i q_{ij} \odot A_{ij}) \cong \bigotimes_i F_m^{m_i} \bigotimes_j F_{m_i}^{n_{ij}}(r_i q_{ij} \odot A_{ij}) \xrightarrow{\bigotimes_i F_m^{m_i}(r_i \cdot e_i)} \bigotimes_i F_m^{m_i}(r_i \odot A_i) \xrightarrow{t} B$$

Before the proof, we begin with a lemma.

Lemma C.2 *In the situation of the above lemma, suppose that $s \in R_m$ and write f for the composite*

$$\bigotimes_{ij} F_m^{n_{ij}}(r_i q_{ij} \odot A_{ij}) \cong \bigotimes_i F_m^{m_i} \bigotimes_j F_{m_i}^{n_{ij}}(r_i q_{ij} \odot A_{ij}) \xrightarrow{\bigotimes_i F_m^{m_i}(r_i \cdot e_i)} \bigotimes_i F_m^{m_i}(r_i \odot A_i) \xrightarrow{t} B.$$

The morphism $s \cdot f$ is equal to the composite

$$\bigotimes_{ij} F_m^{n_{ij}}(sr_i q_{ij} \odot A_{ij}) \cong \bigotimes_i F_m^{m_i} \bigotimes_j F_{m_i}^{n_{ij}}(sr_i q_{ij} \odot A_{ij}) \xrightarrow{\bigotimes_i F_m^{m_i}(sr_i \cdot e_i)} \bigotimes_i F_m^{m_i}(sr_i \odot A_i) \xrightarrow{(s \cdot t)} (s \odot B)$$

Proof. Consider the following diagram in which unlabelled arrows are assembled from μ , δ and τ , to “move scalars out of the tensor product,” and the vertical arrows labeled “ $\sim$ ” are induced by the isomorphisms $F_m^{n_{ij}} \cong F_m^{m_i} \circ F_{m_i}^{n_{ij}}$ and the strong monoidal structures on the functors $F_m^{m_i}$.

$$\begin{array}{ccc} \bigotimes_{ij} F_m^{n_{ij}}(sr_i q_{ij} \odot A_{ij}) & \longrightarrow & s \odot \bigotimes_{ij} F_m^{n_{ij}}(r_i q_{ij} \odot A_{ij}) \\ \sim \downarrow & & \downarrow \sim \\ \bigotimes_i F_m^{m_i} \bigotimes_j F_{m_i}^{n_{ij}}(sr_i q_{ij} \odot A_{ij}) & \longrightarrow & s \odot \bigotimes_i F_m^{m_i} \bigotimes_j F_{m_i}^{n_{ij}}(r_i q_{ij} \odot A_{ij}) \\ \downarrow & & \downarrow \\ \bigotimes_i F_m^{m_i}(sr_i \odot \bigotimes_j F_{m_i}^{n_{ij}}(q_{ij} \odot A_{ij})) & \longrightarrow & s \odot \bigotimes_i F_m^{m_i}(r_i \odot \bigotimes_j F_{m_i}^{n_{ij}}(q_{ij} \odot A_{ij})) \\ \bigotimes_i F_m^{m_i}(sr_i \odot e_i) \downarrow & & \downarrow s \odot \bigotimes_i F_m^{m_i}(r_i \odot e_i) \\ \bigotimes_i F_m^{m_i}(sr_i \odot A_i) & \longrightarrow & s \odot \bigotimes_i F_m^{m_i}(r_i \odot A_i) \\ & & \downarrow s \odot t \\ & & s \odot B \end{array}$$

The composite of the top and right edge is $s \cdot f$, while the composite of the left and bottom edge is the morphism from the claim. The bottom square commutes by naturality of μ , δ and τ . The middle square

commutes by the coherence conditions on μ ; in fact those conditions imply that all morphisms assembled from δ , μ and τ beginning at the top left of that square at the bottom right of that square are equal. The top square commutes by the naturality of the involved transformations, using that μ is a monoidal natural transformation. Thus the diagram commutes, concluding the proof. $\square$

Proof. [Of Lemma C.1] The proof of Theorem B.2 gives a constructive transformation of the derivations of t and the e_i into a derivation of $[e_i/x_i]t$. We proceed by induction on the derivation of t , inspecting the proof of Theorem B.2.

Case 1 (VAR) In this case we must verify that any morphism

$$\llbracket \rho \mid M \odot \Gamma \rrbracket_m \xrightarrow{t} B$$

is equal to

$$F_m^m \llbracket 1\rho \mid M \odot \Gamma \rrbracket_m \xrightarrow{F_m^m(1 \cdot t)} F_m^m(1 \odot B) \cong 1 \odot B \xrightarrow{\epsilon} B$$

which follows from the following commutative diagram

$$\begin{array}{ccccccc}
 \bigotimes_i F_m^{m_i}(1 \cdot r_i \odot A_i) & \xrightarrow{\bigotimes_i F_m^{m_i} \delta} & \bigotimes_i F_m^{m_i}(1 \odot r_i \odot A_i) & \xrightarrow{\bigotimes_i \mu} & \bigotimes_i 1 \odot F_m^{m_i}(r_i \odot A_i) & \xrightarrow{\tau} & 1 \odot \bigotimes_i F_m^{m_i}(r_i \odot A_i) \\
 \parallel & \swarrow \bigotimes_i F_m^{m_i} \epsilon & & \swarrow \bigotimes_i \epsilon & & \swarrow \epsilon & \downarrow 1 \odot e \\
 \bigotimes_i F_m^{m_i}(r_i \odot A_i) & \xlongequal{\quad} & \bigotimes_i F_m^{m_i}(r_i \odot A_i) & \xlongequal{\quad} & \bigotimes_i F_m^{m_i}(r_i \odot A_i) & & 1 \odot B \\
 & & & & \downarrow e & \swarrow \epsilon & \\
 & & & & B & &
 \end{array}$$

Case 2 (WEAK) Consider a term $\sigma \mid N \odot \Delta \vdash_n e : T$ with categorical interpretation

$$\llbracket \sigma \mid N \odot \Delta \rrbracket_n = \bigotimes_j F_n^{n_j}(s_j \odot Y_j) \xrightarrow{e} T$$

Consider the following morphism:

$$\begin{aligned}
 & \bigotimes_i F_m^{m_i} \llbracket \sigma_i \mid N_i \odot \Delta_i \rrbracket_{m_i} \otimes F_m^n \left(\bigotimes_j F_n^{n_j}(0s_j \odot Y_j) \right) \\
 & \xrightarrow{\bigotimes_i F_m^{m_i}(r_i \cdot e_i) \otimes F_m^n(0 \cdot e)} \llbracket \rho \mid M \odot \Gamma \rrbracket_m \otimes F_m^n(0 \odot T) \xrightarrow{t \otimes F_m^n w_T} B \otimes F_m^n(I) \cong B
 \end{aligned}$$

This morphism is the morphism we claim realizes weakening by a variable of type T and then substituting the term e for that variable. On the other hand, the proof of Theorem B.2, constructs this term as a sequence of weakenings, which has the following interpretation:

$$\begin{aligned}
 & \bigotimes_i F_m^{m_i} \llbracket \sigma_i \mid N_i \odot \Delta_i \rrbracket_{m_i} \otimes \bigotimes_j F_m^{n_j}(0 \cdot s_j \odot Y_j) \\
 & \xrightarrow{\bigotimes_i F_m^{m_i}(r_i \cdot e_i) \otimes \bigotimes_j F_m^{n_j}(w_{Y_j})} \llbracket \rho \mid M \odot \Gamma \rrbracket_m \otimes \bigotimes_j F_m^{n_j} I \xrightarrow{t \otimes \bigotimes_j u^{-1}} B \otimes \bigotimes_j I \cong B
 \end{aligned}$$

Our task is therefore to prove that the two composites above are equal. This boils down to showing the commutation of the diagram below, in which the composition across the left edge is the definition of $0 \cdot e$. The top square commutes by the coherence conditions on exponential actions, the second square due to the coherence conditions on μ , the third square due to the fact that w is a monoidal natural transformation and the last square by naturality of w .

$$\begin{array}{ccc}
 \bigotimes_j F_n^{nj}(0s_j \odot Y_j) & \xrightarrow{\bigotimes_j F_n^{nj}(w)} & \bigotimes_j I \\
 \bigotimes_j F_n^{nj}(\delta) \downarrow & & \parallel \\
 \bigotimes_j F_n^{nj}(0 \odot s_j \odot Y_j) & \xrightarrow{\bigotimes_j F_n^{nj}(w)} & \bigotimes_j I \\
 \bigotimes_j \mu \downarrow & & \parallel \\
 \bigotimes_j 0 \odot F_n^{nj}(s_j \odot Y_j) & \xrightarrow{\bigotimes_j w} & \bigotimes_j I \\
 \tau \downarrow & & \sim \downarrow \\
 0 \odot \bigotimes_j F_n^{nj}(s_j \odot Y_j) & \xrightarrow{w} & I \\
 0 \odot e \downarrow & & \parallel \\
 0 \odot T & \xrightarrow{w} & I
 \end{array}$$

Case 3 (SUB) Suppose we have $\rho' = r'_1, \dots, r'_k \geq r_1, \dots, r_k = \rho$. Consider the diagram

$$\begin{array}{ccc}
 \bigotimes_{i,j} F_m^{nij}(r_i q_{ij} \odot A_{ij}) & \longleftarrow & \bigotimes_{i,j} F_m^{nij}(r'_i q_{ij} \odot A_{ij}) \\
 \sim \downarrow & & \downarrow \sim \\
 \bigotimes_i F_m^{mi} \bigotimes_j F_m^{nij}(r_i q_{ij} \odot A_{ij}) & \longleftarrow & \bigotimes_i F_m^{mi} \bigotimes_j F_m^{nij}(r'_i q_{ij} \odot A_{ij}) \\
 \bigotimes_i F_m^{mi}(\tau \odot \bigotimes_j (\mu \odot F_m^{nij}(\delta))) \downarrow & & \downarrow \bigotimes_i F_m^{mi}(\tau \odot \bigotimes_j (\mu \odot F_m^{nij}(\delta))) \\
 \bigotimes_i F_m^{mi}(r_i \odot \bigotimes_j F_m^{nij}(q_{ij} \odot A_{ij})) & \longleftarrow & \bigotimes_i F_m^{mi}(r'_i \odot \bigotimes_j F_m^{nij}(q_{ij} \odot A_{ij})) \\
 \bigotimes_i F_m^{mi}(r_i \odot e_i) \downarrow & & \downarrow \bigotimes_i F_m^{mi}(r'_i \odot e_i) \\
 \bigotimes_i F_m^{mi}(r_i \odot A_i) & \longleftarrow & \bigotimes_i F_m^{mi}(r'_i \odot A_i) \\
 t \downarrow & \nearrow t & \\
 B & &
 \end{array}$$

Where all horizontal arrows are induced by $\rho' \geq \rho$. By the inductive hypothesis, the composite across the left edge is the interpretation of the term $[e_i/x_i]t$. Since the interpretation of the SUB rule is given by precomposing with the action of $\rho' \geq \rho$, the composite of the left edge with the topmost horizontal morphism is the interpretation of applying SUB to the the interpretation of $[e_i/x_i]t$. The composite across the right edge of the diagram is the morphism from our claim, so we must show that this diagram commutes. The top two squares commute by the naturality of the involved vertical morphisms, the third square commutes by the functoriality of the F_m^{mi} . The bottom triangle commutes by definition of the interpretation of the SUB rule.

Case 4 (CONT) Suppose the derivation is

$$\frac{\rho, q_1, q_2 \mid \mathbf{M}, \mathbf{n}, \mathbf{n} \odot \Gamma, y_1 : T, y_2 : T \vdash_{\mathbf{m}} t : B}{\rho, q_1 + q_2 \mid \mathbf{M}, \mathbf{n}, \mathbf{n} \odot \Gamma, z : T \vdash_{\mathbf{m}} [z/y_1, z/y_2]t : B}$$

and we are given a term $\sigma \mid \mathbf{N} \odot \Delta \vdash_{\mathbf{n}} e : T$. Consider the diagram

$$\begin{array}{ccc}
 \begin{array}{c} \bigotimes_i F_m^{m_i} \llbracket r_i \sigma_i \mid N_i \odot \Delta_i \rrbracket_{m_i} \\ \otimes F_m^n \llbracket q_1 \sigma \mid N \odot \Delta \rrbracket_n \otimes F_m^n \llbracket q_2 \sigma \mid N \odot \Delta \rrbracket_n \end{array} & \xleftarrow{\quad \quad \quad} & \begin{array}{c} \bigotimes_i F_m^{m_i} \llbracket r_i \sigma_i \mid N_i \odot \Delta_i \rrbracket_{m_i} \otimes F_m^n \llbracket (q_1 + q_2) \sigma \mid N \odot \Delta \rrbracket_n \\ \downarrow \bigotimes_i F_m^{m_i} (r_i \cdot e_i) \otimes F_m^n ((q_1 + q_2) \cdot e) \end{array} \\
 \downarrow \bigotimes_i F_m^{m_i} (r_i \cdot e_i) \otimes F_m^n (q_1 \cdot e) \otimes F_m^n (q_2 \cdot e) & & \downarrow \bigotimes_i F_m^{m_i} (r_i \odot A_i) \otimes F_m^n ((q_1 + q_2) \odot T) \\
 \begin{array}{c} \bigotimes_i F_m^{m_i} (r_i \odot A_i) \otimes F_m^n (q_1 \odot T) \otimes F_m^n (q_2 \odot T) \\ \downarrow t \\ B \end{array} & \xleftarrow{\text{id} \otimes m^{-1}} & \begin{array}{c} \bigotimes_i F_m^{m_i} (r_i \odot A_i) \otimes F_m^n (q_1 \odot T \otimes q_2 \odot T) \\ \downarrow \text{id} \otimes F_m^n (c) \\ \bigotimes_i F_m^{m_i} (r_i \odot A_i) \otimes F_m^n (q_1 \odot T \otimes q_2 \odot T) \end{array} \\
 & \searrow \text{to}(\text{id} \otimes m^{-1}) & \\
 & B &
 \end{array}$$

By the inductive hypothesis, the composite across the left edge is the interpretation of the term $[e_i/x_i, e/y_1, e/y_2]t$, while the composition across the right edge is the morphism we claim to be the interpretation of the term $[e_i/x_i, e/z][z/y_1, z/y_2]t$. The proof of Theorem B.2 gives the interpretation of this term as the composite of the left edge with the dashed arrow, where the dashed arrow is a sequence of exchanges followed by a sequence of contractions. So, we need to show that the above diagram commutes. It suffices to show that the following diagram commutes:

$$\begin{array}{ccc}
 \llbracket q_1 \sigma \mid N \odot \Delta \rrbracket_n \otimes \llbracket q_2 \sigma \mid N \odot \Delta \rrbracket_n & \xleftarrow{\quad \quad \quad} & \llbracket (q_1 + q_2) \sigma \mid N \odot \Delta \rrbracket_n \\
 \downarrow q_1 \cdot e \otimes q_2 \cdot e & & \downarrow (q_1 + q_2) \cdot e \\
 (q_1 \odot T) \otimes (q_2 \odot T) & \xleftarrow{c} & (q_1 + q_2) \odot T
 \end{array}$$

Suppose that $\llbracket (q_1 + q_2) \sigma \mid N \odot \Delta \rrbracket_n$ is the object $\bigotimes_i F_n^{n_i}((q_1 + q_2)s_i \odot X_i)$ and consider the diagram in Figure C.1. Its left vertical edge is the dashed arrow, and the bottom edge is $q_1 \cdot e \otimes q_2 \cdot e$. The composite across the top/right edges is $c \circ (q_1 + q_2)$. On that diagram's left side, the top square commutes by the coherence conditions on graded actions, the other two squares commute by the naturality of the vertical morphisms, and the triangle commutes by definition. On the right side, the top area commutes by Lemma 4.9. The middle area commutes since c is a monoidal natural transformation, and the bottom area commutes because c is natural.

Case 5 (DROP.I) Suppose we have $n \leq m$ and $s \in R_m$. Starting at the morphism

$$\bigotimes_{ij} F_m^{n_{ij}}(r_i q_{ij} \odot A_{ij}) \cong \bigotimes_i F_m^{m_i} \bigotimes_j F_m^{n_{ij}}(r_i q_{ij} \odot A_{ij}) \xrightarrow{\bigotimes_i F_m^{m_i}(r_i \cdot e_i)} \bigotimes_i F_m^{m_i}(r_i \odot A_i) \xrightarrow{t} B$$

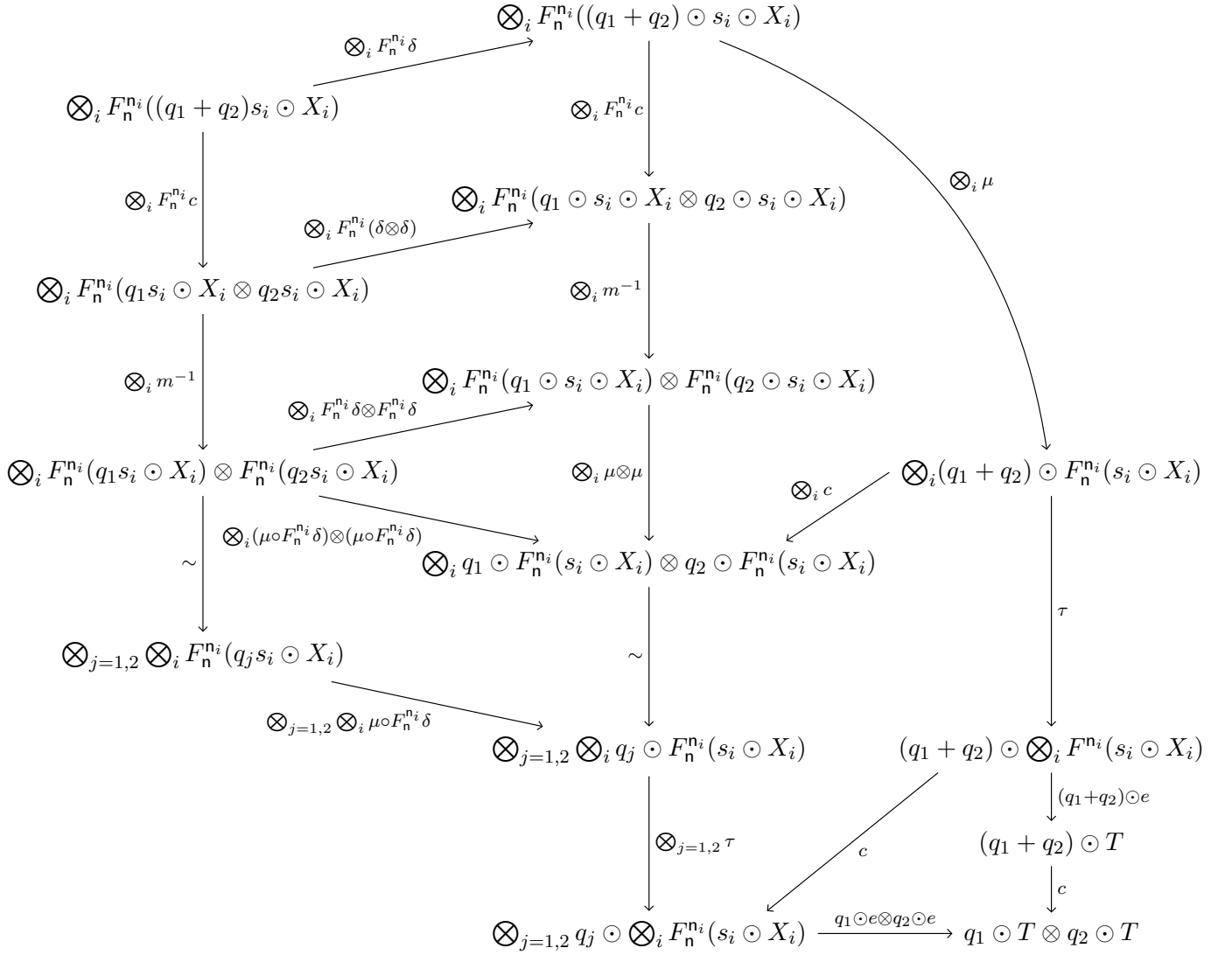
which we will denote by f , and applying DROP.I to it, we obtain the morphism

$$F_n^m \bigotimes_{ij} F_m^{n_{ij}}(sr_i q_{ij} \odot A_{ij}) \xrightarrow{F_n^m(s \cdot f)} F_n^m(s \odot B)$$

We claim that this morphism is equal to

$$\begin{aligned}
 F_n^m \bigotimes_{ij} F_m^{n_{ij}}(sr_i q_{ij} \odot A_{ij}) &\cong F_n^m \bigotimes_i F_m^{m_i} \bigotimes_j F_m^{n_{ij}}(sr_i q_{ij} \odot A_{ij}) \\
 &\xrightarrow{F_n^m \bigotimes_i F_m^{m_i}(sr_i \cdot e_i)} F_n^m \bigotimes_i F_m^{m_i}(sr_i \odot A_i) \xrightarrow{F_n^m(s \cdot t)} F_n^m(s \odot B)
 \end{aligned}$$

which is immediate from Lemma C.2.


 Fig. C.1. Diagram for case CONT. The arrows labeled “ $\sim$ ” are isomorphisms permuting the tensor product.

Case 6 (DROP.E) Let $l \leq n \leq m$. By the inductive hypothesis we start with morphisms

$$\bigotimes_i \llbracket r_i \sigma_i \mid N_i \odot \Delta_i \rrbracket_n \cong \bigotimes_i F_n^{m_i} \llbracket r_i \sigma_i \mid N_i \odot \Delta_i \rrbracket_{m_i} \xrightarrow{\bigotimes_i F_n^{m_i}(r_i \cdot e_i)} \bigotimes_i F_n^{m_i}(r_i \odot A_i) \xrightarrow{t} F_n^m(q \odot A)$$

and

$$\bigotimes_i \llbracket r'_i \sigma'_i \mid N'_i \odot \Delta'_i \rrbracket_l \otimes F_l^m(q \odot A) \cong \bigotimes_i F_l^{m'_i} \llbracket r'_i \sigma'_i \mid N'_i \odot \Delta'_i \rrbracket_{m'_i} \otimes F_l^m(q \odot A)$$

$$\xrightarrow{\bigotimes_i F_l^{m'_i}(r'_i \cdot e'_i) \otimes \text{id}} \bigotimes_i F_l^{m'_i}(r'_i \odot A'_i) \otimes F_l^m(q \odot A) \xrightarrow{t'} B$$

Applying the rule DROP.E to these two morphisms produces the morphism

$$\begin{aligned}
 & \bigotimes_i F_l^{m'_i} \llbracket r'_i \sigma'_i \mid \mathbf{N}'_i \odot \Delta'_i \rrbracket_{\mathbf{m}'_i} \otimes \bigotimes_i F_l^{m_i} \llbracket r_i \sigma_i \mid \mathbf{N}_i \odot \Delta_i \rrbracket_{\mathbf{m}_i} \\
 & \cong \bigotimes_i F_l^{m'_i} \llbracket r'_i \sigma'_i \mid \mathbf{N}'_i \odot \Delta'_i \rrbracket_{\mathbf{m}'_i} \otimes F_l^n \bigotimes_i F_n^{m_i} \llbracket r_i \sigma_i \mid \mathbf{N}_i \odot \Delta_i \rrbracket_{\mathbf{m}_i} \\
 & \xrightarrow{\text{id} \otimes F_l^n (t \odot \bigotimes_i F_n^{m_i} (r_i \cdot e_i))} \bigotimes_i F_l^{m'_i} \llbracket r'_i \sigma'_i \mid \mathbf{N}'_i \odot \Delta'_i \rrbracket_{\mathbf{m}'_i} \otimes F_l^n F_n^m (q \odot A) \\
 & \xrightarrow{\bigotimes_i F_l^{m'_i} (r'_i \cdot e'_i) \otimes \text{id}} \bigotimes_i F_l^{m'_i} (r'_i \odot A'_i) \otimes F_l^n F_n^m (q \odot A) \\
 & \cong \bigotimes_i F_l^{m'_i} (r'_i \odot A'_i) \otimes F_l^m (q \odot A) \xrightarrow{t'} B
 \end{aligned}$$

and we claim that this morphism is equal to

$$\begin{aligned}
 & \bigotimes_i F_l^{m'_i} \llbracket r'_i \sigma'_i \mid \mathbf{N}'_i \odot \Delta'_i \rrbracket_{\mathbf{m}'_i} \otimes \bigotimes_i F_l^{m_i} \llbracket r_i \sigma_i \mid \mathbf{N}_i \odot \Delta_i \rrbracket_{\mathbf{m}_i} \\
 & \xrightarrow{\bigotimes_i F_l^{m'_i} (r'_i \cdot e'_i) \otimes \bigotimes_i F_l^{m_i} (r_i \cdot e_i)} \bigotimes_i F_l^{m'_i} (r'_i \odot A'_i) \otimes \bigotimes_i F_l^{m_i} (r_i \odot A_i) \\
 & \cong \bigotimes_i F_l^{m'_i} (r'_i \odot A'_i) \otimes F_l^n \bigotimes_i F_n^{m_i} (r_i \odot A_i) \\
 & \xrightarrow{\text{id} \otimes F_l^n (t)} \bigotimes_i F_l^{m'_i} (r'_i \odot A'_i) \otimes F_l^n F_n^m (q \odot A) \\
 & \cong \bigotimes_i F_l^{m'_i} (r'_i \odot A'_i) \otimes F_l^m (q \odot A) \xrightarrow{t'} B
 \end{aligned}$$

These morphisms are readily verified to be equal.

Case 7 (RAISE.I) Let $m \leq n$. By the inductive hypothesis, we begin with a morphism

$$\bigotimes_i F_m^{m_i} \llbracket r_i \sigma_i \mid \mathbf{N}_i \odot \Delta_i \rrbracket_{\mathbf{m}_i} \xrightarrow{\bigotimes_i F_m^{m_i} (r_i \cdot e_i)} \bigotimes_i F_m^{m_i} (r_i \odot A_i) \xrightarrow{t} B$$

applying the rule RAISE.I to it yields the morphism

$$\begin{aligned}
 & \bigotimes_i F_n^{m_i} \llbracket r_i \sigma_i \mid \mathbf{N}_i \odot \Delta_i \rrbracket_{\mathbf{m}_i} \\
 & \xrightarrow{\text{unit}} G_m^n F_m^n \bigotimes_i F_n^{m_i} \llbracket r_i \sigma_i \mid \mathbf{N}_i \odot \Delta_i \rrbracket_{\mathbf{m}_i} \\
 & \cong G_m^n \bigotimes_i F_m^n F_n^{m_i} \llbracket r_i \sigma_i \mid \mathbf{N}_i \odot \Delta_i \rrbracket_{\mathbf{m}_i} \\
 & \cong G_m^n \bigotimes_i F_m^{m_i} \llbracket r_i \sigma_i \mid \mathbf{N}_i \odot \Delta_i \rrbracket_{\mathbf{m}_i} \\
 & \xrightarrow{G_m^n \bigotimes_i F_m^{m_i} (r_i \cdot e_i)} G_m^n \bigotimes_i F_m^{m_i} (r_i \odot A_i) \xrightarrow{G_m^n t} G_m^n B
 \end{aligned}$$

We claim that this morphism is equal to

$$\begin{aligned}
 & \bigotimes_i F_n^{m_i} \llbracket r_i \sigma_i \mid N_i \odot \Delta_i \rrbracket_{m_i} \\
 & \xrightarrow{\bigotimes_i F_n^{m_i}(r_i \cdot e_i)} \bigotimes_i F_n^{m_i}(r_i \odot A_i) \\
 & \xrightarrow{\text{unit}} G_m^n F_n^m \bigotimes_i F_n^{m_i}(r_i \odot A_i) \\
 & \cong G_m^n \bigotimes_i F_n^m F_n^{m_i}(r_i \odot A_i) \\
 & \cong G_m^n \bigotimes_i F_m^{m_i}(r_i \odot A_i) \xrightarrow{G_m^n t} G_m^n B
 \end{aligned}$$

The following diagram commutes, since the horizontal morphisms are natural transformations, concluding this case.

$$\begin{array}{ccccc}
 \bigotimes_i F_n^{m_i} \llbracket r_i \sigma_i \mid N_i \odot \Delta_i \rrbracket_{m_i} & \xrightarrow{\text{unit}} & G_m^n F_n^m \bigotimes_i F_n^{m_i} \llbracket r_i \sigma_i \mid N_i \odot \Delta_i \rrbracket_{m_i} & \xrightarrow{\sim} & G_m^n \bigotimes_i F_m^{m_i} \llbracket r_i \sigma_i \mid N_i \odot \Delta_i \rrbracket_{m_i} \\
 \downarrow \bigotimes_i F_n^{m_i}(r_i \cdot e_i) & & & & \downarrow G_m^n \bigotimes_i F_m^{m_i}(r_i \cdot e_i) \\
 \bigotimes_i F_n^{m_i}(r_i \odot A_i) & \xrightarrow{\text{unit}} & G_m^n F_n^m \bigotimes_i F_n^{m_i}(r_i \odot A_i) & \xrightarrow{\sim} & G_m^n \bigotimes_i F_m^{m_i}(r_i \odot A_i)
 \end{array}$$

Case 8 (RAISE.E) Let $n \leq m$. By the inductive hypothesis we begin with a morphism

$$\bigotimes_i \llbracket r_i \sigma_i \mid N_i \odot \Delta_i \rrbracket_m \cong \bigotimes_i F_m^{m_i} \llbracket r_i \sigma_i \mid N_i \odot \Delta_i \rrbracket_{m_i} \xrightarrow{\bigotimes_i F_m^{m_i}(r_i \cdot e_i)} \bigotimes_i F_m^{m_i}(r_i \odot A_i) \xrightarrow{t} G_m^n B$$

Applying the rule DROP.E to this morphism yields

$$\begin{aligned}
 \bigotimes_i \llbracket r_i \sigma_i \mid N_i \odot \Delta_i \rrbracket_n & \cong F_n^m \bigotimes_i \llbracket r_i \sigma_i \mid N_i \odot \Delta_i \rrbracket_m \cong F_n^m \bigotimes_i F_m^{m_i} \llbracket r_i \sigma_i \mid N_i \odot \Delta_i \rrbracket_{m_i} \\
 & \xrightarrow{F_n^m \bigotimes_i F_m^{m_i}(r_i \cdot e_i)} F_n^m \bigotimes_i F_m^{m_i}(r_i \odot A_i) \xrightarrow{F_n^m t} F_n^m G_n^m B \xrightarrow{\text{counit}} B
 \end{aligned}$$

We claim that this is equal to the morphism

$$\begin{aligned}
 \bigotimes_i \llbracket r_i \sigma_i \mid N_i \odot \Delta_i \rrbracket_n & \cong \bigotimes_i F_n^{m_i} \llbracket r_i \sigma_i \mid N_i \odot \Delta_i \rrbracket_{m_i} \\
 & \xrightarrow{\bigotimes_i F_n^{m_i}(r_i \cdot e_i)} \bigotimes_i F_n^{m_i}(r_i \odot A_i) \cong F_n^m \bigotimes_i F_m^{m_i}(r_i \odot A_i) \xrightarrow{F_n^m(t)} F_n^m G_n^m B \xrightarrow{\text{counit}} B
 \end{aligned}$$

So it is sufficient to see that the following diagram commutes

$$\begin{array}{ccccc}
 \bigotimes_i \llbracket r_i \sigma_i \mid N_i \odot \Delta_i \rrbracket_n & \xrightarrow{\sim} & \bigotimes_i F_n^m \llbracket r_i \sigma_i \mid N_i \odot \Delta_i \rrbracket_m & \xrightarrow{\sim} & F_n^m \bigotimes_i \llbracket r_i \sigma_i \mid N_i \odot \Delta_i \rrbracket_m \\
 \downarrow \sim & & \downarrow \sim & & \downarrow \sim \\
 \bigotimes_i F_n^{m_i} \llbracket r_i \sigma_i \mid N_i \odot \Delta_i \rrbracket_{m_i} & \xrightarrow{\sim} & \bigotimes_i F_n^m F_m^{m_i} \llbracket r_i \sigma_i \mid N_i \odot \Delta_i \rrbracket_{m_i} & \xrightarrow{\sim} & F_n^m \bigotimes_i F_m^{m_i} \llbracket r_i \sigma_i \mid N_i \odot \Delta_i \rrbracket_{m_i} \\
 \downarrow \bigotimes_i F_n^{m_i}(r_i \cdot e_i) & & & & \downarrow F_n^m \bigotimes_i F_m^{m_i}(r_i \cdot e_i) \\
 \bigotimes_i F_n^{m_i}(r_i \odot A_i) & \xrightarrow{\sim} & \bigotimes_i F_n^m F_m^{m_i}(r_i \odot A_i) & \xrightarrow{\sim} & F_n^m \bigotimes_i F_m^{m_i}(r_i \odot A_i)
 \end{array}$$

The bottom rectangle commutes by the naturality of the horizontal morphisms, while the top rectangles commute since the system of natural isomorphisms $F_{l_3}^{l_2} \circ F_{l_2}^{l_1} \cong F_{l_3}^{l_1}$ is coherent.

The remaining cases are straightforward using C.2. We demonstrate this using the case for function application as an example.

Case 9 (FUN.E) We begin with morphisms

$$t_1: \bigotimes_i \llbracket r_i \sigma_i \mid N_i \odot \Delta_i \rrbracket_m \cong \bigotimes_i F_m^{m_i} \llbracket r_i \sigma_i \mid N_i \odot \Delta_i \rrbracket_{m_i} \xrightarrow{\bigotimes_i F_m^{m_i}(r_i \cdot e_i)} \bigotimes_i F_m^{m_i}(r_i \odot A_i) \xrightarrow{t} B$$

$$t_2: \bigotimes_i \llbracket r'_i \sigma'_i \mid N'_i \odot \Delta'_i \rrbracket_n \cong \bigotimes_i F_n^{m'_i} \llbracket r'_i \sigma'_i \mid N'_i \odot \Delta'_i \rrbracket_{m'_i} \xrightarrow{\bigotimes_i F_n^{m'_i}(r'_i \cdot e'_i)} \bigotimes_i F_n^{m'_i}(r'_i \odot A'_i) \xrightarrow{t'} F_n^m(q \odot B) \multimap B'$$

applying FUN.E to them yields the morphism

$$\begin{aligned} & \bigotimes_i \llbracket r'_i \sigma'_i \mid N'_i \odot \Delta'_i \rrbracket_n \otimes \bigotimes_i \llbracket q r_i \sigma_i \mid N_i \odot \Delta_i \rrbracket_n \\ & \cong \bigotimes_i F_n^{m'_i} \llbracket r'_i \sigma'_i \mid N'_i \odot \Delta'_i \rrbracket_{m'_i} \otimes F_n^m \bigotimes_i F_m^{m_i} \llbracket q r_i \sigma_i \mid N_i \odot \Delta_i \rrbracket_{m_i} \\ & \xrightarrow{t_2 \otimes F_n^m(q \cdot t_1)} ((q \odot B) \multimap B') \otimes (q \odot B) \xrightarrow{ev} B' \end{aligned}$$

We claim that this morphism is equal to

$$\begin{aligned} & \bigotimes_i \llbracket r'_i \sigma'_i \mid N'_i \odot \Delta'_i \rrbracket_m \otimes \bigotimes_i \llbracket q r_i \sigma_i \mid N_i \odot \Delta_i \rrbracket_m \\ & \cong \bigotimes_i F_m^{m'_i} \llbracket r'_i \sigma'_i \mid N'_i \odot \Delta'_i \rrbracket_{m'_i} \otimes \bigotimes_i F_m^{m_i} \llbracket q r_i \sigma_i \mid N_i \odot \Delta_i \rrbracket_{m_i} \\ & \xrightarrow{(\bigotimes_i F_m^{m'_i}(r'_i \cdot e'_i)) \otimes (\bigotimes_i F_m^{m_i}((q r_i) \cdot e_i))} \bigotimes_i F_m^{m'_i}(r'_i \odot A'_i) \otimes \bigotimes_i F_m^{m_i}((q r_i) \odot A_i) \\ & \xrightarrow{t' \otimes (q \cdot t)} ((q \odot B) \multimap B') \otimes (q \odot B) \xrightarrow{ev} B' \end{aligned}$$

But this follows immediately from Lemma C.2. □

C.2 Semantic soundness of beta-conversion

Theorem C.3 *β -conversion is sound with respect to the categorical semantics: If $\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t_1 \equiv_{\beta} t_2 : A$, then the interpretations of $\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t_1 : A$ and $\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t_2 : A$ are equal in any categorical model.*

Proof. We proceed by induction, inspecting the proof of Theorem B.4, and verifying that the derivations provided there have the same interpretations. We will only consider the cases where the β -conversion happens as a result of an introduction rule being used against an elimination rule, since the other cases are straightforward.

Case 1 (Unit) In this case we have a morphism $\llbracket \rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{m}} \xrightarrow{e} I_{\mathbf{m}}$ and we must show that this morphism is equal to

$$\llbracket \rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{m}} \cong \llbracket \rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{m}} \otimes \llbracket \emptyset \mid \emptyset \odot \emptyset \rrbracket_{\mathbf{m}} \xrightarrow{e \otimes (\iota \circ (q \cdot \text{id}))} I_{\mathbf{m}} \otimes I_{\mathbf{m}} \cong I_{\mathbf{m}}.$$

But $q \cdot \text{id} = \iota^{-1} : I_{\mathbf{m}} \rightarrow q \odot I_{\mathbf{m}}$, from which the claim follows.

Case 2 (Pair) In this case we are given morphisms $\llbracket \sigma_i \mid \mathbf{N}_i \odot \Delta_i \rrbracket_{\mathbf{m}} \xrightarrow{e_i} A_i$ (for $i \in \{1, 2\}$) and

$$\llbracket \rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{n}} \otimes F_{\mathbf{n}}^{\mathbf{m}}(q \odot A_1) \otimes F_{\mathbf{n}}^{\mathbf{m}}(q \odot A_2) \xrightarrow{t} B$$

The interpretation of $\text{let}_{@q}(x_1, x_2) = (e_1, e_2)$ in t is

$$\begin{aligned} & \llbracket \rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{n}} \otimes F_{\mathbf{n}}^{\mathbf{m}} \llbracket q(\sigma_1, \sigma_2) \mid \mathbf{N}_1, \mathbf{N}_2 \odot \Delta_1, \Delta_2 \rrbracket_{\mathbf{m}} \\ & \rightarrow \llbracket \rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{n}} \otimes F_{\mathbf{n}}^{\mathbf{m}} q \odot \llbracket (\sigma_1, \sigma_2) \mid \mathbf{N}_1, \mathbf{N}_2 \odot \Delta_1, \Delta_2 \rrbracket_{\mathbf{m}} \\ & \cong \llbracket \rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{n}} \otimes F_{\mathbf{n}}^{\mathbf{m}} q \odot (\llbracket \sigma_1 \mid \mathbf{N}_1 \odot \Delta_1 \rrbracket_{\mathbf{m}} \otimes \llbracket \sigma_2 \mid \mathbf{N}_2 \odot \Delta_2 \rrbracket_{\mathbf{m}}) \\ & \xrightarrow{\text{id} \otimes F_{\mathbf{n}}^{\mathbf{m}} q \odot (e_1 \otimes e_2)} \llbracket \rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{n}} \otimes F_{\mathbf{n}}^{\mathbf{m}} q \odot (A_1 \otimes A_2) \\ & \cong \llbracket \rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{n}} \otimes F_{\mathbf{n}}^{\mathbf{m}}(q \odot A_1) \otimes F_{\mathbf{n}}^{\mathbf{m}}(q \odot A_2) \\ & \xrightarrow{t} B \end{aligned}$$

where the unlabeled morphism is constructed from μ and τ . On the other hand the interpretation of $[e_1/x_1, e_2/x_2]t$ is given by Lemma C.1 as:

$$\begin{aligned} & \llbracket \rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{n}} \otimes F_{\mathbf{n}}^{\mathbf{m}} \llbracket q(\sigma_1, \sigma_2) \mid \mathbf{N}_1, \mathbf{N}_2 \odot \Delta_1, \Delta_2 \rrbracket_{\mathbf{m}} \\ & \cong \llbracket \rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{n}} \otimes F_{\mathbf{n}}^{\mathbf{m}} \llbracket q\sigma_1 \mid \mathbf{N}_1 \odot \Delta_1 \rrbracket_{\mathbf{m}} \otimes F_{\mathbf{n}}^{\mathbf{m}} \llbracket q\sigma_2 \mid \mathbf{N}_2 \odot \Delta_2 \rrbracket_{\mathbf{m}} \\ & \rightarrow \llbracket \rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{n}} \otimes F_{\mathbf{n}}^{\mathbf{m}} q \odot \llbracket \sigma_1 \mid \mathbf{N}_1 \odot \Delta_1 \rrbracket_{\mathbf{m}} \otimes F_{\mathbf{n}}^{\mathbf{m}} q \odot \llbracket \sigma_2 \mid \mathbf{N}_2 \odot \Delta_2 \rrbracket_{\mathbf{m}} \\ & \xrightarrow{\text{id} \otimes F_{\mathbf{n}}^{\mathbf{m}}(q \odot e_1) \otimes F_{\mathbf{n}}^{\mathbf{m}}(q \odot e_2)} \llbracket \rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{n}} \otimes F_{\mathbf{n}}^{\mathbf{m}}(q \odot A_1) \otimes F_{\mathbf{n}}^{\mathbf{m}}(q \odot A_2) \\ & \xrightarrow{t} B \end{aligned}$$

where the unlabeled arrow is constructed from μ and τ . These morphisms are easily verified to be equal.

Case 3 (Function) We are given morphisms

$$\llbracket \rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{n}} \otimes F_{\mathbf{n}}^{\mathbf{m}}(q \odot A) \xrightarrow{t} B \qquad \llbracket \sigma \mid \mathbf{N} \odot \Delta \rrbracket_{\mathbf{m}} \xrightarrow{e} A$$

By the formula for transposing morphism along adjunctions, the interpretation $(\lambda x.t)$ is

$$\llbracket \rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{n}} \xrightarrow{\text{unit}} F_{\mathbf{n}}^{\mathbf{m}}(q \odot A) \multimap (\llbracket \rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{n}} \otimes F_{\mathbf{n}}^{\mathbf{m}}(q \odot A)) \xrightarrow{F_{\mathbf{n}}^{\mathbf{m}}(q \odot A) \multimap t} F_{\mathbf{n}}^{\mathbf{m}}(q \odot A) \multimap B$$

and thus the interpretation $(\lambda x.t) e$ is

$$\llbracket \rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{n}} \otimes F_{\mathbf{n}}^{\mathbf{m}} \llbracket q\sigma \mid \mathbf{N} \odot \Delta \rrbracket_{\mathbf{m}} \xrightarrow{((F_{\mathbf{n}}^{\mathbf{m}}(q \odot A) \multimap t) \circ \text{unit}) \otimes F_{\mathbf{n}}^{\mathbf{m}}(q \cdot e)} (F_{\mathbf{n}}^{\mathbf{m}}(q \odot A) \multimap B) \otimes F_{\mathbf{n}}^{\mathbf{m}}(q \odot A) \xrightarrow{ev} B$$

We claim that this morphism is equal to

$$\llbracket \rho \mid M \odot \Gamma \rrbracket_n \otimes F_n^m \llbracket q\sigma \mid N \odot \Delta \rrbracket_m \xrightarrow{\text{id} \otimes F_n^m(q \cdot e)} \llbracket \rho \mid M \odot \Gamma \rrbracket_n \otimes F_n^m(q \odot A) \xrightarrow{t} B$$

To see that these morphisms are equal, consider the following diagram:

$$\begin{array}{ccccc}
 \llbracket \rho \mid M \odot \Gamma \rrbracket_n \otimes F_n^m \llbracket q\sigma \mid N \odot \Delta \rrbracket_m & & & & \\
 \downarrow \text{unit} \otimes \text{id} & \searrow \text{id} \otimes F_n^m(q \cdot e) & & & \\
 F_n^m(q \odot A) \multimap (\llbracket \rho \mid M \odot \Gamma \rrbracket_n \otimes F_n^m(q \odot A)) & & \llbracket \rho \mid M \odot \Gamma \rrbracket_n \otimes F_n^m(q \odot A) & & \\
 \otimes F_n^m \llbracket q\sigma \mid N \odot \Delta \rrbracket_m & \searrow \text{id} \otimes F_n^m(q \cdot e) & \downarrow \text{unit} \otimes \text{id} & \searrow \text{id} & \\
 (F_n^m(q \odot A) \multimap t) \otimes \text{id} & & F_n^m(q \odot A) \multimap (\llbracket \rho \mid M \odot \Gamma \rrbracket_n \otimes F_n^m(q \odot A)) & & \\
 & & \otimes F_n^m(q \odot A) & & \\
 \downarrow & \searrow (F_n^m(q \odot A) \multimap t) \otimes \text{id} & \downarrow \text{ev} & & \\
 (F_n^m(q \odot A) \multimap B) \otimes F_n^m \llbracket q\sigma \mid N \odot \Delta \rrbracket_m & & (\llbracket \rho \mid M \odot \Gamma \rrbracket_n \otimes F_n^m(q \odot A)) & & \\
 \searrow \text{id} \otimes F_n^m(q \cdot e) & \downarrow (F_n^m(q \odot A) \multimap t) \otimes \text{id} & \downarrow t & & \\
 & (F_n^m(q \odot A) \multimap B) \otimes F_n^m(q \odot A) & & & \\
 & \searrow \text{ev} & & & \\
 & & B & &
 \end{array}$$

The left/bottom edge is the former morphism, and the top/right edge is the latter morphism. The top left and bottom right squares commute by naturality of the unit and counit, while the bottom left square commutes by functoriality of $\multimap$. Finally the triangle commutes by one of triangle identities for adjunctions.

Case 4 (Drop) We are given $l \leq n \leq m$, and morphisms

$$\llbracket \rho \mid M \odot \Gamma \rrbracket_m \xrightarrow{e} A \qquad \llbracket \sigma \mid N \odot \Delta \rrbracket_l \otimes F_l^m(q \odot A) \xrightarrow{t} B$$

The categorical interpretation $\text{let}_{@q} \downarrow_{n \leq m} x = \downarrow_{n \leq m}^q e$ in t is

$$\begin{aligned}
 & \llbracket \sigma \mid N \odot \Delta \rrbracket_l \otimes \llbracket q\rho \mid M \odot \Gamma \rrbracket_l \\
 & \cong \llbracket \sigma \mid N \odot \Delta \rrbracket_l \otimes F_l^n F_n^m \llbracket q\rho \mid M \odot \Gamma \rrbracket_m \\
 & \xrightarrow{F_l^n F_n^m(q \cdot e)} \llbracket \sigma \mid N \odot \Delta \rrbracket_l \otimes F_l^n F_n^m(q \odot A) \\
 & \cong \llbracket \sigma \mid N \odot \Delta \rrbracket_l \otimes F_l^m(q \odot A) \\
 & \xrightarrow{t} B
 \end{aligned}$$

On the other hand, by Lemma C.1, the interpretation of $[e/x]t$ is

$$\begin{aligned} & \llbracket \sigma \mid \mathbf{N} \odot \Delta \rrbracket_{\mathbf{I}} \otimes \llbracket q\rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{I}} \\ & \cong \llbracket \sigma \mid \mathbf{N} \odot \Delta \rrbracket_{\mathbf{I}} \otimes F_{\mathbf{I}}^{\mathbf{m}} \llbracket q\rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{m}} \\ & \xrightarrow{F_{\mathbf{I}}^{\mathbf{m}}(q \cdot e)} \llbracket \sigma \mid \mathbf{N} \odot \Delta \rrbracket_{\mathbf{I}} \otimes F_{\mathbf{I}}^{\mathbf{m}}(q \odot A) \\ & \xrightarrow{t} B \end{aligned}$$

And it is straightforward verify that these morphisms are equal.

Case 5 (Raise) We are given $\mathbf{m} \leq \mathbf{n} \leq \mathbf{M}$ and a morphism

$$\llbracket \rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{m}} \xrightarrow{t} A$$

We have a commutative diagram

$$\begin{array}{ccccccc} \llbracket \rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{m}} & \xrightarrow{\sim} & F_{\mathbf{m}}^{\mathbf{n}} \llbracket \rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{n}} & \xrightarrow{F_{\mathbf{m}}^{\mathbf{n}}(\text{unit})} & F_{\mathbf{m}}^{\mathbf{n}} G_{\mathbf{m}}^{\mathbf{n}} F_{\mathbf{m}}^{\mathbf{n}} \llbracket \rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{n}} & \xrightarrow{\sim} & F_{\mathbf{m}}^{\mathbf{n}} G_{\mathbf{m}}^{\mathbf{n}} \llbracket \rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{m}} & \xrightarrow{F_{\mathbf{m}}^{\mathbf{n}} G_{\mathbf{m}}^{\mathbf{n}} t} & F_{\mathbf{m}}^{\mathbf{n}} G_{\mathbf{m}}^{\mathbf{n}} A \\ & & & & \downarrow & & \downarrow & & \downarrow \\ & & & & F_{\mathbf{m}}^{\mathbf{n}} \llbracket \rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{n}} & \xrightarrow{\sim} & \llbracket \rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{m}} & \xrightarrow{t} & A \end{array}$$

where unlabeled vertical arrows are the counit of $F_{\mathbf{n}}^{\mathbf{m}} \dashv G_{\mathbf{n}}^{\mathbf{m}}$. The composition across the top and right edges is the interpretation of $\uparrow_{\mathbf{m} \leq \mathbf{n}}^{-1}(\uparrow_{\mathbf{m} \leq \mathbf{n}} t)$, and this diagram shows that that morphism is equal to t , concluding this case. $\square$

C.3 Semantic soundness of eta-conversion

Theorem C.4 *η -conversion is sound with respect to the categorical semantics: If $\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t_1 \equiv_{\eta} t_2 : A$, then the interpretations of $\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t_1 : A$ and $\rho \mid \mathbf{M} \odot \Gamma \vdash_{\mathbf{m}} t_2 : A$ are equal in any categorical model.*

Proof. We have five cases to consider.

Case 1 (Pair) In this case we are given a morphism $\bigotimes_i F_{\mathbf{m}}^{\mathbf{m}_i}(q_i \odot B_i) = \llbracket \sigma \mid \mathbf{N} \odot \Delta \rrbracket_{\mathbf{m}} \xrightarrow{e} A_1 \otimes A_2$ and we must show that it is equal to the composite

$$\llbracket \sigma \mid \mathbf{N} \odot \Delta \rrbracket_{\mathbf{m}} \cong F_{\mathbf{m}}^{\mathbf{m}} \llbracket \sigma \mid \mathbf{N} \odot \Delta \rrbracket_{\mathbf{m}} \xrightarrow{m^{-1} \circ F_{\mathbf{m}}^{\mathbf{m}}(\tau^{-1} \circ (1 \cdot e))} F_{\mathbf{m}}^{\mathbf{m}}(1 \odot A_1) \otimes F_{\mathbf{m}}^{\mathbf{m}}(1 \odot A_2) \cong (1 \odot A_1) \otimes (1 \odot A_2) \xrightarrow{\epsilon \otimes \epsilon} A_1 \otimes A_2$$

with the morphism $F_{\mathbf{m}}^{\mathbf{m}}(1 \odot A_1) \otimes F_{\mathbf{m}}^{\mathbf{m}}(1 \odot A_2) \cong (1 \odot A_1) \otimes (1 \odot A_2) \xrightarrow{\epsilon \otimes \epsilon} A_1 \otimes A_2$ being the interpretation of $1, 1 \mid \mathbf{m}, \mathbf{m} \odot x_1 : A_1, x_2 : A_2 \vdash_{\mathbf{m}} (x_1, x_2) : A_1 \otimes A_2$. This follows from the following commutative diagram

$$\begin{array}{ccccc}
 \bigotimes_i F_{\mathbf{m}}^{\mathbf{m}_i}(q_i \odot B_i) & \xrightarrow{\sim} & F_{\mathbf{m}}^{\mathbf{m}} \bigotimes_i F_{\mathbf{m}}^{\mathbf{m}_i}(1 \cdot q_i \odot B_i) & \xrightarrow{F_{\mathbf{m}}^{\mathbf{m}} \bigotimes_i \mu \circ F_{\mathbf{m}}^{\mathbf{m}_i} \delta} & F_{\mathbf{m}}^{\mathbf{m}} \bigotimes_i 1 \odot F_{\mathbf{m}}^{\mathbf{m}_i}(q_i \odot B_i) \\
 \downarrow e & & \parallel & \swarrow F_{\mathbf{m}}^{\mathbf{m}} \bigotimes_i \epsilon & \downarrow F_{\mathbf{m}}^{\mathbf{m}} \tau \\
 & & F_{\mathbf{m}}^{\mathbf{m}} \bigotimes_i F_{\mathbf{m}}^{\mathbf{m}_i}(q_i \odot B_i) & \xleftarrow{F_{\mathbf{m}}^{\mathbf{m}} \epsilon} & F_{\mathbf{m}}^{\mathbf{m}} 1 \odot \bigotimes_i F_{\mathbf{m}}^{\mathbf{m}_i}(q_i \odot B_i) \\
 & & \downarrow F_{\mathbf{m}}^{\mathbf{m}} e & & \downarrow F_{\mathbf{m}}^{\mathbf{m}}(1 \odot e) \\
 & & F_{\mathbf{m}}^{\mathbf{m}}(A_1 \otimes A_2) & \xleftarrow{F_{\mathbf{m}}^{\mathbf{m}} \epsilon} & F_{\mathbf{m}}^{\mathbf{m}}(1 \odot (A_1 \otimes A_2)) \\
 & \nearrow \sim & \downarrow m^{-1} & \swarrow F_{\mathbf{m}}^{\mathbf{m}}(\epsilon \otimes \epsilon) & \downarrow F_{\mathbf{m}}^{\mathbf{m}}(\tau^{-1}) \\
 & & F_{\mathbf{m}}^{\mathbf{m}}(A_1) \otimes F_{\mathbf{m}}^{\mathbf{m}}(A_2) & \xleftarrow{F_{\mathbf{m}}^{\mathbf{m}}(\epsilon) \otimes F_{\mathbf{m}}^{\mathbf{m}}(\epsilon)} & F_{\mathbf{m}}^{\mathbf{m}}((1 \odot A_1) \otimes (1 \odot A_2)) \\
 & & \nearrow \sim & & \downarrow m^{-1} \\
 A_1 \otimes A_2 & \xleftarrow{\epsilon \otimes \epsilon} & (1 \odot A_1) \otimes (1 \odot A_2) & \xleftarrow{\sim} & F_{\mathbf{m}}^{\mathbf{m}}(1 \odot A_1) \otimes F_{\mathbf{m}}^{\mathbf{m}}(1 \odot A_2)
 \end{array}$$

Case 2 (Unit) Similar to the Pair case, using the fact that $\iota : I \rightarrow 1 \odot I$ and $\epsilon : 1 \odot I \rightarrow I$ are inverse morphisms.

Case 3 (Raise) Given a morphism $e : \llbracket \rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{n}} \rightarrow G_{\mathbf{m}}^{\mathbf{n}} A$, applying RAISE.E and RAISE.I yields the morphism

$$\llbracket \rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{n}} \xrightarrow{\text{unit}} G_{\mathbf{m}}^{\mathbf{n}} F_{\mathbf{m}}^{\mathbf{n}} \llbracket \rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{n}} \xrightarrow{G_{\mathbf{m}}^{\mathbf{n}} F_{\mathbf{m}}^{\mathbf{n}} e} G_{\mathbf{m}}^{\mathbf{n}} F_{\mathbf{m}}^{\mathbf{n}} G_{\mathbf{m}}^{\mathbf{n}} A \xrightarrow{G_{\mathbf{m}}^{\mathbf{n}} \text{counit}} G_{\mathbf{m}}^{\mathbf{n}} A$$

and we must show that this morphism is equal to e , which is straightforward.

For the final two cases we make the following observation: Write $v_A : F_{\mathbf{m}}^{\mathbf{m}}(1 \odot A) \cong 1 \odot A \xrightarrow{\epsilon} A$. Then for any q , the composite $q \odot A \cong F_{\mathbf{m}}^{\mathbf{m}}(q \odot A) \xrightarrow{q \odot v_A} q \odot A$ is the identity. This follows from the diagram

$$\begin{array}{ccccccc}
 q \odot A & \xrightarrow{\sim} & F_{\mathbf{m}}^{\mathbf{m}}(q \odot A) & \xrightarrow{F_{\mathbf{m}}^{\mathbf{m}} \delta} & F_{\mathbf{m}}^{\mathbf{m}}(q \odot 1 \odot A) & \xrightarrow{\mu} & q \odot F_{\mathbf{m}}^{\mathbf{m}}(1 \odot A) & \xrightarrow{q \odot v_A} & q \odot 1 \odot A \\
 & \searrow & \downarrow \sim & & \downarrow \sim & & \downarrow \sim & \nearrow q \odot \epsilon & \\
 & & q \odot A & \xrightarrow{\delta} & q \odot 1 \odot A & \xlongequal{\quad} & q \odot 1 \odot A & &
 \end{array}$$

The composite across the bottom edge is the identity by the laws for colax monoidal functors. The right square commutes by Lemma 4.10 applied to the isomorphism $G_{\mathbf{m}}^{\mathbf{m}} \cong \text{id}$.

Case 4 (Drop) In this case we are given a morphism $e : \llbracket \rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{n}} \rightarrow F_{\mathbf{n}}^{\mathbf{m}}(q \odot A)$. We must show that it is equal to the composite

$$\llbracket \rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{n}} \cong F_{\mathbf{n}}^{\mathbf{n}} \llbracket \rho \mid \mathbf{M} \odot \Gamma \rrbracket_{\mathbf{n}} \xrightarrow{F_{\mathbf{n}}^{\mathbf{n}} e} F_{\mathbf{n}}^{\mathbf{n}} F_{\mathbf{n}}^{\mathbf{m}}(q \odot A) \cong F_{\mathbf{n}}^{\mathbf{m}}(q \odot A) \cong F_{\mathbf{n}}^{\mathbf{m}} F_{\mathbf{m}}^{\mathbf{m}}(q \odot A) \xrightarrow{F_{\mathbf{n}}^{\mathbf{m}}(q \odot v_A)} F_{\mathbf{n}}^{\mathbf{m}}(q \odot A)$$

but this is straightforward by the above observation.

Case 5 (Functions) In this case we are given a morphism $e: \llbracket \rho \mid M \odot \Gamma \rrbracket_n \rightarrow F_n^m(q \odot A) \multimap B$. Applying the rule to this morphism and v_A produces the morphism

$$\llbracket \rho \mid M \odot \Gamma \rrbracket_n \otimes F_n^m(q \odot A) \cong \llbracket \rho \mid M \odot \Gamma \rrbracket_n \otimes F_n^m F_m^m(q \odot A) \xrightarrow{e \otimes F_n^m(q \cdot v_A)} (F_n^m(q \odot A) \multimap B) \otimes F_n^m(q \odot A) \xrightarrow{ev} B$$

By the above observation this morphism is equal to $ev \circ (e \otimes F_n^m(q \odot A))$, that is, the transpose of e along the adjunction $(-) \otimes F_n^m(q \odot A) \dashv F_n^m(q \odot A) \multimap (-)$. Applying to this morphism therefore yields e , which is what we needed to show. □